

Limit Interfaces and Resource Laws in Hypercomputation

A Working Theory of Zeno Machines, Readout Discontinuity, and Endpoint Cost

Research Draft

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Abstract

This paper develops a draft framework for analyzing hypercomputation as a problem of *limit interfaces*. The central hypothesis is that finite-time hypercomputation is not produced by acceleration alone. It is produced by coupling an ordinary computable trace generator to a singular observation functional, a singular spacetime interface, a noncomputable physical input, or an unbounded resource profile. The paper introduces fifteen linked components: a trace/readout collapse theorem, a Zeno aperture calculus for accelerating schedules, a horizon-energy law derived from a generic distinguishable-action assumption, a reliability summability law for supertasks with infinitely many error opportunities, a finite signal-budget law for robust endpoint observation, a reliability-corrected endpoint-energy law, a statistical endpoint bound for noisy readout, a de-Zenoization theorem for finite robust shadows, finite-power endpoint-latch laws, Zeno-utilization bounds, quantifier-shadow pivotal volume, Gaussian look-elsewhere debt for bounded jump-like observers, Bayesian endpoint waterline allocation, endpoint-address capacity shadows, online aggregation/frontier-width laws, and quantum endpoint entropy shadows. These components are presented as candidate theory rather than as a settled impossibility theorem. The mathematical claims that are proved here are elementary but expose a useful invariant: any proposed Zeno-type hypercomputer must pay for its non-Turing power through discontinuity, unbounded endpoint cost, non-robust observation, infinite signal budget, summably vanishing logical error, or imported noncomputability. Version 0.3 adds Gaussian endpoint experiments, a robust shadow equation, distributional finite-shadow tests, and a theorem showing that once robustness collapses an ideal supertask to a finite certified depth, uniform finite scheduling is peak-energy optimal. Version 0.4 adds endpoint latch/impulse complementarity, a finite-power latch horizon, a Zeno utilization theorem proving that every summable infinite schedule has vanishing finite-power utilization, and stress tests identifying a log-slow schedule that is near-linear in finite-power writable depth while catastrophically slow in endpoint convergence. Version 0.5 adds quantifier-shadow volume, certificate/pivotal separation, Gaussian look-elsewhere debt, and average pivotality experiments for bounded quantifier trees. Version 0.6 adds Bayesian finite-shadow theory: an endpoint waterline theorem, prior-tail active-horizon laws, and numerical allocation experiments showing that a finite-resource distributional endpoint optimally develops an explicit blind tail. Version 0.7 adds coded endpoint and online aggregation theory: a Fano endpoint-address capacity shadow, a witness-address debt law, residual-width lower bounds for online aggregators, and an alternating-frontier theorem showing that fair interleavings of bounded $\exists\forall$ shadows require $\Omega(N)$ reliable state bits while row-major shadows use constant width at the cost of unfairness in the infinite limit. Version 0.8 adds causal certificate-flow laws. Version 0.9 adds finite-bandwidth analog endpoint shadows. Version 0.10 adds quantum endpoint shadows: a Mandelstam–Tamm/Helstrom event-time horizon, a Holevo–Fano entropy address bound, Bekenstein and holographic audit equations, and finite-dimensional quantum crowding experiments.

Keywords: hypercomputation; Zeno machine; accelerated Turing machine; supertask; computable analysis; Malament-Hogarth spacetime; quantum speed limit; reliability; finite precision; robust observation.

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1 Purpose and status of the draft

The aim is to isolate blindspots in current discussions of hypercomputation and to propose new equations, laws, and conjectures that can guide further formal and computational work. The theory developed here is intentionally modular. It does not assert that every hypercomputational model is physically impossible. Instead it asks a diagnostic question:

Where exactly does the non-Turing power enter the device?

The answer proposed here is that non-Turing power appears at a *limit interface*. A limit interface is the point where an infinite internal trace is converted into a finite observable answer. A Zeno machine compresses infinitely many steps into finite external time. An infinite time Turing machine moves the interface to ordinal limit stages. A Malament-Hogarth construction moves the interface into causal geometry. An oracle or real-valued physical parameter imports the interface as noncomputable information.

The literature already contains many warnings about the interpretation of accelerated machines. Potgieter reviews Zeno machines as Turing machines with accelerating clocks and emphasizes that claims of having broken the Turing barrier should be treated carefully [1]. Hamkins and Lewis develop infinite time Turing machines as a transfinite model with substantial decidability strength over reals [2]. Calude and Staiger show, among other results, that finite-space accelerated Turing machines do not solve uncomputable problems and that natural accelerated-machine acceptance

semantics yield Boolean closures of low arithmetical classes [3]. Welch analyzes the computational reach of Malament-Hogarth spacetimes, including hyperarithmetical and Borel bounds [4]. Physical limits to speed and storage are framed by quantum speed-limit and finite-resource work such as Margolus–Levitin and Lloyd [6, 7]. A recent preprint by Borsten and Kim argues that relativistic acceleration schemes are restricted by quantum field theory and quantum-gravity effects [8]. This paper builds a bridge between these threads.

1.1 Draft contributions

The proposed contributions are:

1. A **limit-interface conservation law**: if the internal trace is computable and the observation functional is effectively finite, then the whole device computes only a Turing-computable predicate.
2. A **Zeno aperture calculus**: a schedule is represented by

$$\rho_n = \frac{\tau_n}{R_n}, \quad R_n = \sum_{k=n}^{\infty} \tau_k,$$

which separates geometric, polynomial, and stretched-exponential supertasks by their approach to the endpoint.

3. A **horizon-energy law**:

$$E_n \geq \frac{Q}{\rho_n R_n},$$

under the assumption that each distinguishable logical transition requires action at least Q .

4. A **reliability summability law**: for independent uncorrected step errors,

$$P_{\text{clean}} = \prod_n (1 - p_n) > 0 \iff \sum_n p_n < \infty.$$

5. A **robustness-collapse lemma**: any readout strong enough to decide a halting-style existential tail property is non-robust at the nonhalting trace.
6. A **finite signal-budget law**: with total endpoint signal budget B and additive uncertainty η , robust single-event detection over N possible event times requires $N < B/(2\eta)$ in the scalar budget model.
7. A **reliability-corrected horizon-energy law**: if logical error is suppressed exponentially in a correction cost, summable reliability imposes a logarithmic correction on the endpoint-energy lower bound.
8. A **statistical endpoint law**: finite endpoint signal energy plus nonzero Gaussian observation noise imposes a finite uniform detection horizon.
9. A **de-Zenoization theorem**: once robustness reduces an ideal supertask to a finite certified depth, the peak-energy-optimal schedule is uniform rather than accelerated.
10. A **finite-power latch theory**: persistent output cells do not evade endpoint resource laws; they transfer the burden to late writes. For any summable infinite schedule, the finite-power utilization obeys

$$\frac{Q_m N_{\text{lat}}(P; m)}{PT} \rightarrow 0.$$

11. A **quantifier-shadow theory**: bounded jump-like observers pay for pivotal volume and Gaussian look-elsewhere debt, not merely for the size of one certificate.
12. A **Bayesian finite-shadow waterline**: under a computable prior and an exponential endpoint miss model, the optimal finite-budget allocation is

$$b_n^* = \kappa^{-1}[\log(\pi_n/\theta)]_+, \quad \sum_n [\log(\pi_n/\theta)]_+ = \kappa B,$$

so the protected active set is finite for every finite budget.

13. An **endpoint-address capacity shadow**: if an endpoint must return one of M auditable witness addresses through a channel of capacity C bits with error at most α , then

$$\log_2 M \leq \frac{C + h_2(\alpha)}{1 - \alpha}.$$

14. An **online aggregation/frontier law**: if a bounded shadow F is streamed in order π , exact aggregation requires at least

$$W_\pi(F) = \max_t |\{F(a, \cdot) : a \in \{0, 1\}^t\}|$$

residual states. For $F_N^{\exists \forall} = \bigvee_i \bigwedge_j x_{ij}$, $2^{f_\pi} \leq W_\pi \leq 2^{f_\pi} + 1$, where f_π is the maximum open-row frontier.

15. A **quantum endpoint shadow**: finite quantum speed, finite entropy, or finite Hilbert dimension imposes finite auditable event-address horizons. In particular,

$$N_{\text{QSL}}(\Delta E, \alpha) = \#\{n : \tau_n \geq \hbar \arcsin(1 - 2\alpha)/\Delta E\}$$

and

$$\log_2 M \leq \frac{S_{\max} + h_2(\alpha)}{1 - \alpha}.$$

The likely novelty is not each individual mathematical ingredient. The novelty being developed is the *synthesis*: hypercomputation can be studied by decomposing every proposal into trace generation, limit observation, endpoint resource divergence, and infinite reliability.

2 Trace/readout model

Let Σ be a finite alphabet. A computation that runs through countably many internal stages produces a trace

$$z = (z_0, z_1, z_2, \dots) \in \Sigma^\omega.$$

The trace may record machine states, output-cell values, emitted signals, or any finite symbolic abstraction of the process.

Definition 2.1 (Trace/readout architecture). A limit computation architecture is a pair

$$\mathcal{A} = (G, O)$$

where G is a trace generator and $O : \Sigma^\omega \rightarrow A$ is a finite-valued readout, with A usually $\{0, 1\}$. On input x , the computed value is

$$f_{\mathcal{A}}(x) = O(G(x)).$$

For a Zeno halting machine, $G(e, x)$ can be the trace

$$b_n = \begin{cases} 1, & \text{if machine } e \text{ has halted on } x \text{ by stage } n, \\ 0, & \text{otherwise,} \end{cases}$$

and the readout is

$$O_{\exists}(b) = \sup_n b_n.$$

The trace generator is ordinary simulation. The nonordinary part is the conversion of an infinite sequence into a final bit at the Zeno endpoint.

2.1 Limit-interface collapse

Definition 2.2 (Effectively finite readout). A readout $O : \Sigma^\omega \rightarrow A$ is effectively finite if there is a Turing functional Φ such that, for every trace z , Φ^z halts after finitely many oracle queries and outputs $O(z)$.

This is the formal version of an observer that can determine its finite answer from finitely much of the trace.

Theorem 2.3 (Trace-continuity collapse). *Let G be a computable trace generator and let O be effectively finite. Then $x \mapsto O(G(x))$ is Turing-computable.*

Proof. On input x , simulate the oracle functional $\Phi^{G(x)}$. Whenever Φ queries the n -th symbol of its oracle, compute the n -th symbol of $G(x)$ and return it. Because Φ^z halts on every trace z , this simulation halts and outputs $O(G(x))$. Therefore the composite function is Turing-computable. $\square$

Corollary 2.4 (Discontinuity requirement). *If a trace/readout architecture with computable G computes a non-Turing predicate, then the readout is not effectively finite. In topological terms, some part of the readout behaves like a discontinuous or non-effectively-continuous functional on the relevant trace space.*

Proof. Immediate from [Theorem 2.3](#). Every total oracle computation that halts after finitely many trace queries induces a continuous finite-valued map on Cantor space. $\square$

The Zeno halting readout

$$O_{\exists}(b) = 1 \iff \exists n \, b_n = 1$$

is continuous at every trace that contains a 1, but it is discontinuous at 0^ω . No finite prefix of 0^ω certifies that all future entries are zero. Thus the Zeno device's decisive operation is not ordinary simulation. It is endpoint evaluation of a discontinuous existential tail property.

Candidate law 2.5 (Limit-interface conservation). *For computable internal dynamics,*

$$\text{computational power of the device} \leq \text{computable trace generation} + \text{degree of the readout interface.}$$

Equivalently, in Weihrauch-style terms,

$$f_A \leq_W O$$

whenever the preprocessing map $x \mapsto G(x)$ is computable.

The displayed inequality is a diagnostic, not a deep theorem: the trace is a computable preprocessing of the input, and the final answer is a postprocessing of the readout. Its value is conceptual. It says that a proposed hypercomputer should be classified by the degree of its readout, not only by the speed of its internal steps.

3 Robustness and the nonhalting trace

Physical observation is never merely a set-theoretic map. A readout should also be stable under small perturbations. For Σ^ω , use the usual ultrametric

$$d(z, y) = 2^{-k},$$

where k is the first index at which z and y differ; $d(z, z) = 0$.

Definition 3.1 (Pointwise robustness). A readout $O : \Sigma^\omega \rightarrow A$ is robust at z if there exists an m such that all traces sharing the prefix $z \upharpoonright m$ have the same output as z .

Lemma 3.2 (Robustness equals continuity for finite outputs). *A finite-valued readout on Σ^ω is robust at z exactly when it is continuous at z .*

Proof. Basic open neighborhoods in the product topology are prefix cylinders. Robustness is precisely local constancy on some prefix cylinder. For a discrete finite codomain, local constancy is equivalent to continuity at the point. $\square$

Proposition 3.3 (Nonhalting instability of existential readout). *The readout $O_\exists(b) = 1 \iff \exists n b_n = 1$ is not robust at 0^ω . Moreover, every neighborhood of 0^ω contains traces with output 0 and traces with output 1.*

Proof. Fix any prefix length m . The all-zero trace extends that prefix and has output 0. The trace with zeros through position m and a 1 at position $m + 1$ also extends that prefix and has output 1. Thus no finite prefix neighborhood of 0^ω has constant output. $\square$

This gives a sharp interpretation of the Zeno halting construction. Halting instances are one-sided stable, because a finite witness eventually appears. Nonhalting instances are not stable, because they require the endpoint to certify an infinite absence. Any physical realization must therefore implement an infinite negative certificate at finite time.

4 Zeno aperture calculus

Let the n -th internal step occupy external duration $\tau_n > 0$. A Zeno schedule satisfies

$$T = \sum_{n=0}^{\infty} \tau_n < \infty.$$

Define the remaining external time before step n by

$$R_n = \sum_{k=n}^{\infty} \tau_k.$$

The basic variable of the proposed calculus is the aperture

$$\boxed{\rho_n = \frac{\tau_n}{R_n}.$$

It measures the fraction of all remaining time spent on the next step.

Because $R_{n+1} = R_n - \tau_n$,

$$\boxed{R_{n+1} = R_n(1 - \rho_n).}$$

Let

$$a_n = -\log(1 - \rho_n), \quad A_n = \sum_{k < n} a_k.$$

Then

$$R_n = T \exp(-A_n).$$

Theorem 4.1 (Aperture completion criterion). *Let $0 < \rho_n < 1$. The schedule generated by*

$$R_0 = T, \quad R_{n+1} = R_n(1 - \rho_n), \quad \tau_n = \rho_n R_n$$

has infinitely many positive steps completed in finite external time T if and only if

$$A_n = \sum_{k < n} -\log(1 - \rho_k) \rightarrow \infty.$$

Proof. The partial sum through step $N - 1$ is $T - R_N$. The infinite sum is T exactly when $R_N \rightarrow 0$. Since $R_N = T \exp(-A_N)$, this is equivalent to $A_N \rightarrow \infty$. $\square$

The number of steps visible before an endpoint resolution ε is

$$N(\varepsilon) = \min\{n : R_n \leq \varepsilon\} = \min\{n : A_n \geq \log(T/\varepsilon)\}.$$

This equation separates the mathematical idealization from observational precision. A finite-precision observer does not see “infinity”; it sees $N(\varepsilon)$.

4.1 Canonical aperture families

Constant aperture. If $\rho_n = \rho$, then

$$R_n = T(1 - \rho)^n, \quad N(\varepsilon) \sim \frac{\log(T/\varepsilon)}{-\log(1 - \rho)}.$$

The usual halving schedule has $\rho = 1/2$.

Polynomial tail. If

$$R_n = \frac{T}{(n+1)^a}, \quad a > 0,$$

then

$$\rho_n = 1 - \left(\frac{n+1}{n+2}\right)^a \sim \frac{a}{n},$$

and

$$N(\varepsilon) \sim (T/\varepsilon)^{1/a}.$$

Polynomial tails expose far more steps before a given endpoint precision, but they have smaller aperture near the horizon.

Stretched-exponential tail. If

$$R_n = T \exp(-cn^b), \quad c > 0,$$

then

$$N(\varepsilon) \sim \left(\frac{\log(T/\varepsilon)}{c}\right)^{1/b}.$$

For $0 < b < 1$, the aperture decays slowly to zero; for $b = 1$, it is asymptotically constant; for $b > 1$, it tends to one.

5 Endpoint resource laws

Acceleration by itself is kinematic. To connect the schedule to physics, introduce a minimal transition-cost assumption.

Assumption 5.1 (Distinguishable-action lower bound). There exists $Q > 0$ such that each logically distinguishable transition of the computing substrate satisfies

$$E_n \tau_n \geq Q,$$

where E_n is the characteristic available energy scale during step n .

This assumption is deliberately abstract. In an ideal quantum orthogonalization setting, a Margolus–Levitin-type speed limit gives a constant of order $\hbar$ for a distinct state transition [6]. The arguments below only need a positive lower action per distinguishable operation.

Theorem 5.2 (Horizon-energy law). *Under Assumption 5.1, a Zeno schedule satisfies*

$$E_n \geq \frac{Q}{\tau_n} = \frac{Q}{\rho_n R_n}.$$

Consequently, every infinite finite-time schedule with distinguishable steps has

$$\sup_n E_n = \infty$$

and

$$\sum_{n=0}^{\infty} E_n \tau_n = \infty.$$

Proof. Since $\tau_n = \rho_n R_n$, the first inequality is Assumption 5.1 rewritten. For an infinite finite-time schedule, $R_n \rightarrow 0$ and $0 < \rho_n < 1$, so $\rho_n R_n \rightarrow 0$. Hence $Q/(\rho_n R_n) \rightarrow \infty$ along the sequence. The action sum satisfies

$$\sum_n E_n \tau_n \geq \sum_n Q = \infty.$$

□

Corollary 5.3 (Bounded-resource collapse). *If $E_n \leq E_{\max} < \infty$, then at most $TE_{\max}/Q$ distinguishable steps can occur in external time T . If the total action budget satisfies $\sum_n E_n \tau_n \leq A_{\max}$, then at most $A_{\max}/Q$ distinguishable steps can occur.*

Proof. Each step consumes at least Q action. With bounded energy, each step takes at least $Q/E_{\max}$ time. With bounded total action, the count is bounded by $A_{\max}/Q$. □

5.1 Endpoint-energy exponent

Define the endpoint lower bound at precision ε by

$$E_\varepsilon = \frac{Q}{\rho_{N(\varepsilon)} R_{N(\varepsilon)}}.$$

Since $R_{N(\varepsilon)} \lesssim \varepsilon$, the residual schedule dependence is carried by $\rho_{N(\varepsilon)}$. A useful index is

$$\chi = \limsup_{\varepsilon \downarrow 0} \frac{\log(E_\varepsilon/Q)}{\log(1/\varepsilon)}.$$

The horizon-energy law gives

$$\chi \geq 1 + \limsup_{\varepsilon \downarrow 0} \frac{-\log \rho_{N(\varepsilon)}}{\log(1/\varepsilon)}.$$

Thus constant-aperture schedules have the optimal exponent $\chi = 1$, while polynomial tails have

$$R_n \asymp n^{-a} \implies E_\varepsilon = \Omega\left(\varepsilon^{-(1+1/a)}\right).$$

This is a counterintuitive design conclusion: schedules that make many more steps visible before a fixed endpoint precision can demand worse endpoint energy scaling.

Candidate law 5.4 (Aperture-resource tradeoff). *A finite-time infinite computation with distinguishable transitions cannot make $R_n \rightarrow 0$ and $\rho_n R_n \not\rightarrow 0$ simultaneously. The endpoint cost is governed by*

$$\text{endpoint cost} \sim \frac{1}{\text{aperture} \times \text{remaining time}}.$$

6 Reliability summability

A supertask also contains infinitely many opportunities for error. Let p_n be the probability that step n causes an uncorrected error that corrupts the final answer.

Theorem 6.1 (Independent reliability summability). *Assume independent uncorrected step errors with probabilities $0 \leq p_n < 1$. The probability of a completely clean infinite run is*

$$P_{\text{clean}} = \prod_{n=0}^{\infty} (1 - p_n).$$

Moreover,

$$P_{\text{clean}} > 0 \iff \sum_{n=0}^{\infty} p_n < \infty.$$

Proof. The finite clean probability through step N is $\prod_{n < N} (1 - p_n)$. Taking $N \rightarrow \infty$ gives the infinite product. For $0 \leq p_n < 1$, the standard infinite-product criterion says $\prod_n (1 - p_n) > 0$ if and only if $\sum_n p_n < \infty$, ignoring only the trivial case where some $p_n = 1$. $\square$

Corollary 6.2 (Error floor impossibility). *If $p_n \geq p_* > 0$ for infinitely many n , then $P_{\text{clean}} = 0$. More generally, if $p_n \sim c/n^s$, then positive clean probability requires $s > 1$.*

This result is logically independent from the energy law. Even if one could arrange the accelerating clock and endpoint readout, the per-step logical error rate must decay summably. A constant error rate is fatal. A harmonic error decay is also fatal.

7 Observer-depth hierarchy

The trace/readout model suggests a hierarchy based on how many tail quantifiers the observer supplies. Let $R(x, \vec{n})$ be a computable predicate over finite trace data.

Definition 7.1 (Quantifier readout of depth k). A depth- k readout is generated by a formula

$$Q_1 n_1 Q_2 n_2 \cdots Q_k n_k R(x, n_1, \dots, n_k),$$

where each $Q_i \in \{\exists, \forall\}$ and R is computable from finite trace data.

Proposition 7.2 (Arithmetical stratification). *A depth- k quantifier readout yields a predicate in the k -th level of the arithmetical hierarchy, up to the initial quantifier polarity. Conversely, every arithmetical predicate at that level can be encoded as a computable trace plus such a readout.*

Proof. Expanding the readout gives an arithmetical formula with k unbounded number quantifiers and computable matrix. Conversely, given any such arithmetical formula, the trace generator can dovetail finite evaluations of the computable matrix and expose the truth of the matrix instances as finite trace events. The readout supplies the same quantifier pattern. $\square$

The first two natural examples are:

$$O_\exists(z) = 1 \iff \exists n z_n = 1,$$

which is a Σ_1^0 -type observer, and

$$O_\infty(z) = 1 \iff \forall N \exists n \geq N z_n = 1,$$

which is a Π_2^0 -type observer expressing infinitely many occurrences.

Conjecture 7.3 (Natural observer-depth principle). *For physically interpretable accelerated machines whose final output is obtained from finite Boolean combinations of depth- d tail observations, the accepted language class is exactly the corresponding Boolean closure of the $d + 1$ -st arithmetical level, after normalization of the trace alphabet and coding.*

This conjecture is intended to generalize the pattern seen in accelerated-machine acceptance semantics, including the Boolean-closure results reported by Calude and Staiger [3]. It should be treated as a classification program, not as a proved theorem in this draft.

8 Virtual experiments

The formulas above were tested in a small deterministic numerical environment. The code computed $R_n, \rho_n, N(\varepsilon)$, and $E_\varepsilon/Q = 1/(\rho_{N(\varepsilon)} R_{N(\varepsilon)})$ for four schedule families:

$$\begin{aligned} \text{constant aperture:} & \quad \rho_n = 1/2, \\ \text{polynomial tail:} & \quad R_n = (n + 1)^{-a}, \\ \text{stretched exponential:} & \quad R_n = \exp(-n^{1/2}). \end{aligned}$$

The purpose is not empirical validation; it is stress-testing the formulas and exposing scaling behavior.

Reliability experiments used independent error models and evaluated $\prod_{n=1}^{10^5} (1 - p_n)$. The finite value can be misleading: $p_n = 0.1/n$ still has a moderate clean probability at 10^5 steps, but the infinite product vanishes.

Table 1: Endpoint diagnostics with $T = Q = 1$. $N(\epsilon)$ is the first step index for which $R_N \leq \epsilon$; E_ϵ is the lower bound $1/(\rho_N R_N)$.

ϵ	schedule	$N(\epsilon)$	E_ϵ/Q
10^{-6}	constant aperture $\rho = 1/2$	20	2.10×10^6
10^{-6}	polynomial tail $a = 1$	999,999	1.00×10^{12}
10^{-6}	polynomial tail $a = 2$	999	5.01×10^8
10^{-6}	stretched exponential $b = 1/2$	191	2.83×10^7
10^{-12}	constant aperture $\rho = 1/2$	40	2.20×10^{12}
10^{-12}	polynomial tail $a = 1$	1.00×10^{12}	1.00×10^{24}
10^{-12}	polynomial tail $a = 2$	999,999	5.00×10^{17}
10^{-12}	stretched exponential $b = 1/2$	764	5.63×10^{13}
10^{-18}	constant aperture $\rho = 1/2$	60	2.31×10^{18}
10^{-18}	polynomial tail $a = 1$	1.00×10^{18}	1.00×10^{36}
10^{-18}	polynomial tail $a = 2$	1.00×10^9	5.00×10^{26}
10^{-18}	stretched exponential $b = 1/2$	1,718	8.36×10^{19}
10^{-24}	constant aperture $\rho = 1/2$	80	2.42×10^{24}
10^{-24}	polynomial tail $a = 1$	$\approx 1.00 \times 10^{24}$	1.00×10^{48}
10^{-24}	polynomial tail $a = 2$	1.00×10^{12}	5.00×10^{35}
10^{-24}	stretched exponential $b = 1/2$	3,054	1.11×10^{26}

Table 2: Reliability diagnostics for independent error events. The finite column reports $\prod_{n=1}^{10^5} (1 - p_n)$; the last column gives the infinite-product classification.

error model	clean probability through 10^5 steps	infinite clean probability
$p_n = 10^{-3}$	3.54×10^{-44}	zero
$p_n = 0.1/n$	0.296	zero
$p_n = 0.1/n^{1.2}$	0.597	positive
$p_n = 10^{-3}2^{-n}$	0.999	positive

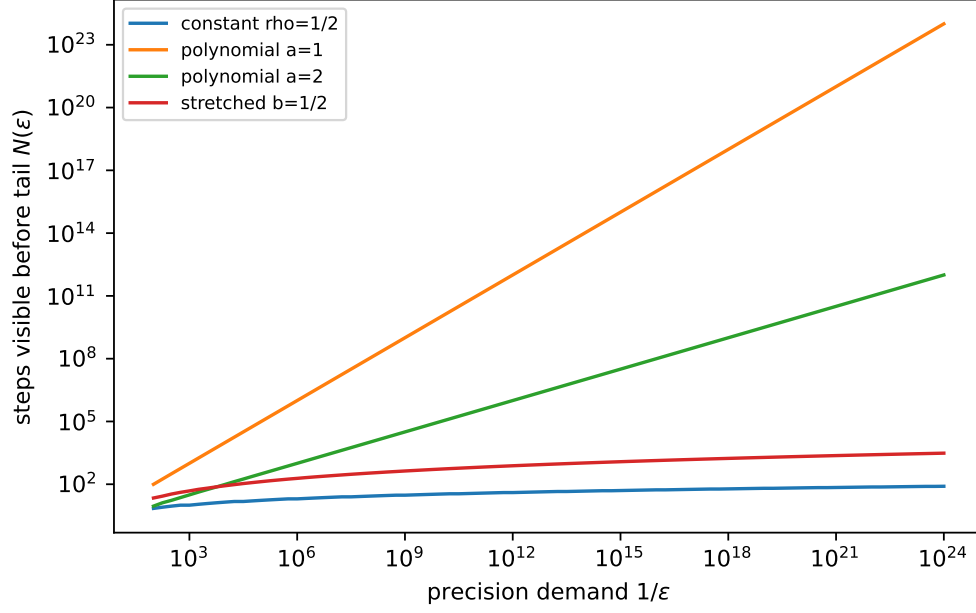


Figure 1: Number of visible steps before the remaining tail is below ε . Polynomial tails expose many more finite-resolution steps than constant-aperture schedules.

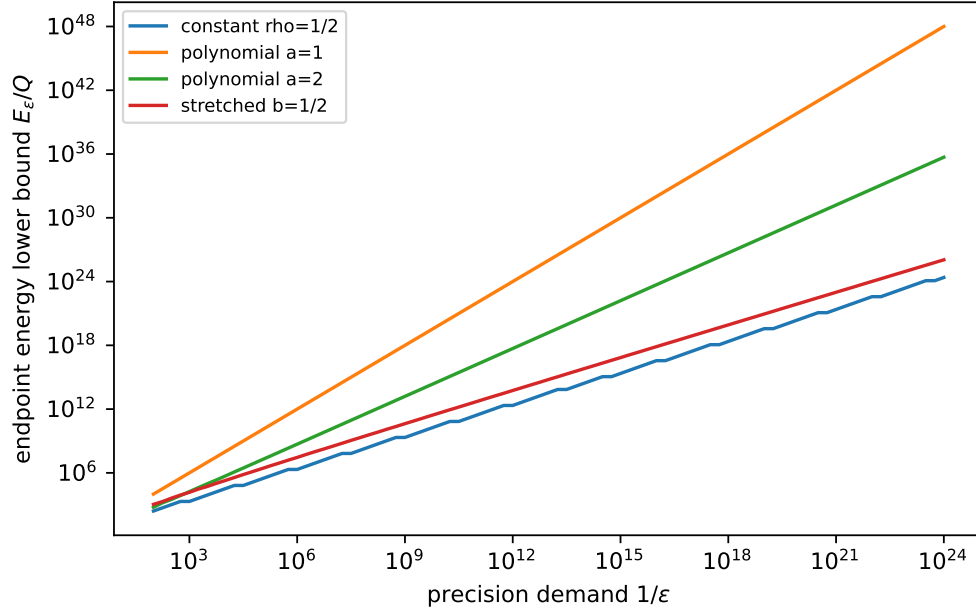


Figure 2: Endpoint energy lower bound E_ε/Q . The same schedules that expose many more steps can have worse endpoint energy scaling.

Version 0.2 appendage: robust endpoint observation and reliability-corrected cost

The following sections record a second investigation. The aim is to replace the phrase “exact endpoint readout” by quantitative resource statements. The main result is a simple but strong obstruction: a finite signal budget and any fixed nonzero readout uncertainty turn an infinite halting-time horizon into a finite robust horizon.

The core new equations are

$$N_{\text{robust}} \leq \frac{B}{2\eta}, \quad E_n^{\text{rel}} \geq \frac{Q + c \log(n+2)}{\rho_n R_n}. \quad (1)$$

9 Robust endpoint observation and finite signal budget

The first draft isolated a qualitative fact: the usual Zeno solution of the halting problem depends on a discontinuous readout at the nonhalting trace. This section turns that fact into a quantitative obstruction. The obstruction applies before any detailed physics is added. It only assumes that late events must be coupled to a finite observable with a finite signal budget and a nonzero uncertainty radius.

9.1 Robust separation at a limit interface

Let $X \subseteq \Sigma^\omega$ be a trace space, let $P_0, P_1 \subseteq X$ be the trace sets corresponding to answers 0 and 1, and let

$$F : X \rightarrow \mathbb{R}^m$$

be the finite endpoint observable delivered to an experimental observer. Define the observable gap

$$\Gamma_F(P_0, P_1) = \inf\{\|F(x) - F(y)\| : x \in P_0, y \in P_1\}.$$

If the endpoint value is only known within additive error η in norm, robust binary separation requires

$$\Gamma_F(P_0, P_1) > 2\eta.$$

Proposition 9.1 (Closure-gap obstruction). *If F is continuous and $\overline{P_0} \cap \overline{P_1} \neq \emptyset$, then $\Gamma_F(P_0, P_1) = 0$. Consequently no positive uncertainty radius η can be robustly tolerated.*

Proof. Choose $z \in \overline{P_0} \cap \overline{P_1}$. There are sequences $x_k \in P_0$ and $y_k \in P_1$ converging to z . By continuity, $F(x_k) \rightarrow F(z)$ and $F(y_k) \rightarrow F(z)$. Thus $\|F(x_k) - F(y_k)\| \rightarrow 0$, so the infimum defining Γ_F is zero. $\square$

For a latched halting trace, let

$$P_0 = \{0^\omega\}, \quad P_1 = \{0^n 1^\omega : n \in \mathbb{N}\}.$$

The nonhalting trace 0^ω lies in the closure of P_1 . Therefore every continuous finite endpoint observable has zero robust gap between nonhalting and sufficiently late halting. Exact Zeno readout escapes Proposition 9.1 only by using a discontinuity, a noncontinuous physical observable, or an unbounded amplification mechanism at the limit trace. This is the topological form of the blindspot: the difficulty is not merely running the steps, but separating 0^ω from traces that agree with it for an arbitrarily long prefix.

9.2 Scalar signal-budget theorem

A common way to imagine endpoint readout is as an accumulated signal. Let e_n denote the single-event trace with a 1 at time n and 0 elsewhere. Let

$$Y_w(x) = \sum_{n=0}^{\infty} w_n x_n, \quad w_n \geq 0, \quad \sum_n w_n \leq B.$$

Here B is the total endpoint signal budget. The no-event trace has $Y_w(0^\omega) = 0$, and a single event at n has signal $Y_w(e_n) = w_n$.

Theorem 9.2 (Finite signal-budget law). *For every $N \geq 1$,*

$$\sup_{\sum_{n < N} w_n \leq B} \min_{0 \leq n < N} w_n = \frac{B}{N}.$$

Consequently, robust detection of a single event among the first N event times with additive uncertainty η requires

$$N < \frac{B}{2\eta}.$$

For an infinite event horizon with finite B , $\inf_n w_n = 0$; hence the robust gap is zero for every fixed $\eta > 0$.

Proof. The average of $w_0, \dots, w_{N-1}$ is at most B/N , so their minimum is at most B/N . Equality is achieved over the first N event times by taking $w_0 = \dots = w_{N-1} = B/N$. For noisy separation from the no-event signal 0, every event signal must exceed 2η , so $B/N > 2\eta$. If $\sum_n w_n \leq B < \infty$, then an infinite sequence of nonnegative weights cannot have a positive lower bound. Therefore $\inf_n w_n = 0$. $\square$

Corollary 9.3 (Vector-energy version). *If event n emits a vector $v_n \in \mathbb{R}^m$ and the endpoint energy budget satisfies $\sum_{n < N} \|v_n\|^2 \leq B$, then*

$$\min_{n < N} \|v_n\| \leq \sqrt{B/N}.$$

Robust event-versus-no-event detection with additive uncertainty η requires

$$N < \frac{B}{4\eta^2}.$$

Proof. Apply the averaging argument to $\|v_n\|^2$. If every event is to lie farther than 2η from the no-event vector, then $\|v_n\| > 2\eta$ for every $n < N$, so $B > \sum_{n < N} 4\eta^2 = 4\eta^2 N$. $\square$

These bounds are elementary, but they appear to expose a useful invariant: a finite budget cannot maintain a fixed endpoint margin over an unbounded set of possible late events. The Zeno endpoint can still decide the existential event property in an exact mathematical model, but only by making the zero signal exactly distinguishable from every positive signal, no matter how small. This is the same discontinuity as the limited principle of omniscience in computable analysis; Weihrauch complexity provides a natural setting for classifying such discontinuities [9].

9.3 No-free-amplification lemma

One might try to rescue a finite signal budget by postprocessing the endpoint scalar through an amplifier ϕ . The following lemma says that ordinary finite-gain amplification does not help.

Lemma 9.4 (Lipschitz amplification preserves collapse). *Let $Y_w(e_n) = w_n$ with $\sum_n w_n \leq B$, and let $\phi : [0, B] \rightarrow \mathbb{R}$ be L -Lipschitz near 0. Then over the first N event times,*

$$\min_{n < N} |\phi(w_n) - \phi(0)| \leq \frac{LB}{N}.$$

Thus fixed-margin detection for unbounded N requires either unbounded signal budget, unbounded gain near zero, or discontinuity at zero.

Proof. By [Theorem 9.2](#), some $w_n \leq B/N$. Lipschitz continuity gives $|\phi(w_n) - \phi(0)| \leq Lw_n \leq LB/N$. $\square$

This lemma identifies a narrow target for further study: any physically plausible “endpoint amplifier” that claims to make all late halting events visible must be examined for a singularity at the no-event state. If the amplifier has finite gain, it cannot remove the finite-horizon collapse. If it has infinite gain or a discontinuity, then the hypercomputational power has simply moved from the Zeno clock into the endpoint amplifier.

9.4 Numerical stress tests

[Table 3](#) evaluates the scalar bound for $B = 1$. Even an uncertainty of 10^{-12} leaves a finite robust horizon of order 5×10^{11} possible event times. That is enormous for engineering, but it is finite; it is not a solution of the halting problem.

Table 3: Finite signal budget bound for robust single-event detection with total signal budget $B = 1$. Additive uncertainty η requires event margin $> 2\eta$, so no scalar additive endpoint code can robustly distinguish more than $\lfloor B/(2\eta) \rfloor$ possible event times.

uncertainty η	maximum robust event horizon	best worst-case margin at horizon
1×10^{-3}	500	2×10^{-3}
1×10^{-6}	5×10^5	2×10^{-6}
1×10^{-9}	5×10^8	2×10^{-9}
1×10^{-12}	5×10^{11}	2×10^{-12}

A second stress test used two exact-real encodings with total budget 1:

$$w_n = 2^{-(n+1)}, \quad w_n = \frac{6}{\pi^2(n+1)^2}.$$

With a conservative 6σ threshold, geometric coding loses detectability after only logarithmically many event positions, while power-law coding lasts longer but still has a finite threshold horizon for every fixed σ .

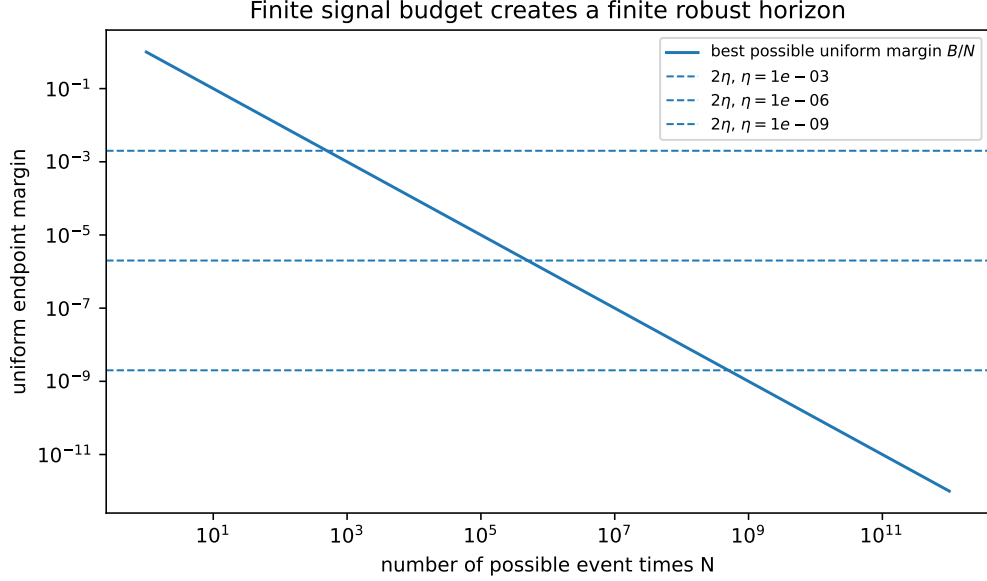


Figure 3: Best possible uniform endpoint margin B/N for scalar single-event detection under total budget $B = 1$. Any fixed readout uncertainty intersects the frontier at a finite horizon.

Table 4: Noisy readout stress test for two exact-real single-event encodings with total signal budget 1. The table reports the last event index whose raw signal exceeds 6σ . Later events are below a conservative six-sigma detection threshold.

noise scale σ	threshold 6σ	geometric code $2^{-(n+1)}$	power code $6/(\pi^2(n+1)^2)$
1×10^{-3}	6×10^{-3}	6	9
1×10^{-6}	6×10^{-6}	16	317
1×10^{-9}	6×10^{-9}	26	1.01×10^4

The exceptional lesson from these tests is not the particular constants. It is the sharp separation between exact and robust endpoint semantics:

exact positivity test: $Y > 0$ has infinite reach,

finite-margin positivity test: $Y > 2\eta$ has finite reach.

Thus exact-real positivity is itself a hypercomputational resource.

10 Reliability-corrected horizon-energy law

The first draft showed that a supertask with independent uncorrected errors has positive clean probability only when the logical errors are summable. This section adds a resource model for making them summable. The result is a logarithmic reliability tax multiplying the endpoint-energy law.

Assumption 10.1 (Exponential logical-error suppression). Let $C_n \geq 0$ be an additional correction cost assigned to logical step n , and suppose that the post-correction logical error has the form

$$p_n = \exp(-\alpha C_n), \quad \alpha > 0.$$

Theorem 10.2 (Logarithmic correction threshold). *Under Assumption 10.1, if*

$$C_n \leq \frac{1}{\alpha} \log(n+1) + O(1)$$

eventually, then $\prod_n (1 - p_n) = 0$. If for some $\delta > 0$,

$$C_n \geq \frac{1+\delta}{\alpha} \log(n+1) + O(1)$$

eventually with the $O(1)$ bounded below, then $\prod_n (1 - p_n) > 0$.

Proof. The first bound implies $p_n \geq c/(n+1)$ for some constant $c > 0$ along the tail, so $\sum_n p_n$ diverges. The second implies $p_n \leq C/(n+1)^{1+\delta}$ for some constant C , so $\sum_n p_n$ converges. The infinite-product criterion in [Theorem 6.1](#) completes the proof. $\square$

Combining this with the horizon-energy law gives a corrected endpoint bound. If the physical action needed for a distinguishable logical transition is Q and the reliability cost contributes additively to the required action, then a summably reliable Zeno schedule requires

$$E_n \geq \frac{Q + c \log(n+2)}{\tau_n} = \frac{Q + c \log(n+2)}{\rho_n R_n} \quad (2)$$

for some positive constant c determined by the desired error exponent.

Corollary 10.3 (Endpoint asymptotics with reliability tax). *For constant aperture $\rho_n = \rho \in (0, 1)$,*

$$E(R) \geq \Omega\left(\frac{1 + \log \log(T/R)}{R}\right)$$

as $R \downarrow 0$. For a polynomial tail $R_n = T(n+1)^{-a}$,

$$E(R) \geq \Omega\left(R^{-(1+1/a)} \log(T/R)\right).$$

Proof. For constant aperture, $R_n = T(1 - \rho)^n$, so $n = \Theta(\log(T/R_n))$ and $\log n = \Theta(\log \log(T/R_n))$. Also $\tau_n = \rho R_n$. Substitution into (2) gives the first bound. For polynomial tail, $n = \Theta((T/R_n)^{1/a})$, $\log n = \Theta(\log(T/R_n))$, and $\tau_n = R_n - R_{n+1} = \Theta(n^{-(a+1)}) = \Theta(R_n^{1+1/a})$, which gives the second. $\square$

The correction is asymptotically mild for geometric schedules but stronger for polynomial schedules. This is a new reason to regard constant-aperture schedules as not merely simple but near-optimal under combined speed and reliability pressure. Existing physical bounds already connect computation rate to energy and available degrees of freedom [\[7\]](#), and recent work argues that relativistic acceleration has an energy-scale bound on e-folds per unit time [\[8\]](#). The logarithmic tax above adds an independent infinite-reliability constraint: even if individual distinguishable steps can be accelerated, their logical error probabilities must be driven down fast enough for an infinite product of correctness probabilities to remain positive.

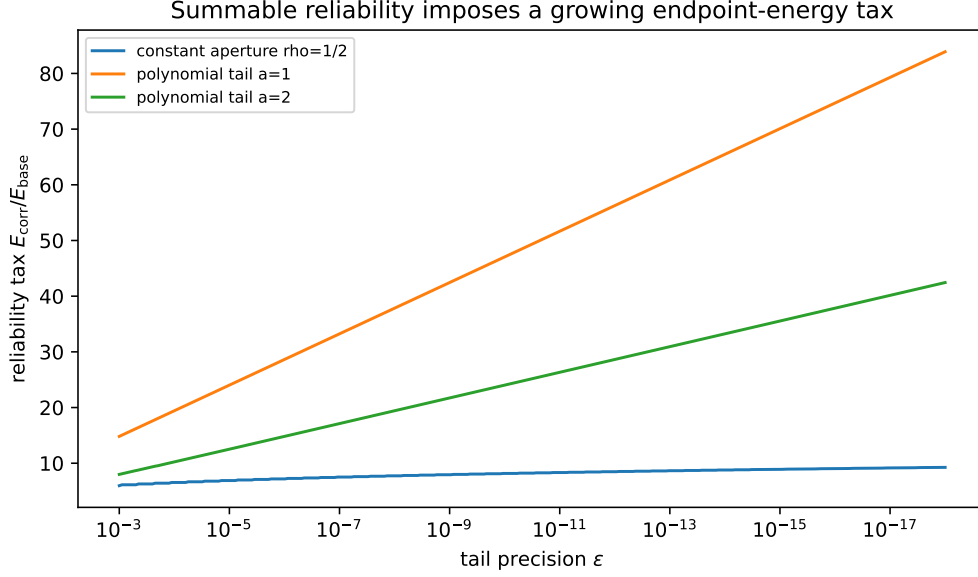


Figure 4: Reliability tax $E_{\text{corr}}/E_{\text{base}} = 1 + 2\log(N(\epsilon) + 2)$ for the logical target $p_n = (n + 2)^{-2}$. Polynomial schedules pay a larger reliability tax because $N(\epsilon)$ grows faster as endpoint precision increases.

Table 5: Reliability-corrected endpoint energy lower bound with $Q = 1$ and logical error target $p_n = (n + 2)^{-2}$. The correction multiplies the baseline bound by $1 + 2\log(N + 2)$.

ϵ	schedule	$N(\epsilon)$	corrected energy lower bound	correction factor
1×10^{-6}	constant aperture $\rho = 1/2$	20	1.51×10^7	7.18
1×10^{-6}	polynomial tail $a = 1$	1×10^6	2.86×10^{13}	28.63
1×10^{-6}	polynomial tail $a = 2$	999	7.42×10^9	14.82
1×10^{-12}	constant aperture $\rho = 1/2$	40	1.86×10^{13}	8.48
1×10^{-12}	polynomial tail $a = 1$	1×10^{12}	5.63×10^{25}	56.26
1×10^{-12}	polynomial tail $a = 2$	1×10^6	1.43×10^{19}	28.63

11 Applications and targets generated by the new laws

11.1 Applications

Audit of hypercomputation proposals. The finite signal-budget law adds a direct diagnostic question to any Zeno-style design: what is the finite-margin separation between no event and the latest event that must still be detected? If the answer relies on distinguishing 0 from an arbitrarily small positive number, the proposal is using exact-real positivity as an oracle-like resource.

Analog and real-number computing. Many analog models hide computational power in exact equality or exact positivity tests. The closure-gap obstruction and the signal-budget law provide a compact way to replace such tests by finite-margin variants. Under finite margin, the infinite decision collapses to a finite horizon.

Anytime algorithms under hard deadlines. A Zeno schedule can still inspire practical scheduling: allocate early time aggressively and reserve a shrinking tail for increasingly unlikely refinements. The aperture calculus gives such algorithms a design variable, while the signal and reliability laws prevent confusing an anytime approximation with hypercomputation.

Relativistic and gravitational computation. In Malament-Hogarth-style settings, causal geometry supplies the limit interface and can increase formal computational strength [4]. The new target is to express finite signal budget, detector noise, and reliability correction along the causal channel from the computer worldline to the observer worldline.

Safety and verification. The robust-horizon equation $N_{\text{robust}} \leq B/(2\eta)$ can be read operationally: a finite detector certifies only a bounded search horizon unless the no-event state is granted exact mathematical status. This can be used to audit claims about unbounded search, formal verification by physical process, and limit-based optimization devices.

11.2 Targets for further study

1. **Robust Weihrauch collapse.** Formalize a robust version of Weihrauch reducibility in which exact discontinuous principles such as LPO are replaced by finite-margin channels. Conjectural target: every finite-margin replacement of LPO over a compact trace space reduces to bounded finite search.
2. **Signal budget for stochastic channels.** Replace deterministic additive error by total variation distance, Kullback–Leibler divergence, or mutual information. The expected theorem is an information version of [Theorem 9.2](#): fixed channel capacity gives finite robust event horizon.
3. **Singular amplifier classification.** Classify endpoint maps that evade [Lemma 9.4](#). The key question is whether every physically regular amplifier has a finite local Lipschitz or finite Fisher-information bound near the no-event state.
4. **Reliability-aware aperture optimization.** Minimize

$$\sup_{n < N} \frac{Q + c \log(n + 2)}{\rho_n R_n}$$

subject to $R_N \leq \epsilon$ and admissible aperture constraints. This is the finite-horizon version of optimal Zeno design.

5. **Malament-Hogarth channel audit.** Generalize $\rho_n = \tau_n/R_n$ to a causal aperture along a spacetime channel, then combine it with detector noise and blueshift bounds. The 2026 acceleration-bound result suggests a concrete comparison target: endpoint e-fold depth should be bounded by an integrated energy scale.
6. **Vector-code and memory-state extensions.** Extend [Corollary 9.3](#) to the problem of identifying the event time, not merely detecting that some event occurred. This should connect signal budget to packing numbers, Bekenstein-style memory bounds, and finite-dimensional channel capacity.

12 Continuation II: statistical endpoints and de-Zenoization

Version 0.3 adds a statistical layer to the limit-interface theory. The finite signal-budget law in the previous section treated deterministic margin. A more physical model treats the endpoint as

a noisy statistical channel. The main finding is that finite statistical information cannot support uniform detection over infinitely many possible halt times. This turns the informal claim “a late signal might be too weak” into an explicit finite horizon.

This continuation is motivated by recent and nearby literature. Muller’s recent analysis of hypercomputing supertasks argues that Zeno-machine output-state semantics face a dilemma unless non-effective or special rules are added [10]. Weihrauch complexity provides a precise language for classifying such non-effective operations, and recent work on jump-like operators indicates that this remains an active area of degree theory [9, 12]. On the physical side, recent reviews of Landauer’s principle emphasize finite-time, finite-bath, and error-correction costs rather than merely the ideal quasistatic erasure bound [11]. The statistical endpoint laws below are intended as a bridge between these mathematical and physical viewpoints.

12.1 Gaussian endpoint model

Let the no-halt observation be a random vector

$$Y \sim P_\infty = \mathcal{N}(0, \sigma^2 I)$$

in a finite or separable Hilbert observation space. A halt at internal step m produces

$$Y \sim P_m = \mathcal{N}(v_m, \sigma^2 I),$$

where v_m is the accessible endpoint signal associated with that halt time. A binary detector $\varphi(Y) \in \{0, 1\}$ says whether a halt occurred. It is a uniform α -detector for halt times $m < N$ when

$$P_\infty(\varphi = 1) \leq \alpha, \quad P_m(\varphi = 0) \leq \alpha \quad (m < N).$$

This is weaker than identifying which m occurred. It only asks for reliable yes/no halting detection.

Theorem 12.1 (Statistical endpoint bound). *In the Gaussian endpoint model, a uniform α -detector with $0 < \alpha < 1/2$ for the first N halt times requires*

$$\|v_m\| \geq 2\sigma z_{1-\alpha} \quad (m < N),$$

where $z_{1-\alpha} = \Phi^{-1}(1 - \alpha)$ is the standard normal quantile. Consequently, if the endpoint signal vectors satisfy the finite budget

$$\sum_{m < N} \|v_m\|^2 \leq B,$$

then

$$N \leq N_{\text{stat}}(B, \sigma, \alpha) := \left\lfloor \frac{B}{4\sigma^2 z_{1-\alpha}^2} \right\rfloor.$$

Proof. Fix $m < N$. Any detector that works uniformly for no halt and for halt time m is, in particular, a test between the two simple Gaussian hypotheses P_∞ and P_m with both type-I and type-II error at most α . For two Gaussians with equal covariance $\sigma^2 I$ and mean separation $d = \|v_m\|$, the optimal equal-prior error is

$$\Phi\left(-\frac{d}{2\sigma}\right).$$

Thus a test with both errors at most α can exist only if

$$\Phi\left(-\frac{d}{2\sigma}\right) \leq \alpha,$$

which is equivalent to $d \geq 2\sigma z_{1-\alpha}$. Summing $\|v_m\|^2 \geq 4\sigma^2 z_{1-\alpha}^2$ over $m < N$ gives the budget inequality. $\square$

Table 6: Gaussian statistical endpoint bound with $\sigma = 1$. Uniform false-positive and false-negative error at most α over N possible halt times requires $N \leq B/(4z_{1-\alpha}^2)$.

α	$4z_{1-\alpha}^2$	B	$N_{\text{stat}}^{\max}$
0.10	6.57	1e+02	15
0.10	6.57	1e+04	1,522
0.10	6.57	1e+06	152,218
0.10	6.57	1e+09	152,218,640
0.05	10.82	1e+02	9
0.05	10.82	1e+04	924
0.05	10.82	1e+06	92,402
0.05	10.82	1e+09	92,402,877
0.01	21.65	1e+02	4
0.01	21.65	1e+04	461
0.01	21.65	1e+06	46,194
0.01	21.65	1e+09	46,194,544

Corollary 12.2 (Infinite statistical signal is necessary). *For any fixed $\sigma > 0$ and $\alpha < 1/2$, uniform Gaussian detection of an unbounded set of possible halt times requires*

$$\sum_{m=0}^{\infty} \|v_m\|^2 = \infty.$$

Equivalently, finite endpoint signal energy plus nonzero observation noise collapses the infinite halting horizon to a finite statistical shadow.

Candidate law 12.3 (Statistical endpoint law). *A Zeno-style yes/no halting readout with nonzero Gaussian noise must pay for uniform reliability over infinitely many possible halt times with at least one of the following:*

$$\begin{aligned} &\text{infinite statistical signal energy} \\ &\vee \text{zero observation noise} \\ &\vee \text{a singular or discontinuous endpoint rule} \end{aligned}$$

Remark 12.4 (Why a shared latch does not evade the law). An ideal symbolic latch maps every halt time to the same final state 1. The statistical model asks how that final state is physically accessed. If the observer literally reads a perfect endpoint bit, then the endpoint bit is the singular readout. If instead the halt must be conveyed through finite-energy, noisy, time-extended signals, the accessible signal vectors v_m for late events cannot remain uniformly separated from the no-halt law without an infinite statistical budget or a nonintegrable endpoint gain.

12.2 Numerical stress test: finite signal budget

The first statistical experiment fixes $B = 10^4$, $\sigma = 1$, and sets the false-positive threshold to $z_{0.95}$. Under equal signal allocation over N possible halt times, each event has amplitude $a = \sqrt{B/N}$. The analytic false-negative probability is

$$\Phi(z_{0.95} - a).$$

The predicted uniform-control boundary is $N \approx 924$. The simulation used independent Gaussian samples with a fixed random seed and matches the analytic curve.

Table 7: Gaussian endpoint detection experiment with $B = 10^4$, $\sigma = 1$, and false-positive threshold chosen for $\alpha = 0.05$. Equal allocation gives event amplitude $a = \sqrt{B/N}$. The statistical bound predicts loss of 5% uniform false-negative control near $N = 924$.

N	a	analytic FN	simulated FN	simulated FP
50	14.142	0.000	0.000	0.050
100	10.000	0.000	0.000	0.049
300	5.774	0.000	0.000	0.050
924	3.290	0.050	0.050	0.050
2,000	2.236	0.277	0.278	0.050
5,000	1.414	0.591	0.591	0.050
10,000	1.000	0.740	0.739	0.050
30,000	0.577	0.857	0.857	0.050

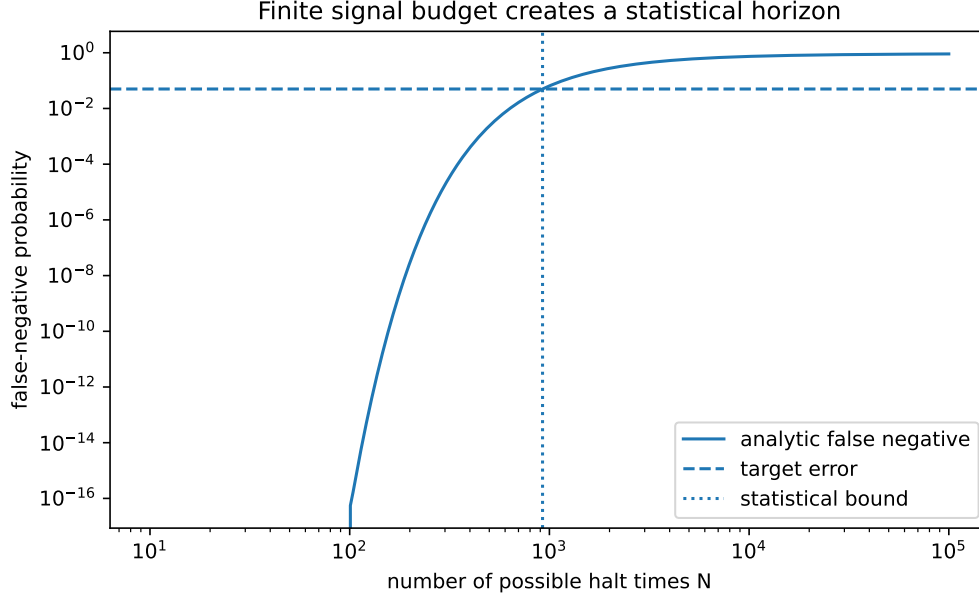


Figure 5: False-negative probability under finite Gaussian endpoint signal budget. The false-positive threshold is fixed at 5%. Once N exceeds the statistical bound, at least one halt alternative must become too weak for uniform 5% false-negative control.

12.3 De-Zenoization theorem for finite robust shadows

The earlier sections show several ways an ideal infinite computation becomes a finite operational shadow. Once this happens, acceleration itself becomes a liability. For a finite certified depth, the energetically best schedule is not a Zeno schedule at all.

Theorem 12.5 (De-Zenoization of finite shadows). *Fix a deadline T , endpoint slack ϵ , certified finite depth N , and per-step action lower bound $E_n \tau_n \geq Q$. Among all schedules for the first N steps with*

$$\sum_{n=0}^{N-1} \tau_n \leq T - \epsilon,$$

the minimal possible peak lower-bound energy is

$$\min_{\tau_0, \dots, \tau_{N-1}} \max_{n < N} \frac{Q}{\tau_n} = \frac{QN}{T - \epsilon}.$$

Equality is attained by the uniform finite schedule

$$\tau_0 = \dots = \tau_{N-1} = \frac{T - \epsilon}{N}.$$

Proof. Since $\sum_{n < N} \tau_n \leq T - \epsilon$, at least one step has

$$\tau_n \leq \frac{T - \epsilon}{N}.$$

For that step, $Q/\tau_n \geq QN/(T - \epsilon)$, so the peak lower-bound energy is at least $QN/(T - \epsilon)$. The uniform schedule uses the full available time and makes every step attain this lower bound. $\square$

Corollary 12.6 (Finite robust hypercomputers are schedule-dominated by ordinary clocks). *If readout, noise, power, or reliability constraints restrict an accelerated machine to a finite certified depth N , then its robust computational behavior is no stronger than a bounded Turing simulation of N steps, and the peak-energy-optimal implementation of that bounded behavior uses a uniform finite schedule.*

For comparison, choose the constant-aperture schedule whose tail after N steps is ϵ , so $R_N = \epsilon$ and $R_0 = T$. Then

$$\rho = 1 - \left(\frac{\epsilon}{T}\right)^{1/N}, \quad \tau_{N-1} = \epsilon \left[\left(\frac{T}{\epsilon}\right)^{1/N} - 1 \right].$$

Its final-step peak lower-bound energy is

$$E_{\max}^{\text{geo}} = \frac{Q}{\epsilon [(T/\epsilon)^{1/N} - 1]} \sim \frac{QN}{\epsilon \log(T/\epsilon)}.$$

Relative to the uniform finite schedule, the asymptotic penalty is

$$\frac{E_{\max}^{\text{geo}}}{QN/(T - \epsilon)} \sim \frac{T}{\epsilon \log(T/\epsilon)}.$$

Thus geometric acceleration can be near-optimal for the idealized infinite-tail problem while being catastrophically suboptimal for the finite robust-shadow problem.

Table 8: Finite-shadow scheduling comparison for $T = 1$, $\epsilon = 10^{-12}$, $Q = 1$. For a finite certified depth N , uniform scheduling minimizes the peak lower-bound energy Q/τ_n . The geometric row uses the constant-aperture schedule chosen to leave tail ϵ after N steps.

N	uniform peak	geometric peak	ratio
10	1.00e+01	6.73e+10	6.73e+09
30	3.00e+01	6.61e+11	2.20e+10
100	1.00e+02	3.14e+12	3.14e+10
300	3.00e+02	1.04e+13	3.46e+10
1,000	1.00e+03	3.57e+13	3.57e+10
10,000	1.00e+04	3.61e+14	3.61e+10
1,000,000	1.00e+06	3.62e+16	3.62e+10

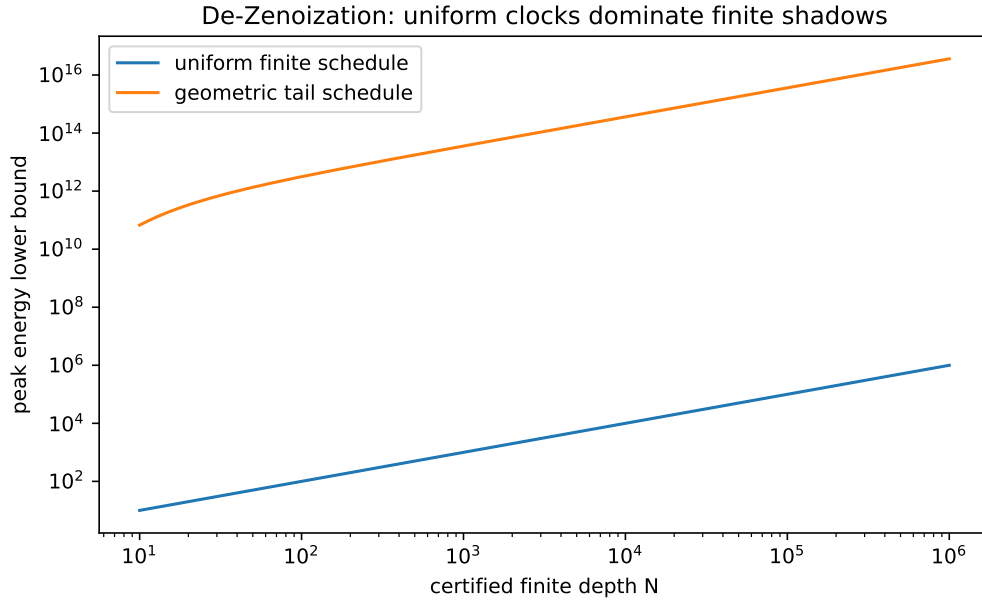


Figure 6: For a finite certified depth, a uniform finite schedule minimizes peak energy. A geometric schedule forced to leave a tiny endpoint tail pays an enormous final-step penalty. This is the de-Zenoization effect: after robustness collapses infinity to a finite shadow, the optimal clock is no longer accelerated.

12.4 Robust shadow equation

The different collapse mechanisms can be combined into a single finite-depth diagnostic. Let

$$\begin{aligned}
N_{\text{clk}} &:= \text{depth allowed by endpoint time precision,} \\
N_{\text{read}} &:= \text{depth allowed by finite-margin endpoint readout,} \\
N_{\text{stat}} &:= \left\lfloor B/(4\sigma^2 z_{1-\alpha}^2) \right\rfloor, \\
N_{\text{pow}} &:= \lfloor E_{\text{max}}(T - \epsilon)/Q \rfloor, \\
N_{\text{rel}} &:= \left\lfloor \frac{\log(1 - \beta)}{\log(1 - p)} \right\rfloor \quad \text{for constant per-step error floor } p.
\end{aligned}$$

Here β is the allowed total failure probability in the finite-shadow run. The certified robust depth is then

$$N_{\text{rob}} = \min\{N_{\text{clk}}, N_{\text{read}}, N_{\text{stat}}, N_{\text{pow}}, N_{\text{rel}}\}.$$

The corresponding robust shadow of the halting predicate is the bounded predicate

$$H_{N_{\text{rob}}}(e, x) = 1 \iff \text{the universal simulator halts on } (e, x) \text{ before } N_{\text{rob}} \text{ steps.}$$

Thus every finite value in the shadow equation has a direct computational interpretation. It is not merely an engineering imperfection; it is a reduction from an infinite limit-interface claim to a finite search problem.

Table 9: Examples of the robust shadow equation for a constant-aperture clock. The certified depth is the minimum of clock, readout, statistical, peak-power, and reliability horizons.

scenario	N_{clk}	N_{read}	N_{stat}	N_{pow}	N_{rel}	N_{rob}	limiter
readout-limited	40	14	92,402	999,999,999,999	51,293,295	14	readout
clock-limited	40	40	92,402	999,999,999,999	51,293,295	40	clock
statistics-limited	60	60	9	1,000,000,000,000	51,293,295	9	statistics
reliability-limited	60	60	92,402,877	1,000,000,000,000	51	51	reliability
power-limited	60	60	92,402,877	25	51,293,295	25	power

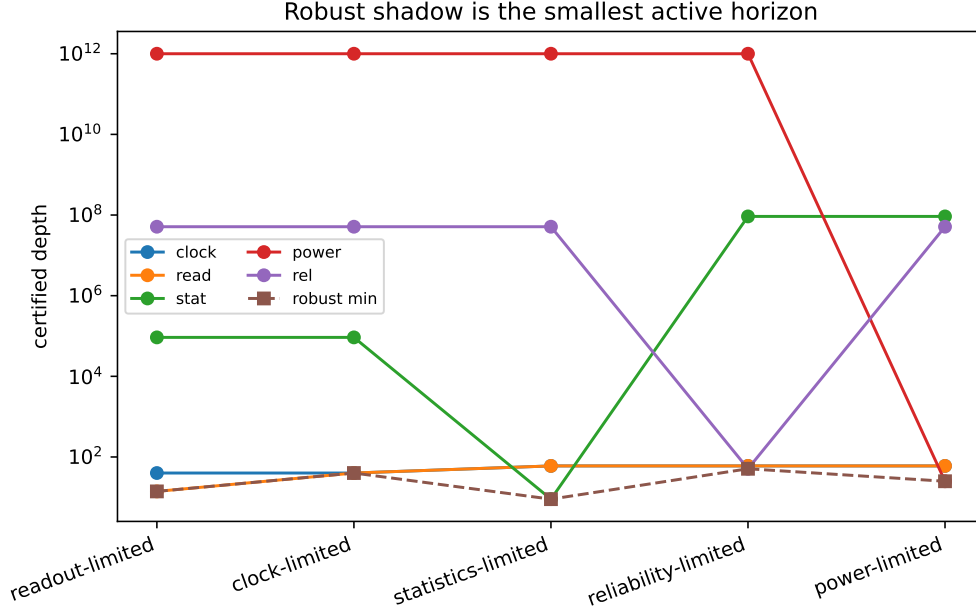


Figure 7: Illustrative robust-shadow components. The active limiter changes across parameter regimes, but the certified depth is always the minimum active horizon. This figure suggests an audit protocol: compute all horizons and identify the bottleneck before claiming super-Turing behavior.

12.5 Distributional use: finite shadows can still be valuable

Uniform hypercomputation fails when $N_{\text{rob}} < \infty$, but finite shadows can be useful under a computable prior over halt times. Let π_n be a prior probability that a halt, if it occurs, first occurs at step n . If no robust evidence is available after depth N , then the unavoidable prior mass of missed late halts is at least

$$M_\pi(N) = \sum_{n>N} \pi_n.$$

This observation separates two goals that are often conflated:

$$\text{uniform solution of halting} \neq \text{high-value distributional proof search.}$$

The first requires a singular limit interface. The second can be optimized as a finite resource-allocation problem.

Table 10: Distributional application: lower bound on missed halt mass after a finite robust depth, for three computable halt-time priors truncated at 5000. Uniform hypercomputation fails when the robust depth is finite, but a rapidly decaying prior can make the operational miss mass small.

prior	depth	tail mass beyond depth
geometric $q=0.01$	10	9.044e-01
geometric $q=0.01$	40	6.690e-01
geometric $q=0.01$	100	3.660e-01
geometric $q=0.01$	924	9.267e-05
geometric $q=0.01$	1,000	4.317e-05
pareto $s=1.2$	10	4.729e-01
pareto $s=1.2$	40	3.150e-01
pareto $s=1.2$	100	2.303e-01
pareto $s=1.2$	924	7.809e-02
pareto $s=1.2$	1,000	7.382e-02
pareto $s=2.0$	10	5.774e-02
pareto $s=2.0$	40	1.489e-02
pareto $s=2.0$	100	5.928e-03
pareto $s=2.0$	924	5.361e-04
pareto $s=2.0$	1,000	4.861e-04

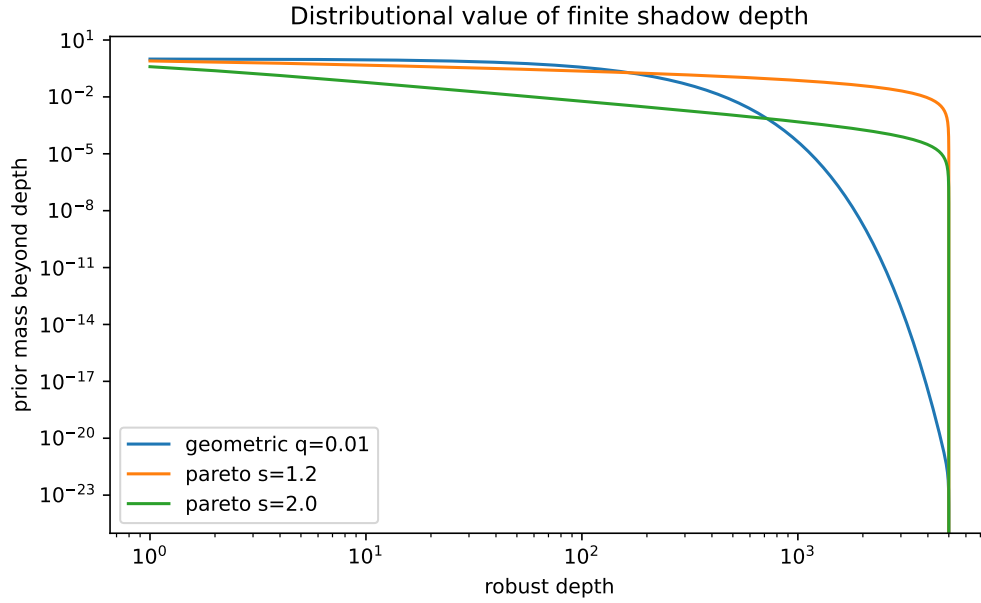


Figure 8: Prior mass beyond finite robust depth for three computable halt-time priors, truncated at 5000 for the experiment. Heavy-tailed priors preserve significant missed mass even at large finite depths; rapidly decaying priors make finite shadows practically useful without producing hypercomputation.

12.6 Exceptional findings from the statistical continuation

The new findings can be summarized as follows.

1. **Uniform statistical halting detection requires infinite signal energy.** Finite endpoint signal budget and nonzero noise impose an explicit finite bound $N_{\text{stat}} = \lfloor B/(4\sigma^2 z_{1-\alpha}^2) \rfloor$.
2. **Finite robust shadows de-Zenoize.** Once a finite certified depth is imposed, the peak-energy-optimal schedule is uniform rather than accelerated.
3. **The robust computational content is a minimum of horizons.** Clock precision, finite-margin readout, statistical signal, peak power, and reliability each define a finite shadow; the smallest one is the operative computation.
4. **Distributional acceleration is not hypercomputation.** A bounded-depth accelerator may be valuable for proof search or verification under a favorable prior, but its miss mass is exactly the prior tail beyond N_{rob} .

12.7 Targets for further study

Non-Gaussian statistical laws. The Gaussian theorem should be generalized using total variation, Hellinger distance, Fisher information, or channel capacity. A likely abstract form is:

$$\text{uniform error } \alpha \implies \sum_{m < N} D(P_m \| P_\infty) \geq c_\alpha N,$$

where D is an information divergence appropriate to the detector.

Optimal prior-weighted endpoint codes. For a computable prior π_m and finite budget B , find signal amplitudes $\|v_m\|$ that minimize Bayes miss probability. The expected result is a water-filling law: high-prior early events receive margin first, while low-prior late events are deliberately allowed to fall into the blind cone.

Quantifier-volume law for bounded jumps. A depth- d finite shadow of an arithmetical predicate produces bounded formulas with up to N^d atomic leaves. A candidate robust-jump cost is

$$B_d(N, \alpha, \sigma) \gtrsim 4\sigma^2 z_{1-\alpha}^2 N^d$$

in independent certificate-channel models. The target is to connect this finite resource law to the Weihrauch jump hierarchy and to recent work on jump operators in the Weihrauch lattice [12].

Relativistic detector budgets. For Malament-Hogarth proposals, compute P_m at the observer for signals emitted from later and later proper times along the computer worldline. The critical question becomes whether the family $\{P_m\}$ has uniformly positive statistical separation from the no-signal law under finite detector energy, redshift, blueshift, and background noise.

Finite-shadow design. Instead of claiming hypercomputation, design practical bounded accelerators by choosing N_{rob} from a target prior, failure probability, and energy budget. Candidate applications include proof search, bounded model checking, finite-horizon verification, anytime theorem proving, and high-confidence search over rare counterexamples.

13 Continuation III: endpoint latches, finite-power shadows, and Zeno utilization

Version 0.4 examines a possible loophole in the finite signal-budget analysis. A simple Zeno halting machine is usually imagined with a persistent output cell: if the simulated program halts, the machine writes a 1 into the output cell and leaves it there. At first sight this seems to evade the signal-budget law, because every halt time produces the same final macroscopic state rather than a different small signal. The new finding is that the loophole is only apparent. A persistent latch moves the limiting burden from endpoint signal accumulation to the ability to perform an arbitrarily late robust write.

This continuation is also motivated by nearby developments in computable analysis and physical computation. Recent Weihrauch work treats loops, inverse limits, and iterated limit operators as first-class structure [13, 14]. On the physical side, current thermodynamic-computing work emphasizes energy-time-accuracy tradeoffs, finite-time costs, and error-dissipation behavior rather than only idealized quasistatic Landauer costs [11, 15, 16]. The formulas below connect those themes to the Zeno endpoint.

13.1 Latch/impulse normal form

There are at least two endpoint implementation primitives.

Impulse endpoint. A halt event at internal time n emits an endpoint signal v_n . If the endpoint is noisy and the total signal energy is finite, the statistical endpoint law in Theorem 12.1 gives a finite uniform detection horizon.

Persistent latch. A halt event writes a stable output bit. In this case the final state can have a fixed margin from the no-halt state, but the write itself must be carried out inside the duration τ_n of the n -th internal step.

The latch case requires a new but very weak resource abstraction.

Assumption 13.1 (Finite-margin write action). For a fixed robust output margin $m > 0$, there is a positive action cost $Q_m > 0$ such that writing the latch during a time interval of length τ under peak available power P requires

$$P\tau \geq Q_m.$$

This is not intended to be a detailed model of CMOS, reversible logic, or a metastable molecular memory. It is only the claim that a robust finite-margin physical transition cannot be made in arbitrarily small time at bounded peak power. More detailed models would replace Q_m by a function that also depends on temperature, allowed error, barrier height, control protocol, and dissipation.

Definition 13.2 (Latch write horizon). For a schedule $\tau = (\tau_n)_{n \geq 0}$ and a peak power bound P , define

$$N_{\text{lat}}(P; m) = \#\{n \in \mathbb{N} : \tau_n \geq Q_m/P\}.$$

This is the number of internal steps at which a robust latch write is physically allowed under Theorem 13.1.

Theorem 13.3 (Finite-power latch horizon). If $\sum_n \tau_n = T < \infty$, $Q_m > 0$, and $P < \infty$, then

$$N_{\text{lat}}(P; m) < \infty.$$

Thus a finite-power persistent latch cannot provide a robust output write for unbounded late halt times in a finite-time supertask.

Proof. Since $\sum_n \tau_n < \infty$, necessarily $\tau_n \rightarrow 0$. For the fixed threshold $Q_m/P > 0$, only finitely many terms can satisfy $\tau_n \geq Q_m/P$. $\square$

The finite statement is useful, but the asymptotic statement is more revealing.

Theorem 13.4 (Zeno utilization theorem). *Let $\tau_n > 0$ and $\sum_n \tau_n = T < \infty$. Define the counting function*

$$D(\delta) = \#\{n : \tau_n \geq \delta\}.$$

Then

$$\lim_{\delta \downarrow 0} \delta D(\delta) = 0.$$

Consequently, for finite-margin latch writes,

$$U(P) := \frac{Q_m N_{\text{lat}}(P; m)}{PT} \rightarrow 0 \quad (P \rightarrow \infty).$$

Proof. Fix $\epsilon > 0$. Choose M so that $\sum_{n \geq M} \tau_n < \epsilon$. Split the count into the first M terms and the tail:

$$D(\delta) \leq M + \#\{n \geq M : \tau_n \geq \delta\}.$$

For the tail,

$$\delta \#\{n \geq M : \tau_n \geq \delta\} \leq \sum_{n \geq M} \tau_n < \epsilon.$$

Therefore $\delta D(\delta) \leq M\delta + \epsilon$. Taking $\delta \downarrow 0$ gives $\limsup_{\delta \downarrow 0} \delta D(\delta) \leq \epsilon$. Since ϵ is arbitrary, $\delta D(\delta) \rightarrow 0$. Substitute $\delta = Q_m/P$ and divide by T . $\square$

Candidate law 13.5 (Finite-power de-Zenoization). *Every genuine infinite finite-time schedule has vanishing finite-power utilization:*

$$\frac{\text{robust latch-writable prefix at peak power } P}{\text{uniform finite prefix possible at peak power } P} \rightarrow 0.$$

Thus once a proposed hypercomputer is reduced to a finite robust prefix, its supertask clock is asymptotically worse than an ordinary uniform finite clock for peak-power use.

This sharpens the de-Zenoization theorem in [Theorem 12.5](#). The earlier result optimized a fixed finite prefix. [Theorem 13.4](#) says that the inefficiency is not an artifact of geometric clocks; it holds for every summable infinite schedule.

13.2 Closed forms and a new schedule family

For the geometric schedule

$$\tau_n = T\rho(1 - \rho)^n,$$

the latch horizon grows only logarithmically:

$$N_{\text{lat}}(P; m) \sim \frac{\log(PT\rho/Q_m)}{-\log(1 - \rho)}.$$

For the polynomial-tail family

$$R_n = T(n+1)^{-a}, \quad \tau_n = R_n - R_{n+1} \sim aTn^{-(a+1)},$$

the latch horizon grows polynomially:

$$N_{\text{lat}}(P; m) \sim \left(\frac{aTP}{Q_m} \right)^{1/(a+1)}.$$

The utilization therefore decays as $P^{-a/(a+1)}$.

A more surprising family is the log-slow summable schedule

$$\tau_n = \frac{c}{(n+2) \log^2(n+2)}, \quad \sum_{n=0}^{\infty} \tau_n = 1.$$

The normalization is $c \approx 0.474$. This schedule is barely summable. Its latch horizon is almost linear:

$$N_{\text{lat}}(P) \sim \frac{cP}{\log^2 P}, \quad U(P) \sim \frac{c}{\log^2 P}.$$

This is exceptional because it nearly saturates the finite-power upper envelope while still being a true infinite finite-time schedule. The price is endpoint convergence:

$$R_N \sim \frac{c}{\log N}, \quad N_{\text{clk}}(\epsilon) \approx \exp(c/\epsilon).$$

Thus a schedule can be made friendly to finite-power robust prefixes only by making the approach to the endpoint extraordinarily slow.

Candidate law 13.6 (Power-clock seesaw). *Within summable Zeno schedules, finite-power writable depth and endpoint-tail convergence pull against each other. Fast endpoint convergence, as in geometric Zeno clocks, gives poor finite-power writable depth. Near-linear writable depth, as in the log-slow schedule, gives extremely poor endpoint convergence.*

13.3 Numerical stress tests

The following experiments set $T = Q_m = 1$ and compare four infinite schedules against the ordinary finite uniform benchmark. They are not simulations of a specific device. They are stress tests of the counting laws above.

At $P_{\text{max}} = 10^{12}$, the geometric clock can robustly write only 39 latch events, while the polynomial $a = 1$ schedule reaches 999,999 events and the log-slow schedule reaches about 1.09×10^9 events. The uniform finite schedule would reach 10^{12} events. This illustrates [Theorem 13.4](#): true supertask schedules can improve dramatically over geometric Zeno clocks, but their finite-power utilization still tends to zero.

[Table 12](#) reverses the question: how much peak power is required to certify the first N latch writes? For $N = 10^6$, the geometric schedule requires $\log_{10} P \approx 301030$, while the log-slow schedule requires only $\log_{10} P \approx 8.60$. This is a genuine design lesson: geometric Zeno clocks are simple but extremely poor if the aim is to maximize robust finite prefix depth under a peak-power constraint.

The seesaw is clearest in [Table 13](#). The log-slow schedule gives over a billion finite-power writable steps at $P_{\text{max}} = 10^{12}$, but to reduce the remaining tail time below 10^{-3} it requires roughly $10^{205.85}$ steps. It is therefore a good finite-prefix accelerator and a bad endpoint hypercomputer. Geometric scheduling is the opposite: it reaches small tail time quickly but gives a tiny finite-power writable prefix.

Table 11: Endpoint write horizons under a peak-power bound $P_{\max}$, with $T = Q = 1$. A robust latch write at step n is allowed only if $\tau_n \geq 1/P_{\max}$. The log-slow schedule uses $\tau_n = c/((n+2)\log^2(n+2))$ normalized to total time one, with $c \approx 0.474$.

$P_{\max}$	uniform finite	geometric	poly $a = 1$	poly $a = 2$	log-slow	log-slow utilization
1e+03	1,000	9	31	12	35	3.500e-02
1e+06	1,000,000	19	999	125	6,212	6.212e-03
1e+09	1,000,000,000	29	31,622	1,259	2,219,718	2.220e-03
1e+12	1,000,000,000,000	39	999,999	12,598	1,094,182,375	1.094e-03
1e+18	1,000,000,000,000,000,000	59	999,999,999	1,259,920	418,180,106,217,419	4.182e-04

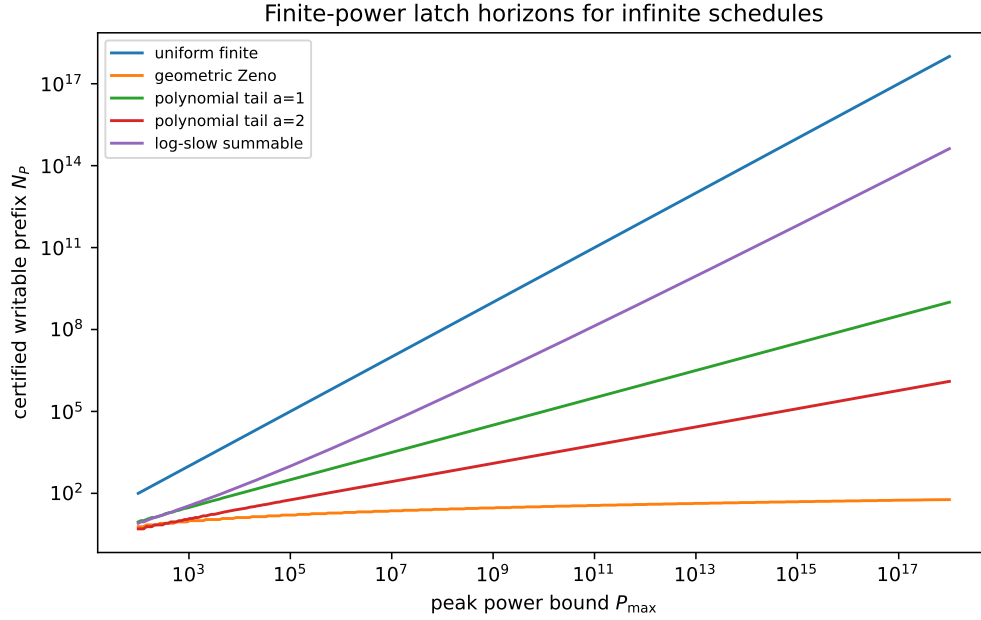


Figure 9: Finite-power latch horizons. The uniform finite line is the finite optimum. Every infinite finite-time schedule lies below it asymptotically, but the log-slow schedule approaches it up to a logarithmic penalty.

Table 12: Peak power required to certify the first N latch writes, reported as $\log_{10} P_N$ for $T = Q = 1$. Uniform finite scheduling is the finite optimum. Among true infinite schedules, the log-slow tail can be much more finite-power efficient than geometric Zeno scheduling, but it pays with extremely slow endpoint convergence.

N	uniform	geometric	poly $a = 1$	poly $a = 2$	log-slow
10	1.00	3.01	2.04	2.76	2.13
40	1.60	12.04	3.21	4.52	3.08
100	2.00	30.10	4.00	5.71	3.66
1,000	3.00	301.03	6.00	8.70	5.00
1,000,000	6.00	301030.00	12.00	17.70	8.60

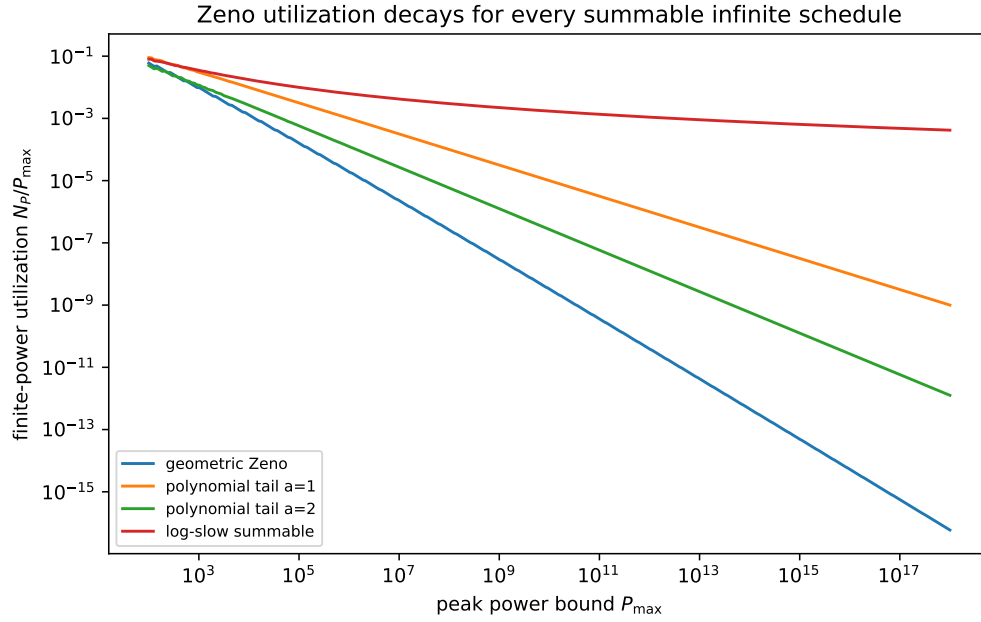


Figure 10: Finite-power utilization $N_P/P_{\max}$ for true infinite schedules. The decay to zero is guaranteed by [Theorem 13.4](#); the log-slow schedule shows that the decay can be made very slow.

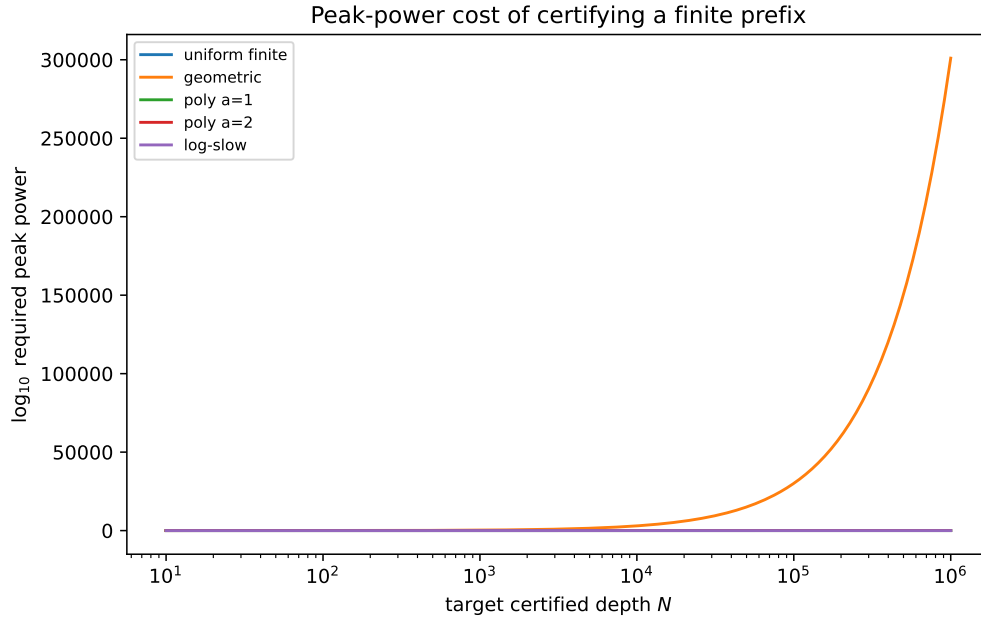


Figure 11: Peak-power cost of certifying a finite prefix. Geometric Zeno scheduling has an exponential peak-power cost in the certified depth. The log-slow schedule has almost linear cost, but only by surrendering fast convergence to the Zeno endpoint.

Table 13: Power-clock seesaw with $P_{\max} = 10^{12}$ and endpoint tail target $R_N \leq 10^{-3}$. The log-slow schedule is exceptionally good for finite-power writable prefixes, but its endpoint convergence is so slow that the clock depth is reported as $\log_{10} N_{\text{clk}}$.

schedule	$N_{\text{pow}}(10^{12})$	utilization $N_{\text{pow}}/P_{\max}$	$\log_{10} N_{\text{clk}}(10^{-3})$
geometric	39	3.900e-11	1.00
poly a=1	999,999	1.000e-06	3.00
poly a=2	12,598	1.260e-08	1.49
log-slow	1,094,182,375	1.094e-03	205.85

13.4 Latch-impulse complementarity

The finite signal-budget law and the finite-power latch law are not competing objections. They are complementary normal forms.

Theorem 13.7 (Latch-impulse complementarity). *Consider a Zeno endpoint implementation in which each possible halt time n is served by one of two mechanisms:*

1. *a latch write requiring $\tau_n \geq Q_m/P$ under finite peak power P , or*
2. *an impulse signal v_n with total signal energy $\sum \|v_n\|^2 \leq B$ observed with a fixed positive uncertainty or Gaussian noise.*

If $P < \infty$, $B < \infty$, and the readout gain is finite near the no-event state, then there is no uniform positive-margin detector for all halt times $n \in \mathbb{N}$.

Proof. Only finitely many indices can be served by the latch mechanism, by [Theorem 13.3](#). If infinitely many halt times remain to be served by impulse signals, then finite signal energy implies that their norms cannot be bounded below by a positive constant. The deterministic finite signal-budget law and the Gaussian endpoint law therefore rule out a uniform positive gap under nonzero uncertainty. Finite-gain postprocessing cannot restore the gap by [Theorem 9.4](#). $\square$

This yields a sharper audit equation. The robust shadow should not merely include clock precision, statistical signal budget, power, and reliability as independent limits. It should identify which endpoint normal form is being used:

$$N_{\text{endpoint}} = \begin{cases} N_{\text{stat}}(B, \sigma, \alpha), & \text{impulse/noisy endpoint,} \\ N_{\text{lat}}(P; m), & \text{persistent latch endpoint,} \\ \infty, & \text{only with singular gain, zero noise, unbounded power, or ideal exact readout.} \end{cases}$$

Then

$$N_{\text{rob}}^{(0.4)} = \min\{N_{\text{clk}}, N_{\text{endpoint}}, N_{\text{rel}}, N_{\text{memory}}, N_{\text{geometry}}\}.$$

13.5 Applications

The latch results suggest several practical uses for the theory even if physical hypercomputation is unavailable.

1. **Auditing Zeno-machine proposals.** A proposal with a persistent output bit should be asked for its latch-write horizon $N_{\text{lat}}(P; m)$, not only its formal schedule τ_n .
2. **Designing bounded accelerators.** If the application values a large finite semi-decision prefix rather than exact halting, log-slow or slowly varying schedules can dominate geometric schedules under peak-power constraints.
3. **Benchmarking physical computation.** The utilization function $U(P) = Q_m N_{\text{lat}}(P)/(PT)$ provides a dimensionless way to compare finite-time accelerators against ordinary finite clocks.
4. **Robust computable-analysis models.** The map from an ideal limit operation to its finite robust shadow may be formalized as a resource-bounded truncation functor on Weihrauch or SCI-style limit constructions.

13.6 Targets generated by Version 0.4

1. Prove optimal schedule shapes for maximizing $N_{\text{lat}}(P)$ subject to both $\sum \tau_n = T$ and a required endpoint-tail profile $R_N \leq r(N)$.
2. Develop a regular-variation atlas for Zeno schedules, relating τ_n , R_n , $N_{\text{lat}}(P)$, and $N_{\text{clk}}(\epsilon)$.
3. Replace [Theorem 13.1](#) by device-specific metastability and error-dissipation models, including finite-time erasure, reversible latch writes, and time-asymmetric control protocols.
4. Extend the latch-impulse complementarity theorem to Malament-Hogarth settings, where the endpoint write may be replaced by causal message reception near a horizon.
5. Define a robust-shadow operator on Weihrauch degrees and test whether it collapses standard discontinuity degrees to computable finite-prefix approximants under finite resource budgets.

14 Continuation IV: quantifier shadows and look-elsewhere debt

The previous sections reduced a single halting-style supertask to a finite robust shadow. This continuation asks what happens when the endpoint observer is not merely deciding one existential event, but is implicitly evaluating a bounded fragment of an arithmetical formula. This is the natural next blindspot. A single Zeno halting observer resembles one limit or one jump. Iterated jumps, iterated limits, and finite-height limit towers are the language of much current computability analysis, Weihrauch complexity, and SCI theory [9, 12, 14]. The question here is not only which degree is represented, but how a finite noisy endpoint pays for the bounded shadow of that degree.

The new finding is that robust finite shadows have two different sizes. The first is a *certificate size*: how many primitive facts suffice to justify one particular yes or no answer. The second is a *pivotal volume*: how many primitive facts could be the fact on which the answer turns. Hypercomputational readout pays for pivotal volume, not only for certificate size.

14.1 Bounded quantifier shadows

Let

$$Q = Q_1 Q_2 \cdots Q_d, \quad Q_i \in \{\exists, \forall\},$$

be a quantifier prefix and let $N_1, \dots, N_d$ be finite horizons. For a Boolean predicate table

$$R : [N_1] \times \cdots \times [N_d] \rightarrow \{0, 1\},$$

define the bounded quantifier shadow

$$F_Q^{\mathbf{N}}(R) = Q_1 n_1 < N_1 \ Q_2 n_2 < N_2 \ \cdots \ Q_d n_d < N_d \ R(n_1, \dots, n_d).$$

The primitive leaves of this finite tree are the entries of R . Their number is

$$V(\mathbf{N}) = \prod_{i=1}^d N_i.$$

For equal horizons $N_i = N$, this becomes $V = N^d$.

Definition 14.1 (Pivotal leaf). A leaf $\ell = (n_1, \dots, n_d)$ is pivotal for $F_Q^{\mathbf{N}}$ if there is an assignment of all other leaves such that changing only $R(\ell)$ changes the value of $F_Q^{\mathbf{N}}$.

Theorem 14.2 (Every leaf is potentially pivotal). *For every nonempty finite horizon vector $\mathbf{N}$ and every quantifier prefix Q , every leaf of $F_Q^{\mathbf{N}}$ is pivotal.*

Proof. Fix a target leaf. Work from the root of the quantifier tree to the target. At every existential node on the target path, set all off-path sibling subformulas to false. Then the existential value is controlled by the target branch. At every universal node on the target path, set all off-path sibling subformulas to true. Then the universal value is again controlled by the target branch. Continue recursively until the target leaf is reached. Under this assignment the value of the whole bounded formula equals the value of the target leaf, so flipping that leaf flips the output. $\square$

Candidate law 14.3 (Pivotal-volume margin law). *Consider an additive finite-margin endpoint architecture in which changing leaf ℓ changes the endpoint state by a vector v_ℓ , and suppose a uniform robust observer requires endpoint states on opposite sides of a pivotal boundary to be separated by at least 2η . If the accessible quadratic signal budget is*

$$B = \sum_{\ell} \|v_\ell\|^2,$$

then uniform robust computation of $F_Q^{\mathbf{N}}$ over all predicate tables requires

$$B \geq 4\eta^2 V(\mathbf{N}).$$

For equal horizons, the resulting quantifier-root bound is

$$N \leq \left(\frac{B}{4\eta^2} \right)^{1/d}.$$

The important feature is the root. With fixed physical budget, each additional bounded quantifier level root-extracts the certified branch horizon. The cost is not merely a constant factor added by a more complicated observer; it is a volume expansion of the finite robust shadow.

14.2 Certificate size is not pivotal volume

The pivotal-volume law could be mistaken for a statement about proof size. It is not. A bounded formula may have a small certificate for the answer that happens to be true, while still having all leaves potentially pivotal across the space of possible predicate tables.

Define $C_+(Q; \mathbf{N})$ and $C_-(Q; \mathbf{N})$ as the worst-case number of primitive predicate leaves needed in a smallest positive or negative certificate. For one outer block of size N and a tail formula φ , the recurrences are

$$\begin{aligned} C_+(\exists_N \varphi) &= C_+(\varphi), & C_-(\exists_N \varphi) &= NC_-(\varphi), \\ C_+(\forall_N \varphi) &= NC_+(\varphi), & C_-(\forall_N \varphi) &= C_-(\varphi), \end{aligned}$$

with $C_+ = C_- = 1$ at a predicate leaf.

Theorem 14.4 (Certificate/pivotal separation). *For every prefix Q and horizon vector $\mathbf{N}$, the certificate volumes are generated by the recurrences above, while the pivotal volume is always $V(\mathbf{N})$. Therefore there are prefixes for which both C_+ and C_- are much smaller than V , even though a uniform yes/no endpoint must remain sensitive to every leaf.*

Proof. The certificate recurrences are the standard introduction rules for finite existential and universal statements. An existential truth is certified by one true branch, while an existential falsehood requires certificates that all branches are false. A universal truth requires all branches, while a universal falsehood requires one false branch. Iterating these rules gives the displayed recurrences. The pivotal claim is [Theorem 14.2](#). $\square$

Table 14: Finite quantifier shadows at branch horizon $N = 30$. $V = N^d$ is the number of primitive predicate leaves. C_+ and C_- are smallest worst-case positive and negative certificate sizes. The pivotal budget is the deterministic finite-margin lower bound $4\eta^2V$ with $\eta = 1$.

Prefix	V	C_+	C_-	$4V$	certificate asymmetry
$\exists$	30	1	30	120	30
$\forall$	30	30	1	120	30
$\exists\forall$	900	30	30	3.6×10^3	1
$\forall\exists$	900	30	30	3.6×10^3	1
$\exists\exists\exists$	2.700×10^4	30	900	1.080×10^5	30
$\forall\forall\forall$	2.700×10^4	900	30	1.080×10^5	30
$\exists\forall\exists\forall$	8.100×10^5	900	900	3.240×10^6	1

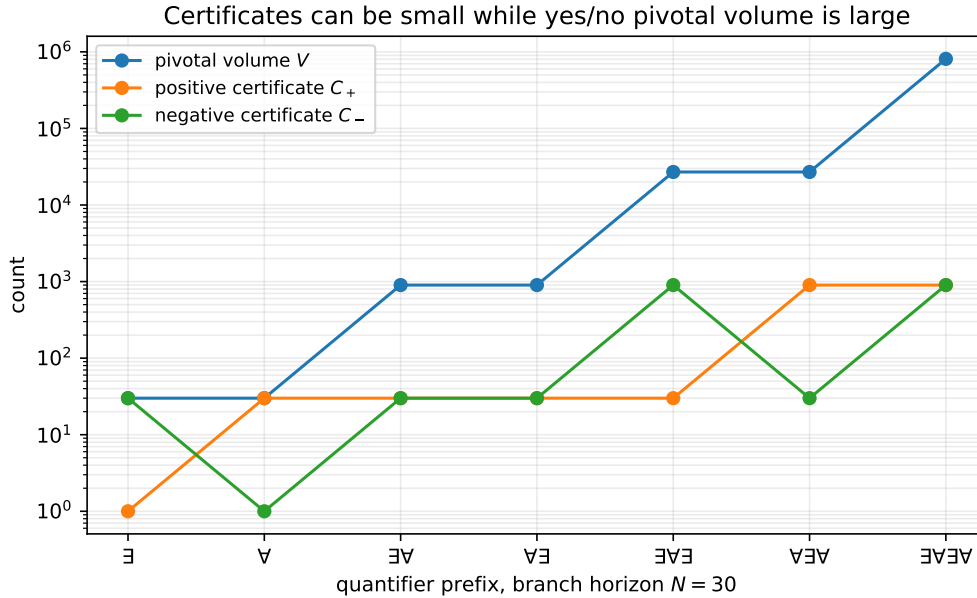


Figure 12: Certificate size versus pivotal volume for bounded quantifier shadows with branch horizon $N = 30$. Certificate sizes can remain far below the leaf volume, but every leaf is still potentially pivotal for the yes/no readout.

14.3 Gaussian look-elsewhere debt

The statistical endpoint theorem in the previous continuation gave a linear finite-horizon bound when each possible halt time needs a fixed Gaussian margin. A sharper blindspot appears when the

endpoint does not know in advance which of V leaves is the pivotal one. Then the no-event state must be protected not from one false alarm, but from the maximum of many false alarms. This is the same mathematical pressure behind family-wise multiple comparison corrections such as the Sidak correction [23].

The cleanest model is the finite OR shadow. Let

$$Y_j = ab_j + \xi_j, \quad j = 1, \dots, V,$$

where $b_j \in \{0, 1\}$, the noise variables ξ_j are independent $N(0, \sigma^2)$, and the observer declares an event when $\max_j Y_j > \theta$.

Theorem 14.5 (Look-elsewhere endpoint law for orthogonal Gaussian events). *For the OR model above, suppose the false-positive probability under $b = 0^V$ is at most α and the false-negative probability is at most α for every single-leaf event $b = e_j$. Then any max-threshold detector must have*

$$\theta \geq \sigma z_{\max}(V, \alpha), \quad z_{\max}(V, \alpha) = \Phi^{-1}((1 - \alpha)^{1/V}),$$

and

$$a \geq \sigma(z_{\max}(V, \alpha) + z_{1-\alpha}).$$

Consequently the total signal budget for uniform single-event detection satisfies

$$B_{\text{LEE}}(V, \alpha, \sigma) \geq V \sigma^2 (z_{\max}(V, \alpha) + z_{1-\alpha})^2.$$

As $V \rightarrow \infty$ with fixed α ,

$$B_{\text{LEE}}(V, \alpha, \sigma) = 2\sigma^2 V \log V + O(V \sqrt{\log V}).$$

Proof. Under $b = 0^V$,

$$\Pr[\max_j Y_j \leq \theta] = \Phi(\theta/\sigma)^V.$$

False-positive probability at most α requires $\Phi(\theta/\sigma)^V \geq 1 - \alpha$, which gives the first inequality. If $b = e_j$, then one coordinate has distribution $N(a, \sigma^2)$. For that coordinate alone to exceed the threshold with probability at least $1 - \alpha$, one needs $(a - \theta)/\sigma \geq z_{1-\alpha}$. Combining this with the lower bound on θ gives the amplitude bound. Multiplying a^2 by V gives the budget bound. The asymptotic formula follows from the standard normal tail inversion $\Phi^{-1}(1 - u) = \sqrt{2 \log(1/u)} + o(\sqrt{\log(1/u)})$ with $u \asymp \alpha/V$. $\square$

Candidate law 14.6 (Quantifier look-elsewhere debt). *For an additive orthogonal endpoint implementation of a bounded quantifier shadow with $V = \prod_i N_i$ potentially pivotal leaves, nonzero Gaussian endpoint noise and uniform family-wise error control impose the scaling*

$$B_{Q,\text{stat}}(\mathbf{N}, \alpha, \sigma) \gtrsim 2\sigma^2 V(\mathbf{N}) \log V(\mathbf{N}).$$

For equal horizons $N_i = N$, the statistical branch horizon is determined implicitly by

$$B \gtrsim 2\sigma^2 d N^d \log N.$$

Thus the earlier linear statistical shadow acquires a logarithmic look-elsewhere debt when the endpoint must protect a whole quantifier tree rather than one known event coordinate.

Table 15: Gaussian look-elsewhere debt for an unknown pivotal leaf. Here $\alpha = 0.05$ and $\sigma = 1$. The threshold $z_{\max}$ satisfies $\Phi(z_{\max})^V = 1 - \alpha$. The final column compares the required budget with the earlier no-look-elsewhere bound $4\sigma^2 z_{1-\alpha}^2 V$.

V	$z_{\max}$	required amplitude	B_{LEE}	ratio
10	2.568	4.213	177	1.64
100	3.283	4.928	2.43×10^3	2.24
1×10^3	3.884	5.529	3.057×10^4	2.82
1.000×10^4	4.412	6.057	3.668×10^5	3.39
1.000×10^6	5.322	6.967	4.854×10^7	4.49
1.000×10^8	6.105	7.750	6.007×10^9	5.55

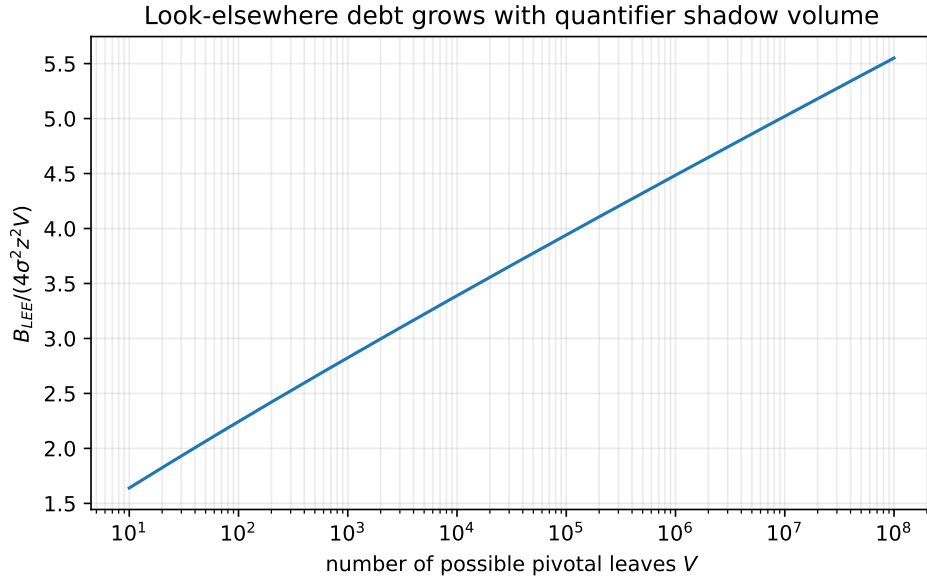


Figure 13: The statistical budget ratio between the look-elsewhere bound and the earlier fixed-coordinate Gaussian bound. The penalty grows slowly, but it is unbounded.

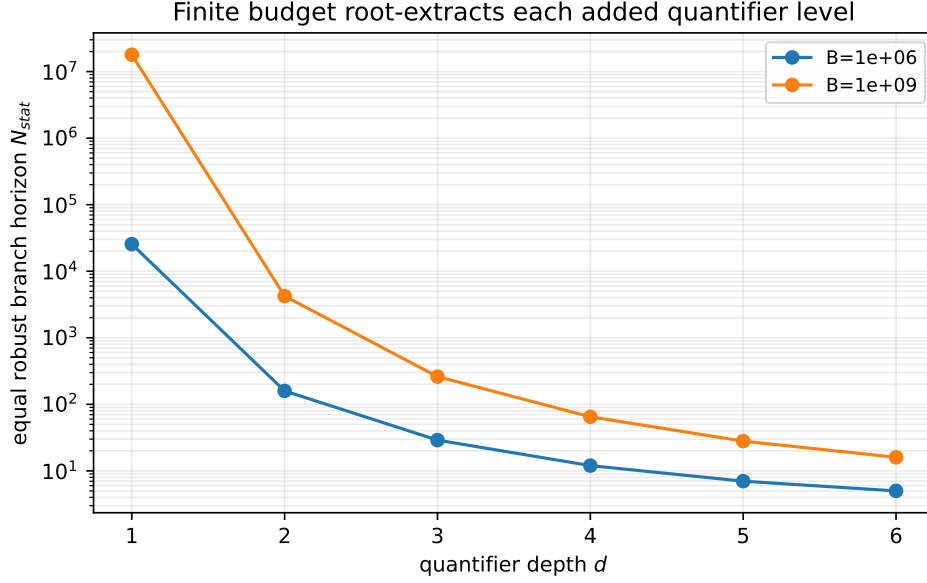


Figure 14: Equal branch horizon under fixed statistical budgets. The robust branch horizon falls approximately as a depth root, with an additional logarithmic penalty from look-elsewhere control.

Table 16: Equal-branch robust horizon erosion. For a depth- d quantifier shadow with equal horizon N and volume $V = N^d$, the deterministic column uses $4V \leq B$ with $\eta = 1$. The statistical column uses the Gaussian look-elsewhere budget with $\alpha = 0.05$ and $\sigma = 1$.

budget B	depth d	N_{det}	N_{stat}	V_{stat}
1.000×10^6	1	2.500×10^5	2.556×10^4	2.556×10^4
1.000×10^6	2	500	159	2.528×10^4
1.000×10^6	3	62	29	2.439×10^4
1.000×10^6	4	22	12	2.074×10^4
1.000×10^6	5	12	7	1.681×10^4
1.000×10^6	6	7	5	1.562×10^4
1.000×10^9	1	2.500×10^8	1.792×10^7	1.792×10^7
1.000×10^9	2	1.581×10^4	4.23×10^3	1.792×10^7
1.000×10^9	3	629	261	1.778×10^7
1.000×10^9	4	125	65	1.785×10^7
1.000×10^9	5	47	28	1.721×10^7
1.000×10^9	6	25	16	1.678×10^7

14.4 Average-case pivotality and threshold zones

Worst-case pivotal volume is unforgiving. Average-case pivotality is more structured. If the primitive predicate leaves are independent Bernoulli variables with $\Pr[R(\ell) = 1] = q$, then the truth probability of a bounded prefix is generated by the recursion

$$P_{\exists_N \varphi}(q) = 1 - (1 - P_{\varphi}(q))^N, \quad P_{\forall_N \varphi}(q) = P_{\varphi}(q)^N,$$

starting with $P_R(q) = q$ at a leaf. For monotone Boolean functions, the derivative dP_Q/dq equals the total influence, i.e. the expected number of pivotal leaves under the product distribution; this is the finite Margulis–Russo identity [24]. By symmetry, the average pivotal probability of a particular leaf is

$$\Pr[\ell \text{ is pivotal}] = \frac{1}{V} \frac{dP_Q(q)}{dq}.$$

This gives a distributional complement to [Theorem 14.3](#). Uniform readout must budget for all leaves, but a prior-weighted bounded accelerator may concentrate resources near the threshold zone where the derivative is large. Outside that zone, most leaves are not pivotal on typical instances.

Table 17: Average pivotality for random predicate tables. The threshold q_{50} is the Bernoulli leaf probability for which the bounded formula is true with probability 1/2. The total influence is $d\Pr[F = 1]/dq$ at q_{50} ; per-leaf pivotality is total influence divided by V .

N	V	prefix	q_{50}	total influence	per-leaf pivotality
10	10	$\exists$	0.0670	5.36	0.536
10	10	$\forall$	0.9330	5.36	0.536
10	100	$\exists\forall$	0.7631	4.7	0.047
10	100	$\forall\exists$	0.2369	4.7	0.047
10	1×10^3	$\exists\forall\exists$	0.1341	12.9	0.0129
10	1×10^3	$\forall\exists\forall$	0.8659	12.9	0.0129
30	30	$\exists$	0.0228	15.4	0.512
30	30	$\forall$	0.9772	15.4	0.512
30	900	$\exists\forall$	0.8816	11.9	0.0133
30	900	$\forall\exists$	0.1184	11.9	0.0133
30	2.700×10^4	$\exists\forall\exists$	0.0687	45.5	0.00168
30	2.700×10^4	$\forall\exists\forall$	0.9313	45.5	0.00168
100	100	$\exists$	0.0069	50.3	0.503
100	100	$\forall$	0.9931	50.3	0.503
100	1.000×10^4	$\exists\forall$	0.9515	36.6	0.00366
100	1.000×10^4	$\forall\exists$	0.0485	36.6	0.00366
100	1.000×10^6	$\exists\forall\exists$	0.0298	183	1.829×10^{-4}
100	1.000×10^6	$\forall\exists\forall$	0.9702	183	1.829×10^{-4}

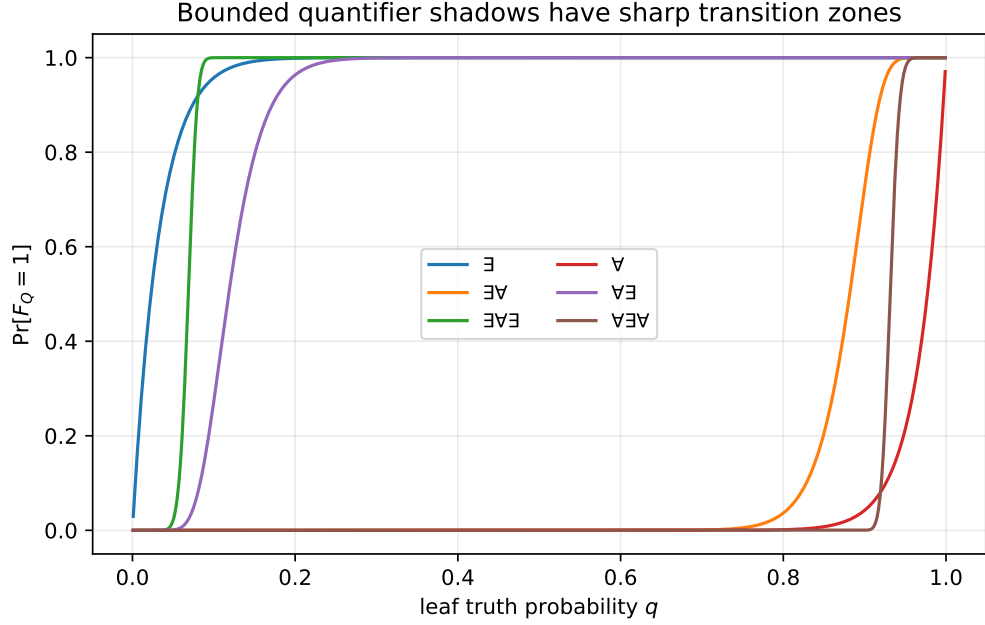


Figure 15: Truth-probability transition curves for bounded quantifier shadows with branch horizon $N = 30$. Alternating prefixes push the transition toward extreme predicate probabilities.

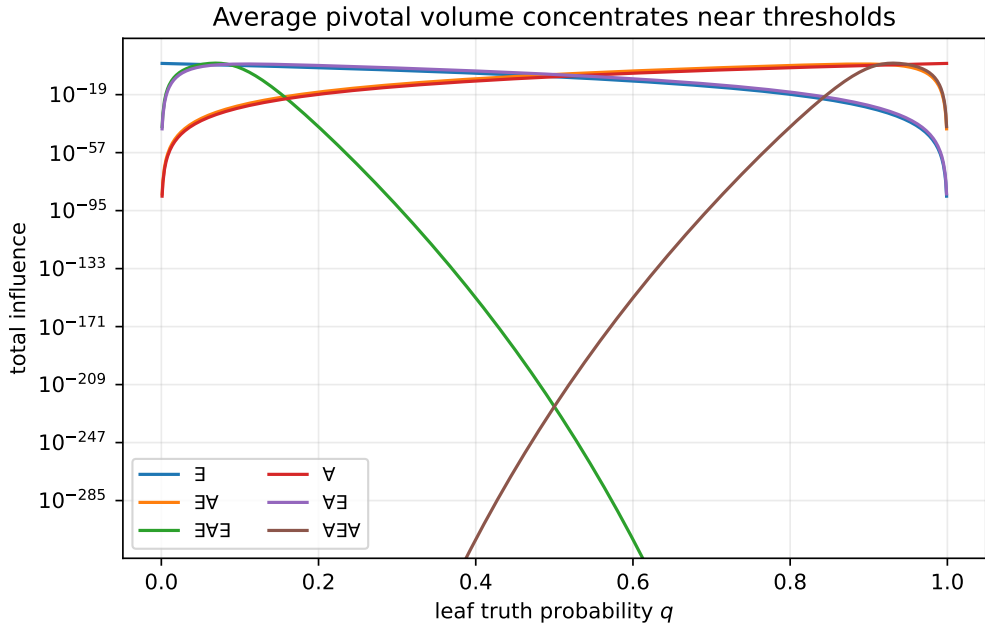


Figure 16: Total influence for the same bounded shadows. Average pivotality concentrates in narrow transition zones; away from these zones, typical instances are stable even though worst-case pivotal volume remains V .

14.5 Exceptional findings from the quantifier-shadow continuation

The new tests suggest five useful principles.

1. **Pivotal volume is the robust yes/no cost.** Every primitive leaf of a bounded quantifier shadow is potentially decisive, so worst-case robust readout pays for $V = \prod_i N_i$ possible pivotal locations.
2. **Certificate size can understate endpoint cost.** A formula such as $\exists i \forall j R(i, j)$ has certificates of size N for both answers, but it has N^2 potentially pivotal leaves. Returning only the answer is therefore not necessarily cheaper than certificate search at the endpoint.
3. **Multiple comparisons add a logarithmic debt.** Under Gaussian endpoint noise, unknown pivotal location changes the statistical budget from order V to order $V \log V$ in an orthogonal event-channel architecture.
4. **Each additional bounded jump root-extracts the branch horizon.** With equal horizons and deterministic finite-margin budget B , the branch horizon scales like $B^{1/d}$. With statistical look-elsewhere control it is smaller, approximately solving $B \asymp 2\sigma^2 d N^d \log N$.
5. **Worst-case and distributional shadows diverge.** Under a Bernoulli prior, average pivotality is concentrated near sharp transition zones. This creates an application space for prior-weighted finite-shadow accelerators without restoring worst-case hypercomputation.

14.6 Applications suggested by quantifier shadows

1. **Bounded arithmetic auditing.** A proposed device claiming a bounded fragment of a jump hierarchy can be audited by its quantifier volume V and its look-elsewhere budget, rather than only by its clock rate.
2. **Verification and model checking.** Safety properties often contain alternating bounded quantifiers: for all environments, there exists a controller response, for all time steps a bad state is absent. The root law estimates how much branch depth is lost when robust physical margins are imposed.
3. **Search over rare counterexamples.** Counterexample search is often an existential shadow; robust universal safety is its dual. The certificate/pivotal split explains why finding a counterexample can be much cheaper than certifying its absence.
4. **Statistical theorem-proving portfolios.** Influence curves identify regions of a problem distribution where additional finite-shadow resources are most likely to affect the final answer. This suggests allocating computational effort by pivotality rather than by raw search depth.
5. **SCI/Weihrauch rank diagnostics.** Finite-height limit towers can be assigned a resource shadow: depth becomes quantifier volume, while discontinuous post-processing becomes a readout budget. This may provide a practical companion to degree-theoretic rank.

14.7 Targets for further study

1. **Information-theoretic lower bounds beyond orthogonal channels.** Prove or falsify the conjecture that every finite-dimensional noisy endpoint implementation of an arbitrary bounded quantifier shadow requires information budget at least proportional to a metric entropy of the pivotal boundary.

2. **Optimal nonuniform branch horizons.** Given a budget B and different priors or costs at different quantifier levels, choose $N_1, \dots, N_d$ to maximize certified prior mass subject to $\prod_i N_i \log(\prod_i N_i)$ constraints.
3. **Influence-weighted endpoint coding.** Replace worst-case leaf margins by a Bayes risk objective weighted by pivotal probabilities. The expected optimizer should be a water-filling law in which leaves with negligible influence receive no margin.
4. **Adaptive quantifier trees.** The current model uses full rectangular horizons. Search trees in verification and theorem proving are adaptive and pruned. The target is a robust-shadow law in terms of explored tree boundary size rather than N^d .
5. **Connection to jump operators.** Develop a finite-resource analogue of Weihrauch jumps in which the jump operation increases both logical degree and endpoint pivotal volume. This could sharpen the earlier bound $\deg(f) \leq \deg_W(O)$ into a joint logical/resource invariant.

15 Continuation V: Bayesian finite shadows and endpoint waterlines

The preceding sections are mostly worst-case. They ask for uniform correctness over every halt time, every pivotal leaf, or every endpoint address. This continuation studies a more practical but easily confused variant: a finite device may be designed for a *computable prior* over possible decisive event times. The device then no longer claims to solve an uncomputable predicate uniformly. It becomes a finite statistical experiment whose resource allocation can be optimized.

The blindspot is that distributional usefulness is often presented informally as if it were a softened form of hypercomputation. The calculation below suggests a sharper statement. Finite resources do not merely make the machine less certain. Under a broad soft-error model, the Bayes-optimal endpoint develops a finite *waterline*: high-prior events receive endpoint margin, while sufficiently low-prior events receive exactly zero endpoint resource. The optimized finite shadow therefore contains an explicit blind tail even when the loss function is smooth.

This section uses the notation of statistical signal allocation. The role of classical water-filling in information theory is only an analogy; the problem here is a detection allocation problem for endpoint events rather than a communication-rate maximization problem [21, 22]. The physical motivation is also consistent with recent finite-time thermodynamic computing results, where time, dissipation, and accuracy must be traded rather than treated as independent ideal limits [15].

15.1 Soft endpoint miss model

Let π_n be a computable prior over the event “the decisive event first occurs at stage n ”, sorted so that

$$\pi_1 \geq \pi_2 \geq \dots > 0, \quad \sum_{n=1}^{\infty} \pi_n = 1, \quad \pi_n \rightarrow 0.$$

Assign endpoint signal energy

$$b_n = \|v_n\|^2 \geq 0, \quad \sum_n b_n \leq B.$$

The parameter B is the finite endpoint resource available for making event n statistically visible. The exponential miss surrogate is

$$M_n(b_n) = e^{-\kappa b_n}, \quad \kappa > 0.$$

For an additive Gaussian endpoint, κ can be read as a Chernoff-rate constant proportional to $1/\sigma^2$; in this draft it is kept as an abstract normalized separation rate. The Bayes miss objective is

$$L(b) = \sum_{n=1}^{\infty} \pi_n e^{-\kappa b_n}.$$

The finite-shadow design problem is therefore

$$L^*(B) = \inf_{b_n \geq 0, \sum b_n \leq B} \sum_{n=1}^{\infty} \pi_n e^{-\kappa b_n}.$$

This is not a hypercomputational objective: it is an ordinary computable optimization problem once the prior and budget are finite descriptions. Its value is diagnostic: it shows what a rational finite endpoint would do instead of pretending to cover an infinite horizon uniformly.

Theorem 15.1 (Endpoint waterline theorem). *Let π_n be nonincreasing, positive, summable, and tend to zero. For every finite $B > 0$, the optimizer of the exponential endpoint miss problem is*

$$b_n^* = \frac{1}{\kappa} [\log(\pi_n/\theta)]_+,$$

where $\theta > 0$ is the unique waterline satisfying

$$\sum_{n=1}^{\infty} [\log(\pi_n/\theta)]_+ = \kappa B.$$

The active protected set is finite:

$$A_B = \{n : \pi_n > \theta\}, \quad K_B = |A_B| < \infty.$$

The optimized residual miss has the closed form

$$L^*(B) = K_B \theta + \sum_{n > K_B} \pi_n,$$

when the prior is strictly decreasing and the active set is the first K_B events.

Proof. For a finite truncation, the problem is convex. The Lagrange conditions for an active coordinate give

$$-\kappa \pi_n e^{-\kappa b_n} + \lambda = 0,$$

so

$$b_n = \frac{1}{\kappa} \log \frac{\kappa \pi_n}{\lambda}.$$

Writing $\theta = \lambda/\kappa$ gives the displayed formula, and inactive coordinates are exactly those with $\pi_n \leq \theta$. The budget equation determines θ because the left-hand side is continuous and strictly decreasing between prior atoms, tends to zero as $\theta \uparrow \pi_1$, and tends to infinity as $\theta \downarrow 0$. Since $\pi_n \rightarrow 0$, any positive θ leaves only finitely many active coordinates. The infinite-dimensional result follows by applying the finite KKT solution to longer prefixes and passing to the monotone limit. For each active coordinate,

$$\pi_n e^{-\kappa b_n^*} = \theta,$$

which gives the residual-miss formula. □

Candidate law 15.2 (Bayesian finite-shadow law). *A finite endpoint resource budget plus a computable prior turns a halting-style endpoint into a finite active shadow:*

$$\boxed{\text{finite } B \implies \text{Bayes-optimal protected set } A_B \text{ is finite.}}$$

The unprotected tail is not a defect of a bad schedule; it is the optimal solution of the finite-resource statistical design problem under the exponential miss model.

This law is a distributional counterpart to the robust-shadow equation. Worst-case robustness gives a hard finite horizon. Bayesian design gives a soft finite horizon with a waterline and a prior tail.

15.2 Prior-tail scaling laws

The waterline equation can be solved asymptotically for common monotone priors. Let K_B denote the active horizon and let constants be suppressed when they do not affect scaling.

Theorem 15.3 (Prior-tail active-horizon asymptotics). *For the endpoint waterline optimizer:*

1. *If $\pi_n \asymp Ce^{-sn}$, $s > 0$, then*

$$\boxed{K_B \sim \sqrt{2\kappa B/s}.$$

2. *If $\pi_n \asymp Ce^{-an^p}$, $a, p > 0$, then*

$$\boxed{K_B \sim \left(\frac{\kappa(p+1)B}{ap} \right)^{1/(p+1)}.$$

3. *If $\pi_n \asymp Cn^{-\beta}$, $\beta > 1$, then*

$$\boxed{K_B \sim \kappa B/\beta.}$$

The associated residual-risk envelopes are

$$\boxed{L^*(B) \asymp \begin{cases} \sqrt{B} e^{-\sqrt{2\kappa s B}}, & \pi_n \asymp e^{-sn}, \\ B^{1/(p+1)} e^{-cB^{p/(p+1)}}, & \pi_n \asymp e^{-an^p}, \\ B^{1-\beta}, & \pi_n \asymp n^{-\beta}. \end{cases}}$$

Proof. The active set is determined by $\pi_n > \theta$, so θ is comparable to π_{K_B} . The budget equation is

$$\kappa B = \sum_{n \leq K_B} \log(\pi_n/\theta).$$

For a geometric tail, $\log(\pi_n/\theta) \sim s(K_B - n)$, giving $\kappa B \sim sK_B^2/2$. For a stretched exponential tail,

$$\log(\pi_n/\theta) \sim a(K_B^p - n^p),$$

so

$$\kappa B \sim a \int_0^{K_B} (K_B^p - x^p) dx = \frac{ap}{p+1} K_B^{p+1}.$$

For a power law,

$$\log(\pi_n/\theta) \sim \beta \log(K_B/n),$$

and Stirling’s formula gives

$$\sum_{n \leq K_B} \log(K_B/n) = K_B \log K_B - \log(K_B!) \sim K_B.$$

Thus $\kappa B \sim \beta K_B$. The risk formula follows from $L^*(B) = K_B \theta + \sum_{n > K_B} \pi_n$ and standard tail estimates. $\square$

The exceptional feature is a tail-shape inversion. Exponentially decaying priors produce very small residual risk, but the active horizon grows only like $\sqrt{B}$. Power-law priors activate linearly many addresses, but the residual risk falls only polynomially. Thus “how far the machine looks” and “how reliable the answer is” are not the same resource curve.

Table 18: Asymptotic active-horizon and residual-risk envelopes for monotone prior families under the exponential miss model. Constants are omitted in the risk column.

prior family	active horizon K_B	risk envelope
geometric $\pi_n \propto e^{-sn}$	$\sqrt{2\kappa B/s}$	$\sqrt{B} e^{-\sqrt{2\kappa s B}}$
stretched exp. $\pi_n \propto e^{-an^p}$	$\left(\frac{\kappa(p+1)B}{ap}\right)^{1/(p+1)}$	$B^{1/(p+1)} e^{-cB^{p/(p+1)}}$
power law $\pi_n \propto n^{-\beta}$	$\kappa B/\beta$	$B^{1-\beta}$

15.3 Numerical waterline experiments

The experiments set $\kappa = 1$ and solve the waterline equation for five priors: geometric tails with $r = 0.95$ and $r = 0.995$, a stretched exponential tail with exponent $p = 1/2$, and power laws with $\beta = 1.5$ and $\beta = 2.5$. The scripts use deterministic finite arrays with negligible numerical truncation at the reported active depths.

Table 19: Bayesian endpoint waterline experiment with normalized Chernoff rate $\kappa = 1$. The active set K_B is finite for every finite budget and matches the asymptotic laws even at moderate budgets.

prior	B	K_B	K_B/K_{asym}	Bayes miss $L^*(B)$	inactive miss
geometric $r = 0.95$	100	62	0.993	0.171	0.0416
geometric $r = 0.95$	1,000	197	0.998	4.44×10^{-4}	4.09×10^{-5}
geometric $r = 0.95$	1.0×10^4	624	0.999	4.06×10^{-13}	1.24×10^{-14}
geometric $r = 0.995$	100	200	1.001	0.735	0.367
geometric $r = 0.995$	1,000	632	1.001	0.176	0.0421
geometric $r = 0.995$	1.0×10^4	1,997	1.000	4.94×10^{-4}	4.49×10^{-5}
stretched $\beta_s = 1/2$	100	113	1.001	0.508	0.259
stretched $\beta_s = 1/2$	1,000	525	1.002	0.0762	0.0221
stretched $\beta_s = 1/2$	1.0×10^4	2,433	1.000	3.99×10^{-4}	5.96×10^{-5}
power law $\beta = 1.5$	100	69	1.035	0.137	0.0918
power law $\beta = 1.5$	1,000	670	1.005	0.0443	0.0296
power law $\beta = 1.5$	1.0×10^4	6,671	1.001	0.0141	0.00937
power law $\beta = 2.5$	100	42	1.050	0.00441	0.00179
power law $\beta = 2.5$	1,000	403	1.008	1.53×10^{-4}	6.13×10^{-5}
power law $\beta = 2.5$	1.0×10^4	4,005	1.001	4.90×10^{-6}	1.96×10^{-6}

The agreement with the asymptotic laws is striking. At $B = 1000$, the active horizons are 197, 632, 525, 670, and 403 for the five priors, and the ratios to the asymptotic predictions are already

within about one percent except for the small-budget cases. This supports the use of the closed-form laws as design rules rather than only as limiting statements.

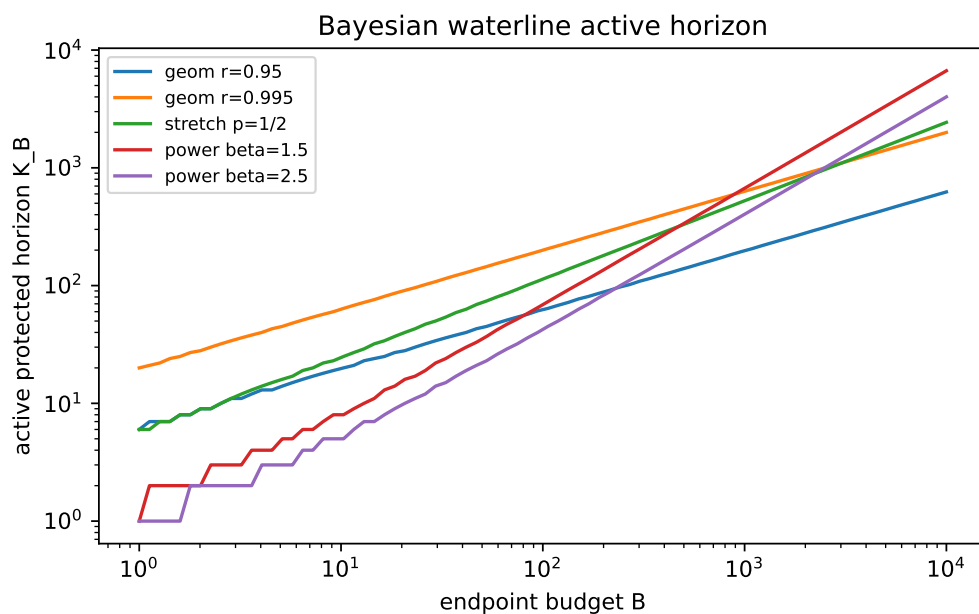


Figure 17: Active protected horizon K_B under the Bayesian endpoint waterline. Tail shape controls how quickly finite endpoint budget can extend the distributional shadow.

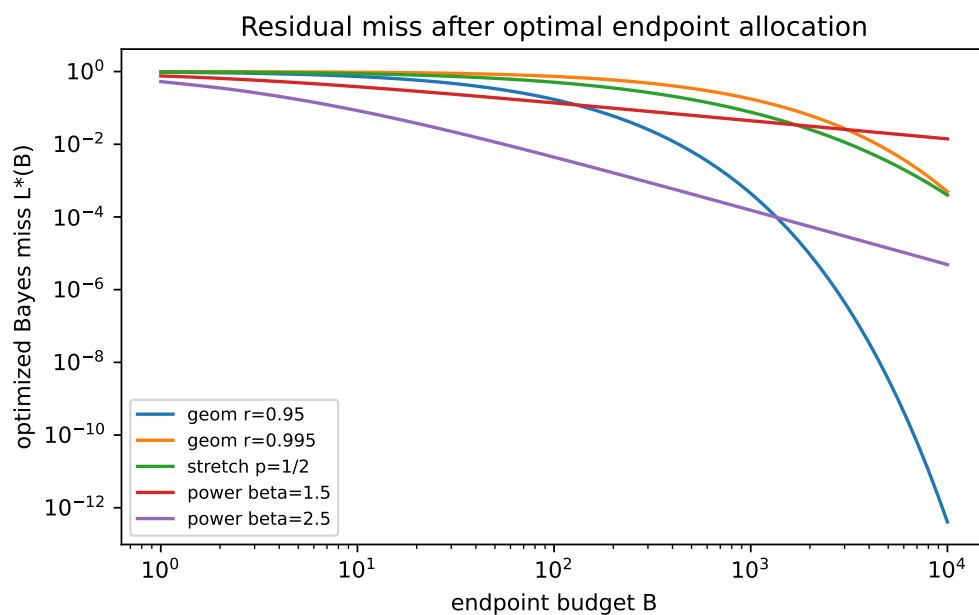


Figure 18: Optimized Bayes miss probability after waterline allocation. Priors with broader active horizons need not have lower residual risk; support depth and answer reliability separate.

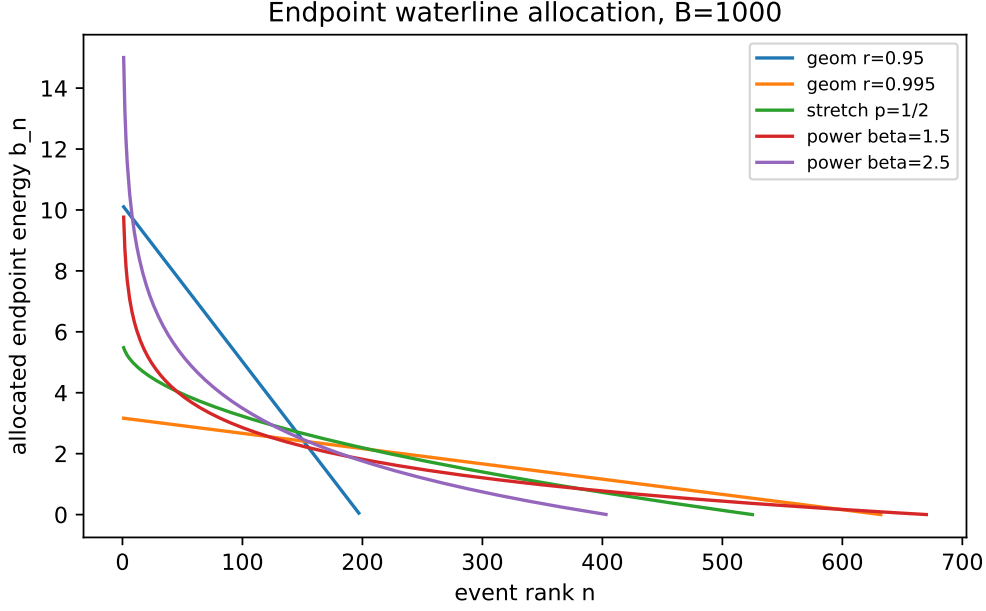


Figure 19: Endpoint energy allocations b_n^* at $B = 1000$. The waterline equalizes active weighted miss contributions: $\pi_n e^{-\kappa b_n^*} = \theta$ for all protected events.

A hard-margin design with unit cost per protected address behaves differently from the soft waterline design. Hard coverage minimizes tail miss for a fixed number of protected prefix addresses. The soft waterline spends extra margin on high-prior early events and may protect fewer addresses than a hard prefix. The two designs answer different questions: hard coverage asks which events can be certified above a fixed margin; soft waterline allocation asks how to minimize expected miss under a smooth statistical loss.

Table 20: Hard prefix coverage versus soft Bayesian waterline allocation at $B = 1000$ and $\kappa = 1$. Hard coverage uses unit margin on the first B addresses; the waterline spreads unequal margins to minimize the exponential Bayes miss surrogate.

prior	hard prefix $M = B$ miss	waterline active K_B	waterline miss	b_1	b_{K_B}
geometric $r = 0.95$	0	197	4.44×10^{-4}	10.1	0.0494
geometric $r = 0.995$	0.00665	632	0.176	3.16	8.22×10^{-4}
stretched $\beta_s = 1/2$	0.00332	525	0.0762	5.48	7.13×10^{-4}
power law $\beta = 1.5$	0.0242	670	0.0443	9.76	0.00188
power law $\beta = 2.5$	1.57×10^{-5}	403	1.53×10^{-4}	15	0.0057

The table shows why finite-shadow claims must specify their loss function. At $B = 1000$, the hard prefix covers the first 1000 events. For the geometric $r = 0.95$ prior this leaves essentially no tail in the displayed precision, while the soft waterline still accepts a small miss probability in order to make early detections more reliable. For the power-law $\beta = 1.5$ prior, the hard prefix miss is about 2.4×10^{-2} , while the soft waterline miss is about 4.4×10^{-2} , but the latter is optimizing a different exponential detection-risk objective rather than hard prefix coverage.

15.4 Coupling to Zeno schedules

The waterline allocation can be combined with the de-Zenoization and latch laws. Let a protected event require endpoint energy b_n and action time τ_n . Under a peak power limit P , a necessary compatibility condition is

$$\boxed{b_n \leq P\tau_n.}$$

If the robust finite shadow is already active only on A_B , then the peak-power optimal schedule for those active events is not Zeno. It is the finite schedule

$$\boxed{\tau_n^* = \frac{b_n^*}{\sum_{m \in A_B} b_m^*} T, \quad n \in A_B,}$$

when the goal is to realize the waterline endpoint allocation over a finite external time T with constant peak power. This is a second de-Zenoization principle: after a prior and a finite budget are specified, the optimal physical implementation is a finite resource-matched schedule, not an infinite accelerating tail.

Candidate law 15.4 (Waterline de-Zenoization). *A distributional finite shadow decomposes an ideal supertask into two ordinary finite optimization layers:*

$$\boxed{\text{endpoint waterline } b_n^* \quad + \quad \text{resource-matched finite schedule } \tau_n^* \propto b_n^* .}$$

Any residual infinite tail is deliberately unprotected under the specified prior and loss.

15.5 Applications

The waterline theory suggests applications that do not require claims of physical hypercomputation.

1. **Theorem-proving portfolios.** Empirical proof-length or proof-search priors can be converted into endpoint margins. The waterline identifies which proof depths merit extra checking or redundant verification.
2. **Counterexample search.** In model checking or fuzzing, likely short counterexamples receive high margin, while long low-prior cases remain in the explicit blind tail. The blind tail can be reported as a calibrated miss probability rather than hidden behind an idealized acceleration claim.
3. **Anytime verification.** A finite-shadow verifier can publish $L^*(B)$, K_B , and the inactive prior mass as an audit trail for how much distributional risk remains.
4. **Thermodynamic or stochastic accelerators.** For noisy hardware, the waterline gives a target energy profile for making early high-prior events statistically separable without pretending that all late events are equally protected.

15.6 Targets for further study

General miss functions. Replace $e^{-\kappa b}$ by a measured detector miss curve $M(b)$. The expected general law is

$$b_n^* = (M')^{-1}(-\lambda/\pi_n),$$

with finite active support whenever $|M'(0)| < \infty$ and $\pi_n \rightarrow 0$. A singular slope at zero may be the only way to obtain a genuinely infinite active set under finite budget.

Robust-prior design. The prior itself may be uncertain. A minimax or distributionally robust waterline should replace π_n by an ambiguity set and may produce thicker tails than the Bayes-optimal allocation above.

Empirical halt-time priors. Run proof search, SAT solving, bounded model checking, and fuzzing corpora to estimate realistic decisive-event priors. Fit the observed tails to geometric, stretched exponential, lognormal, or power-law families, then compare measured performance with [Theorem 15.3](#).

Quantum endpoint channels. Replace the scalar energy allocation by quantum states ρ_n with an energy constraint and optimize Bayes risk using trace-distance or quantum Chernoff bounds. The candidate analogue is again a waterline, but now over distinguishability exponents rather than Euclidean signal energies.

Degree theory of distributional shadows. Classical Weihrauch degrees classify uniform computational content. The finite-shadow waterline suggests a complementary distributional invariant:

$$B \mapsto (K_B, L^*(B), \sum_{n > K_B} \pi_n).$$

This invariant does not replace Weihrauch degree. It measures how quickly a discontinuous observer becomes useful when restricted to finite resources and a computable prior.

16 Continuation VI: coded endpoints and online aggregation

The previous continuation studied Bayesian waterlines: finite endpoint budget plus a prior creates an optimal blind tail. This continuation studies a different possible escape route. Perhaps the endpoint need not allocate an independent signal to every possible late event. It might instead use a code, an internal aggregator, or a symbolic state that compresses many leaves into one final answer. This is a serious objection to any purely leaf-counting resource law.

The result is a cost-migration principle. Coding can enlarge a finite shadow, and online aggregation can compress independent leaf signals dramatically. But the missing cost does not disappear. It reappears as endpoint channel capacity, witness-address information, state-space width, reliable memory, update energy, or an ordering/fairness constraint. This section therefore separates three tasks that are often conflated:

1. deciding one final bit;
2. returning an auditable witness or pivotal address;
3. aggregating an interleaved family of obligations into a final bit.

The first can sometimes be compressed to a single latch. The second has an information capacity debt. The third has an online state-width debt.

This viewpoint connects the limit-interface framework to classical information theory and to ordered binary decision diagrams. Shannon’s channel model makes exact recovery of arbitrary continuous information require unbounded capacity in ordinary noisy settings [21]. Fano-type converses, as presented for example in standard information theory texts, lower-bound the information needed to recover one of many messages with small error [22]. Finite-blocklength information theory makes the same issue non-asymptotic rather than merely asymptotic [17]. On the symbolic side,

ordered binary decision diagrams show that variable order can change representation size from linear to exponential [18, 19]. The new contribution here is to reinterpret these facts as *finite-shadow laws for hypercomputational endpoints*.

16.1 Endpoint address capacity

Suppose the endpoint is asked not merely to say that a bounded event occurred, but to return which event occurred. Let

$$J \in \{1, \dots, M\}$$

be a uniformly distributed witness address, and let the endpoint observation be a random variable or physical state Y . A decoder returns $\hat{J}(Y)$. If the endpoint channel has mutual information at most

$$I(J; Y) \leq C$$

and the error probability obeys $\Pr(\hat{J} \neq J) \leq \alpha < 1$, then Fano's inequality gives, in bits,

$$H(J | Y) \leq h_2(\alpha) + \alpha \log_2(M - 1),$$

where h_2 is binary entropy. Since

$$I(J; Y) = \log_2 M - H(J | Y),$$

one obtains the finite-address bound

$$\log_2 M \leq \frac{C + h_2(\alpha)}{1 - \alpha}.$$

This yields the following capacity shadow.

Theorem 16.1 (Endpoint address-capacity shadow). *Let a finite endpoint channel have accessible information capacity C bits for a witness address. Uniform recovery of one of M equiprobable addresses with error at most $\alpha < 1$ requires*

$$M \leq 2^{(C + h_2(\alpha))/(1 - \alpha)}.$$

Thus any fixed finite endpoint capacity gives a finite witness-address horizon.

Proof. The displayed inequality is the standard Fano converse rearranged for M . The data processing inequality also implies that no post-processing of the endpoint observation can increase $I(J; Y)$, so any physical decoder is subject to the same bound. $\square$

Candidate law 16.2 (Witness-address debt). *A finite endpoint can hide a witness address only by not returning it. If an endpoint claims to return an auditable witness, certificate, pivotal leaf, or finite trace prefix, then it must pay at least the information needed to identify that object:*

$$\text{auditable address extraction} \implies \text{endpoint capacity } \Omega(\log M).$$

At fixed nonzero error, an unbounded address catalog requires unbounded endpoint capacity.

Table 21: Endpoint-address capacity shadows from Fano’s inequality. A finite channel capacity can support an enormous but finite number of reliably distinguishable witness addresses; it cannot support an unbounded address catalog at fixed error.

error target α	capacity C (bits)	upper bound $\log_2 M$	equiv. $\log_{10} M$
0.01	5	5.13	1.54
0.01	10	10.18	3.07
0.01	20	20.28	6.11
0.01	50	50.59	15.23
0.01	100	101.09	30.43
0.05	5	5.56	1.68
0.05	10	10.83	3.26
0.05	20	21.35	6.43
0.05	50	52.93	15.93
0.05	100	105.56	31.78
0.10	5	6.08	1.83
0.10	10	11.63	3.50
0.10	20	22.74	6.85
0.10	50	56.08	16.88
0.10	100	111.63	33.60

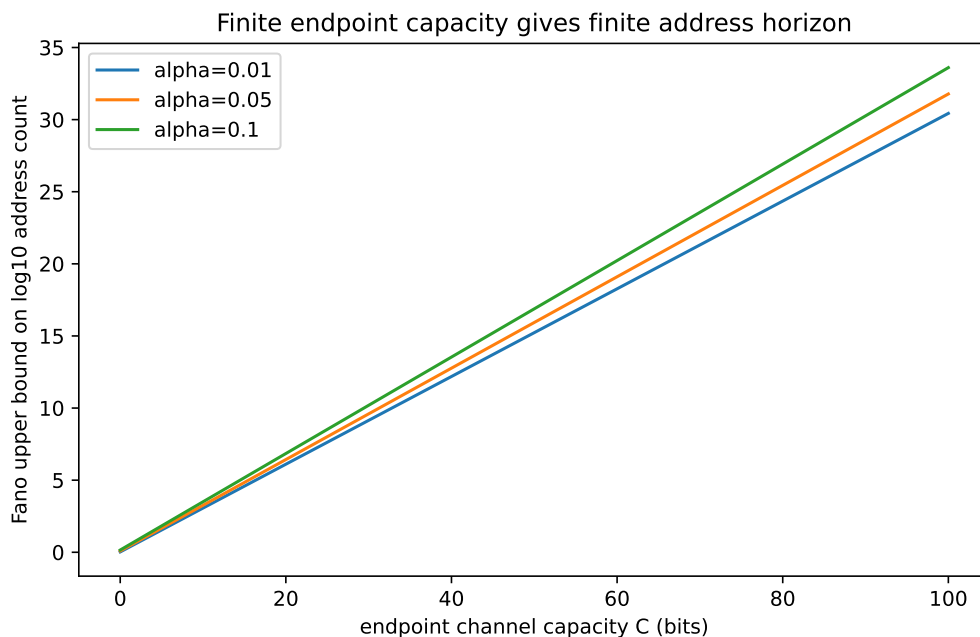


Figure 20: Fano capacity shadow for endpoint address extraction. The bound grows exponentially with capacity, so coding can greatly enlarge a finite shadow, but no finite capacity supports an unbounded address catalog at fixed error.

Table 22: Semantic compression at the endpoint. Decision-only endpoints can hide the witness address; certificate-returning endpoints must pay an explicit communication debt.

task	endpoint output	auditable object	address load	finite-shadow risk
decide $\exists n < N$ $R(n)$	one bit	none	0	non-verifiable yes/no
return witness for $\exists n < n$ $N R(n)$		witness index	$\lceil \log_2 N \rceil$	capacity/address debt
return pivotal leaf of leaf address bounded formula		local predicate query	$\lceil \log_2 V \rceil$	look-elsewhere debt
return full finite trace prefix	trace code	replayable transcript	$\Omega(N)$ bits generally	ordinary finite communication

The semantic distinction in Table 22 is important. A pure halting decision can compress all positive witnesses into one symbolic output. That does not violate the capacity law because the endpoint is not returning the address. However, then the endpoint has given up auditability: a positive answer cannot be checked by inspecting a reported witness. If the device instead returns a witness, the address capacity debt appears. If it returns only a latch, the finite-power latch and late-write laws from Section 12 and the later latch section apply.

Remark 16.3 (Coding versus orthogonal catalogs). The earlier Gaussian orthogonal endpoint bound used one independent channel per possible late event. General coding can be more efficient than an orthogonal catalog. For example, a finite-dimensional codebook in an m -dimensional ball can pack exponentially many points in m at fixed margin. But this merely replaces a linear signal-budget law by a metric-entropy or channel-capacity law. The robust horizon remains finite unless dimension, capacity, precision, or reliability becomes singular.

Table 23: Endpoint resource classes. Coded endpoints can enlarge a finite shadow, but every finite class retains an explicit finite horizon under fixed nonzero resolution and reliability requirements.

endpoint model	finite resource	robust horizon	singular escape
scalar/orthogonal impulse catalog	$B = \sum_n \ v_n\ ^2$	$N = O(B/\eta^2)$	infinite catalog energy or zero noise
finite-dimensional coded states	dimension m , radius A	$N + 1 \leq (1 + A/\eta)^m$	unbounded metric dimension or zero resolution
noisy channel at the limit	capacity C_{lim}	$N + 1 \leq 2^{(C_{\text{lim}} + h_2(\epsilon))/(1-\epsilon)}$	infinite capacity or zero-error oracle channel
persistent latch	peak power P	$N_{\text{lat}}(P; m) = \#\{n : \tau_n \geq Q_m/P\}$	unbounded late-write power
online aggregation	residual width $W_\pi(F)$	$\log_2 W_\pi(F)$ reliable bits	unbounded memory or singular ordering

16.2 Online aggregation and residual width

A second escape route is internal aggregation. Instead of sending every primitive event to the endpoint, the device may update a finite internal state and send only the final value. This is exactly what a latch does for a bounded OR. The right finite-shadow question is therefore not merely how many primitive leaves are pivotal. It is how many different *residual futures* the internal state must distinguish as the stream is processed.

Let $F : \{0, 1\}^M \rightarrow \{0, 1\}$ be a bounded Boolean shadow of a limit predicate, and let π be the order in which the M primitive bits are presented to the aggregator. After a prefix $a \in \{0, 1\}^t$, the remaining suffix $u \in \{0, 1\}^{M-t}$ is evaluated by the residual function

$$F_{a,t}^\pi(u) = F(a, u)$$

where the variables are understood in the order π . Define the residual family and its width by

$$\mathcal{R}_t(F, \pi) = \{F_{a,t}^\pi : a \in \{0, 1\}^t\}, \quad W_\pi(F) = \max_{0 \leq t \leq M} |\mathcal{R}_t(F, \pi)|.$$

Theorem 16.4 (Online state lower bound). *Any deterministic online aggregator that reads variables in order π and computes F exactly must have at least $W_\pi(F)$ distinguishable internal states at some time. Equivalently, it must carry at least*

$$\log_2 W_\pi(F)$$

reliable state bits.

Proof. If two prefixes a and b produce different residual functions, then there is a suffix u such that $F(a, u) \neq F(b, u)$. If the aggregator placed a and b in the same internal state, then feeding the same suffix u afterward would force the same final answer in both cases, contradicting correctness. Thus all residual functions present at a cut require pairwise distinct states. $\square$

Candidate law 16.5 (Readout-state migration). *Endpoint compression conserves difficulty. Reducing independent endpoint leaf readout for a bounded shadow F requires one of the following payments:*

$$\begin{aligned} & \text{leaf signal volume} \vee \log_2 W_\pi(F) \text{ reliable state bits} \\ & \vee \text{late-write/latch cost} \vee \text{restricted ordering or unfairness.} \end{aligned}$$

This is not a no-compression theorem; it is a cost-location theorem.

16.3 Frontier shadow for a bounded alternating predicate

Consider the finite Σ_2^0 -like shadow

$$F_N^{\exists\forall}(x) = \bigvee_{i=1}^N \bigwedge_{j=1}^N x_{ij}.$$

This is the finite question: “is there a row whose every entry is one?” It is a bounded version of an existential choice over universal obligations. It is also the smallest example where online order becomes decisive.

For an order π , let $f_\pi(t)$ be the number of rows that have been opened but not completed after t variables have been read:

$$f_\pi(t) = \#\{i : 0 < \#\{j : x_{ij} \text{ has appeared in the first } t \text{ positions of } \pi\} < N\},$$

and let

$$f_\pi = \max_t f_\pi(t).$$

This is the row-frontier of the stream.

Theorem 16.6 (Alternating-frontier shadow). *For $F_N^{\exists\forall} = \bigvee_i \bigwedge_j x_{ij}$ under order π ,*

$$2^{f_\pi} \leq W_\pi(F_N^{\exists\forall}) \leq 2^{f_\pi} + 1.$$

Consequently, any exact online aggregator for this bounded alternating shadow requires

$$\log_2 W_\pi(F_N^{\exists\forall}) = \Theta(f_\pi)$$

reliable state bits.

Proof. Fix a cut with $f = f_\pi(t)$ open rows. A prefix can make each open row either still live (all entries seen so far in that row are one) or dead (some seen entry is zero). Completed rows can be forced not to satisfy the formula by placing a zero in each completed row; unopened rows are fixed by the suffix and do not depend on the prefix. For every subset A of the open rows, choose a prefix in which exactly the rows in A remain live. If $A \neq B$, choose a row in the symmetric difference and complete exactly that row with ones while forcing all other live or unopened rows to fail in the suffix. The two residual functions then differ. Hence at least 2^f residuals occur.

For the upper bound, a prefix matters only through whether a completed satisfying row has already occurred, in which case the residual is the constant-one function, and otherwise through the subset of open rows that remain live. There are at most 2^f such subsets plus the constant-one residual. $\square$

The theorem explains why independent pivotal-volume accounting can be too pessimistic and also why aggregation is not free. The formula has N^2 primitive leaves. A row-major order has $f_\pi = 1$, so it can be evaluated with constant state width. A column-major or round-robin order has $f_\pi = N$, so it requires N reliable bits and 2^N residual states. Thus the cost drops from leaf volume N^2 to frontier memory N , but it does not vanish.

Table 24: Exact residual-width check for $F_N^{\exists\forall} = \bigvee_i \bigwedge_j x_{ij}$. Width is the number of distinct suffix functions an online aggregator must distinguish after some prefix.

N	order	max frontier f_π	lower 2^{f_π}	upper $2^{f_\pi} + 1$	brute W_π
2	row-major	1	2	3	3
2	column-major	2	4	5	4
3	row-major	1	2	3	3
3	column-major	3	8	9	8
4	row-major	1	2	3	3
4	column-major	4	16	17	16

The brute-force experiment computes exact residual functions for $N = 2, 3, 4$. The row-major width is always 3; the column-major width is exactly 2^N . The calculation validates the frontier theorem on complete truth tables and makes the cost migration visible at small sizes.

Table 25: Cost migration for $F_N^{\exists\forall}$. Independent leaf readout pays pivotal volume N^2 . Online aggregation can compress row-major order to constant state width, but fair/column interleaving pays N reliable memory bits and 2^N residual states.

N	pivotal leaves N^2	row states	fair memory bits	fair residual states
4	16	3	4	2^4
8	64	3	8	2^8
16	256	3	16	2^{16}
32	1024	3	32	2^{32}
64	4096	3	64	2^{64}

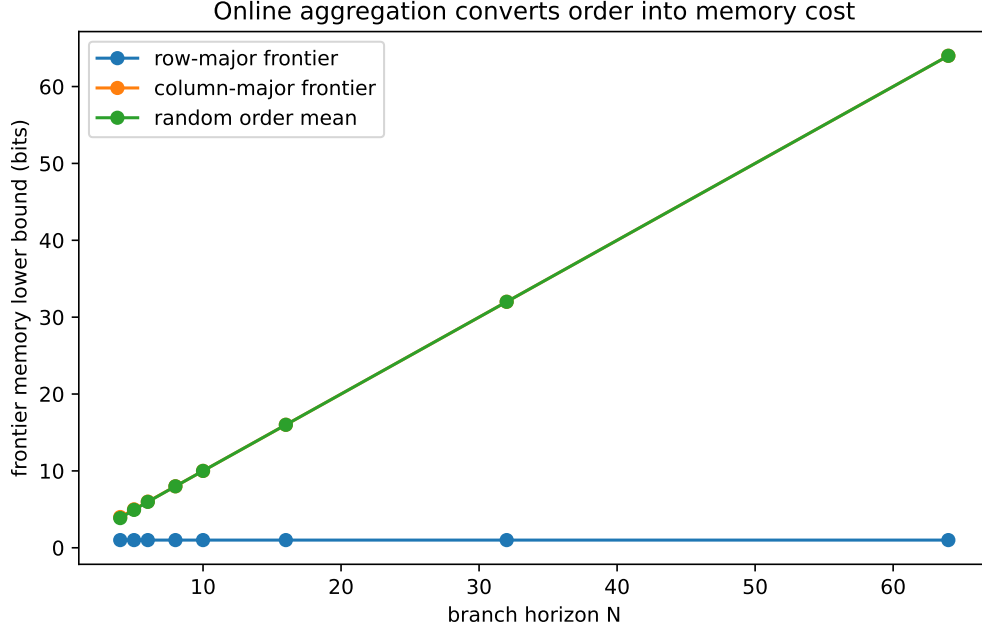


Figure 21: Frontier memory lower bound for row-major, column-major, and random orders. A random interleaving almost immediately behaves like the fair column-major case: nearly all rows are simultaneously open.

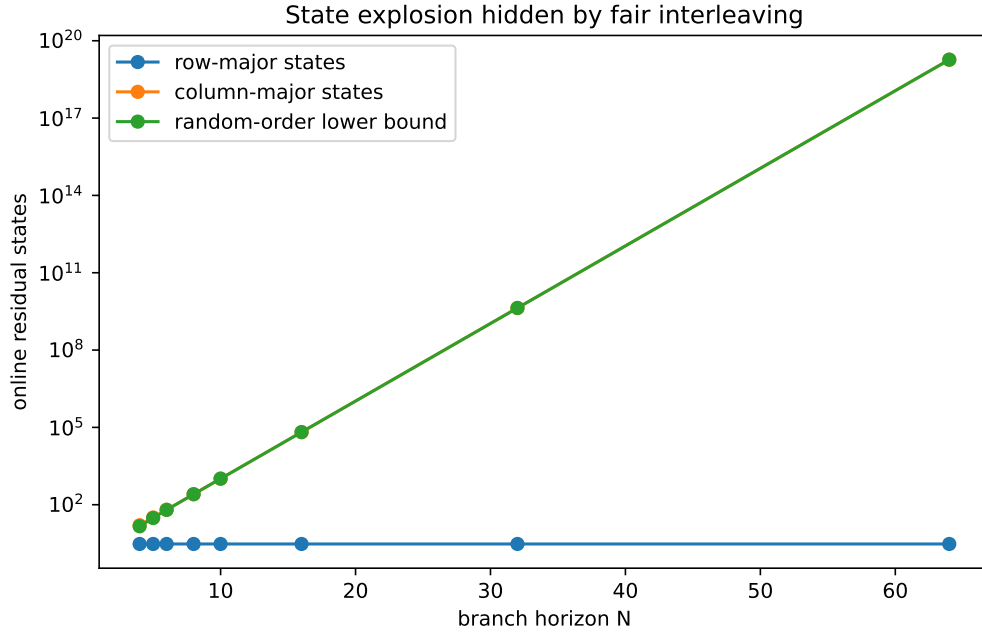


Figure 22: Residual-state explosion for fair interleavings of the bounded $\exists\forall$ shadow. Row-major aggregation uses constant state width, while fair or random order has state count exponential in the branch horizon.

Table 26: Random interleavings of an $N \times N$ bounded $\exists\forall$ shadow almost always expose the full row frontier. Since fair column-major order has memory lower bound N bits, random order behaves like the fair case rather than the row-major case.

N	trials	mean max frontier	5%	95%	$P(f_\pi = N)$
4	1000	3.863	3	4	0.865
5	1000	4.929	4	5	0.930
6	1000	5.973	6	6	0.973
8	500	7.996	8	8	0.996
10	500	10.000	10	10	1.000
16	200	16.000	16	16	1.000
32	100	32.000	32	32	1.000
64	50	64.000	64	64	1.000

The random-order experiment is striking. For $N = 10$ the full row frontier appeared in 99.9% of random orders; for $N \geq 16$ it appeared in every sampled order. This is not a claim of a deep probability theorem, but it identifies a useful practical invariant: unless an order is deliberately blocked, a bounded alternating shadow behaves as though all branches are simultaneously open. Random fairness is memory-expensive.

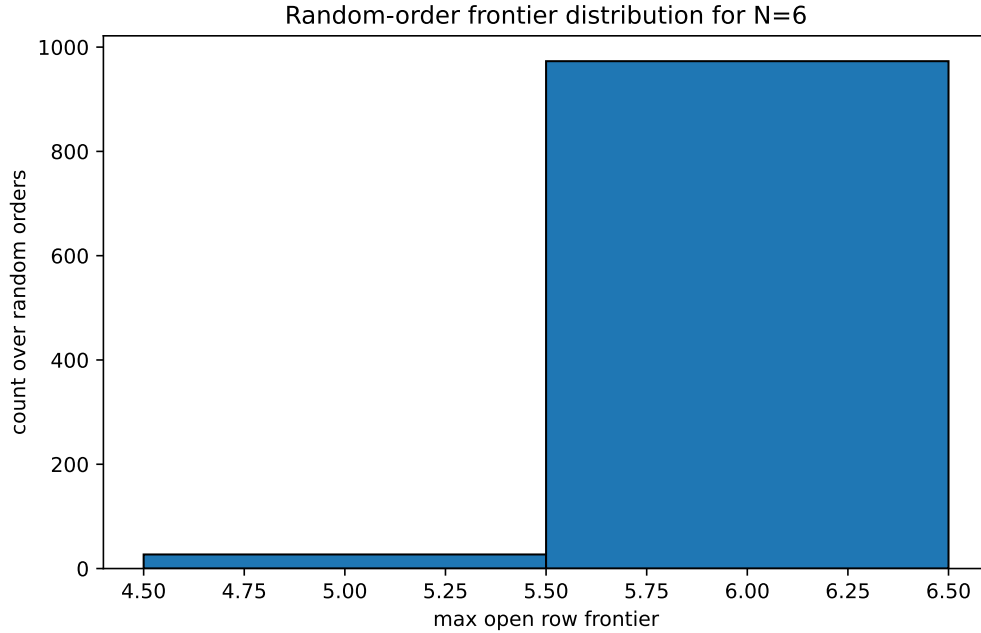


Figure 23: Distribution of the maximum open-row frontier for random orders at $N = 6$. Even at this small size, almost all random orders expose the full frontier.

16.4 Fairness-memory trilemma for alternating supertasks

The frontier calculation suggests a new obstruction for hypercomputation beyond energy, noise, and endpoint readout. Alternating quantifiers require either finishing one branch before opening many others or maintaining a growing live frontier. In a finite problem, row-major processing is legal and

cheap in memory. In an infinite problem such as

$$\exists i \forall j R(i, j),$$

row-major processing is branch-starving: if the first branch has infinitely many universal checks, the process never reaches later i without another nested limit interface. Fair interleaving avoids starvation but opens more and more branches, creating unbounded frontier memory.

Candidate law 16.7 (Fairness-memory trilemma). *A Zeno-style implementation of an alternating infinite predicate must pay with at least one of the following:*

$$\boxed{\text{nested Zeno completion} \quad \vee \quad \text{unbounded frontier memory} \quad \vee \quad \text{branch starvation/unfairness}.}$$

For finite $N \times N$ shadows of $\exists\forall$, fair interleaving requires $\Omega(N)$ reliable state bits by [Theorem 16.6](#); in the unbounded limit this becomes an infinite-memory requirement unless another singular interface is introduced.

This law gives a structural reason why finite-space accelerated machines have limited power. It is not just that the endpoint is noisy or late writes are hard. For alternating properties, the internal aggregation problem itself develops an unbounded frontier under fair scheduling. This connects the present resource theory to finite-space results for accelerated Turing machines [\[3\]](#).

16.5 Applications and targets generated by Version 0.7

Hypercomputation audits. A proposed endpoint should be asked whether it returns a witness, a certificate, a full trace prefix, or only a decision bit. If it returns a witness, apply the address-capacity shadow. If it returns only a decision bit, audit the latch, reliability, and nonverifiable readout assumptions.

Bounded verification and model checking. The residual-width invariant is directly useful for finite verification. It identifies when a bounded liveness/safety shadow can be streamed with small memory and when a fair order will force exponential state width. This overlaps with the practical role of OBDDs in symbolic model checking, but adds a physical interpretation: OBDD width is a reliable memory lower bound for a compressed endpoint.

Theorem-proving portfolios. A proof-search accelerator that reports only “a proof exists” may be hard to audit. A proof-search accelerator that reports the proof index pays address capacity. A portfolio that interleaves many partially checked obligations pays frontier memory. These tradeoffs can guide finite-shadow theorem provers without claiming hypercomputation.

Scheduling design. The frontier law turns ordering into a resource. Blocked orders reduce memory but risk starvation in infinite settings. Fair orders prevent starvation but increase frontier memory. A new optimization target is to minimize a weighted cost

$$\boxed{J(\pi) = a f_{\pi} + b T_{\text{starvation}}(\pi) + c N_{\text{late}}(\pi)}$$

for finite shadows, where the terms penalize memory, fairness delay, and late-write cost.

Further mathematical targets. The next formal targets are:

1. develop a robust Weihrauch/streaming hybrid in which a finite-shadow problem is classified by both its readout degree and its residual-width growth;
2. prove probabilistic frontier laws for random interleavings of $N \times N$ grids;
3. extend [Theorem 16.6](#) to deeper bounded quantifier trees, where the expected state width should be governed by a hierarchy of open-frontier sets;
4. include noisy or thermodynamic memory, replacing $\log_2 W_\pi(F)$ by a reliability- corrected memory/action cost;
5. study quantum endpoint channels, where Holevo-type information bounds should replace classical capacity while the witness-address debt remains.

16.6 Summary of the continuation

Version 0.7 adds two conservation principles. First, endpoint coding can replace a linear orthogonal signal catalog by a capacity-limited code, but Fano’s inequality still gives a finite address horizon for every finite channel capacity. Second, internal aggregation can replace primitive leaf readout by a finite-state stream, but exact aggregation requires $W_\pi(F)$ residual states. For bounded alternating shadows, the decisive parameter is not leaf count alone; it is the open-branch frontier induced by the order. Thus finite-shadow hypercomputation audits must track not only endpoint readout and energy, but also witness semantics, channel capacity, stream order, and reliable state width.

17 Version 0.8: causal certificate flow and preloaded address memory

Version 0.7 separated a boolean endpoint from an auditable endpoint. A boolean endpoint may only assert that some event occurred. An auditable endpoint must also return enough information to locate or verify the event. This section studies a loophole: perhaps the address need not be written after the event occurs. A device could maintain a running address register before the event and then merely freeze or select the preloaded address when the event occurs.

The result is a conservation law. Certificate information can be moved between a pre-event memory channel and a post-event tail channel, but it cannot be created at the endpoint. Finite pre-event address memory and finite post-event communication rate imply only a finite certified horizon. This statement is independent of the detailed coding scheme; it follows from information flow. Shannon–Hartley capacity gives one physical model for finite analog communication rate, while finite-blocklength theory warns that the asymptotic capacity formula is optimistic for short endpoint tails [\[21, 20, 22, 17\]](#).

17.1 The causal certificate-flow law

Let H denote the halt time or witness address. Suppose that, at stage n , the endpoint contains at most S_n robust bits of pre-event address information. After the event at stage n , the endpoint has only the remaining time interval $[t_n, T]$ in which to send, write, or refine further address information. Let the cumulative post-event address capacity of that interval be

$$C_n = \int_{t_n}^T c(t) dt.$$

If the endpoint has a finite rate bound $c(t) \leq c_{\max}$, then

$$C_n \leq c_{\max} R_n, \quad R_n = T - t_n.$$

Candidate law 17.1 (Causal certificate-flow law). *Assume that an endpoint must identify one of M possible event addresses with error probability at most $\alpha < 1/2$. If the pre-event address memory carries at most S_n bits and the post-event tail channel carries at most C_n bits, then*

$$(1 - \alpha) \log_2 M - h_2(\alpha) \leq S_n + C_n.$$

Here h_2 is binary entropy in bits.

Proof sketch. The final decoded address is a function of the pre-event memory state and the post-event channel output. By assumption, these together carry at most $S_n + C_n$ bits of mutual information about H . Fano's inequality gives $H(H | \hat{H}) \leq h_2(\alpha) + \alpha \log_2(M - 1)$, and therefore $I(H; \hat{H}) \geq (1 - \alpha) \log_2 M - h_2(\alpha)$ up to the usual harmless $M - 1$ replacement. Combining the lower and upper bounds yields the claim. $\square$

For an address among the first $n + 1$ possible stages, the zero-error design rule becomes

$$\log_2(n + 1) \leq S_n + c_{\max} R_n.$$

Thus the event address can be split between a preloaded register and a late tail channel, but the sum of these two resources must cover the address entropy.

Corollary 17.2 (Finite causal certified horizon). *If $S_n \leq S < \infty$, $c_{\max} < \infty$, and $R_n \rightarrow 0$, then the uniformly auditable address horizon is finite. In the simplified zero-error model it is bounded by*

$$N_{\text{caus}}(S, c_{\max}) = \max\{n : \log_2(n + 1) \leq S + c_{\max} R_n\}.$$

Consequently a certified finite-shadow horizon obeys

$$N_{\text{cert}} \leq \min\{N_{\text{caus}}, N_{\text{lat}}, N_{\text{cap}}, N_{\text{pack}}, N_{\text{rel}}, N_{\text{clk}}\}.$$

Remark 17.3. The corollary is not a no-go theorem for all finite uses of acceleration. It says that a claim of auditable unbounded witness return must identify which resource becomes unbounded: preloaded address memory, post-event channel capacity, endpoint packing capacity, late-write power, reliability, or clock precision.

17.2 Post-event tail-address horizons

With no preloaded address memory, define

$$N_{\text{tail}}(c) = \max\{n : \log_2(n + 1) \leq c R_n\}.$$

This is a deliberately generous bound: it lets the entire remaining tail after the event be used for address transmission.

Several normalized schedules expose a sharp phase difference. For geometric tails $R_n = \lambda^n$, N_{tail} grows only logarithmically in c . For polynomial tails $R_n = (n + 1)^{-a}$,

$$\ln(N_{\text{tail}} + 1) \lesssim \frac{1}{a} W(ac \ln 2), \quad N_{\text{tail}} \asymp \left(\frac{ac \ln 2}{W(ac \ln 2)} \right)^{1/a},$$

where W is the Lambert W function. For the log-slow tail $R_n = (1 + \ln(n + 1))^{-1}$,

$$\ln(N_{\text{tail}} + 1) \leq \frac{-1 + \sqrt{1 + 4c \ln 2}}{2},$$

and therefore

$$\log_{10}(N_{\text{tail}} + 1) \sim \frac{\sqrt{c \ln 2}}{\ln 10}.$$

Table 27: Post-event address horizons for four normalized schedules, using the generous condition $\log_2(n + 1) \leq cR_n$ with no preloaded address memory. Geometric acceleration has a surprisingly small certificate horizon even at enormous endpoint bit-rates.

tail bit-rate c	geometric	$R_n = (n + 1)^{-1}$	$R_n = (n + 1)^{-2}$	log-slow
1,000	8	139	14	165,532,916,380
1,000,000	17	62,745	343	$10^{361.36}$
1.0×10^9	27	39,620,076	8,738	$10^{11433.75}$
1.0×10^{12}	37	28,781,593,876	236,672	$10^{361573.66}$

The exceptional numerical result in Table 27 is the geometric case. Even with $c = 10^{12}$ address bits per unit time, a geometric Zeno schedule with no preloaded address memory has a post-event address horizon of only 37 stages. The reason is simple but easy to miss: the event address grows like $\log n$, while the remaining tail capacity decays exponentially.

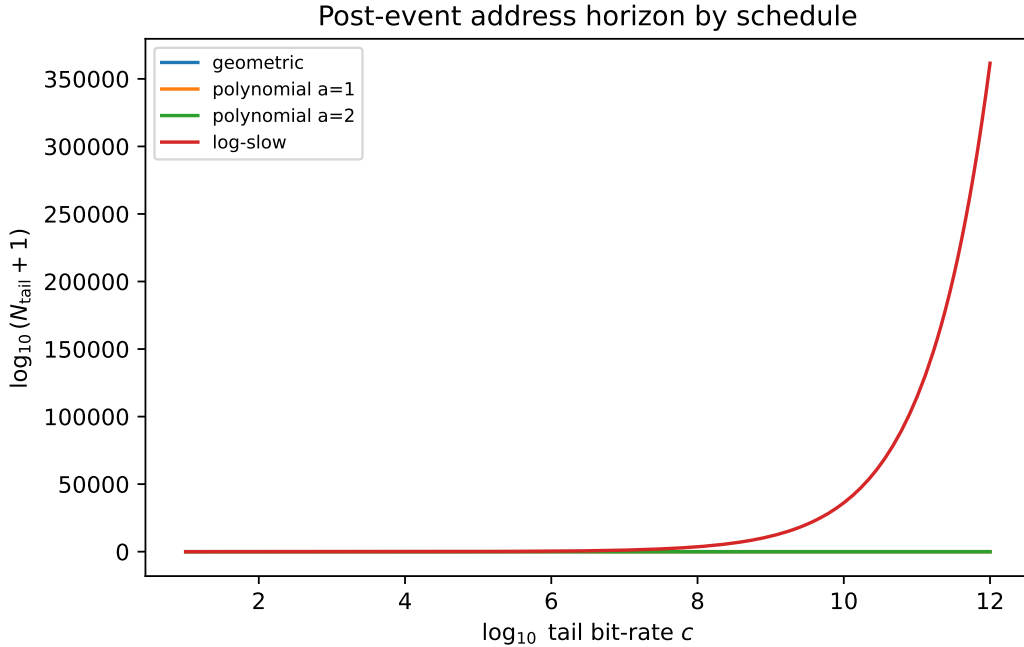


Figure 24: Post-event address horizon as a function of tail bit-rate. Geometric acceleration is excellent for placing many steps before a deadline, but it is poor for returning a late certificate after the event because the remaining channel capacity collapses exponentially.

17.3 Preload deficit and the certificate–clock seesaw

The causal-flow law can be inverted. For a desired certified event horizon N and a post-event bit-rate c , the minimum robust preloaded address memory is

$$S_{\min}(N; c) = [\log_2(N + 1) - cR_N]_+.$$

This equation measures how much address information must already be present before the halt occurs.

Table 28: Minimum preloaded address memory $S_{\min}(N; c) = [\log_2(N + 1) - cR_N]_+$ for $c = 10^{12}$ bits per unit time. Slow-tail schedules leave substantial post-event capacity but do so by sacrificing approach to the endpoint.

target N	$\log_2(N + 1)$	geometric	$a = 1$	$a = 2$	log-slow
1,000,000	19.9	19.9	0	18.9	0
1.0×10^{12}	39.9	39.9	38.9	39.9	0
1.0×10^{30}	99.7	99.7	99.7	99.7	0
1.0×10^{100}	332	332	332	332	0

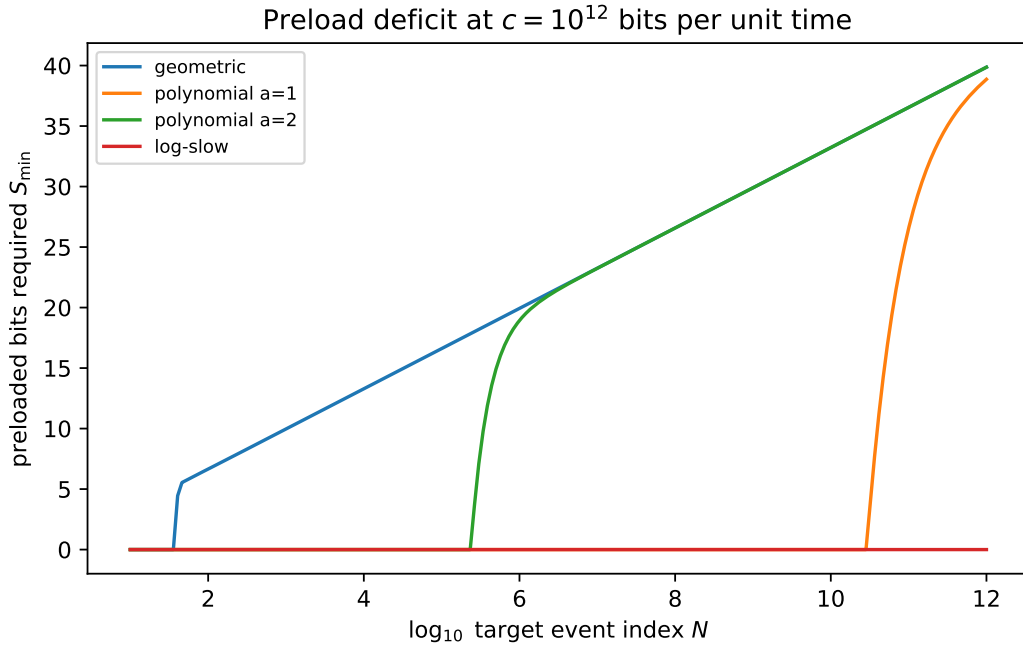


Figure 25: Preloaded address-memory deficit at $c = 10^{12}$ bits per unit time. Fast endpoint convergence erases post-event certificate capacity, forcing nearly all address information into preloaded memory. Log-slow convergence keeps tail capacity available but pays by remaining far from the endpoint.

This produces a new design principle, the *certificate–clock seesaw*. Schedules that rapidly approach the endpoint are good for a boolean supertask but bad for late witness return. Schedules

that preserve tail capacity can return much larger certificates, but only by postponing endpoint convergence. The two objectives are not aligned.

Table 29: Causal address horizon as $\log_{10}(N + 1)$ for $c = 10^6$ bits per unit time when S robust preloaded address bits are available before the halt. Finite preloaded memory shifts the horizon, but it does not remove finiteness unless the memory itself grows without bound.

schedule	$S = 0$	$S = 16$	$S = 32$	$S = 64$
geometric	1.26	4.82	9.63	18.06
polynomial a=1	4.80	5.59	9.63	18.06
polynomial a=2	2.54	4.82	9.63	18.06
log-slow	361.36	363.77	366.21	371.12

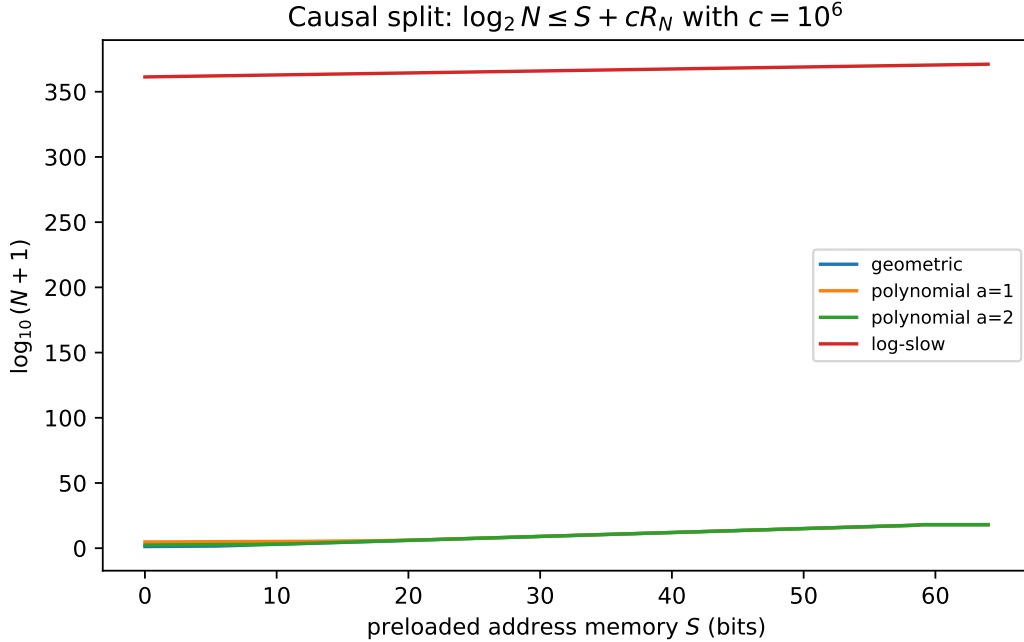


Figure 26: Causal split between preloaded address memory and post-event tail capacity for $c = 10^6$. Finite memory can create enormous finite certified shadows, but the horizon remains finite unless the memory capacity itself grows without bound.

17.4 Prefix-free certificates and Kraft-tail knapsacks

A returned certificate is often self-delimiting or prefix-free: the decoder must know where the address ends and the next field begins. This adds a second finite-shadow law. Let A be an active set of event addresses. Suppose address n can receive at most

$$b_n = \lfloor cR_n \rfloor$$

post-event bits. A prefix-free post-event address code with lengths $\ell_n \leq b_n$ exists if and only if the longest allowed lengths already satisfy Kraft’s budget:

$$\boxed{\sum_{n \in A} 2^{-b_n} \leq 1.}$$

Indeed, choosing $\ell_n = b_n$ gives the smallest Kraft expenditure, and the Kraft–McMillan theorem gives sufficiency and necessity for feasible prefix lengths [22].

Thus a prior-weighted finite-shadow problem becomes a knapsack:

$$\boxed{\max_{x_n \in \{0,1\}} \sum_n \pi_n x_n \quad \text{subject to} \quad \sum_n 2^{-\lfloor cR_n \rfloor} x_n \leq 1.}$$

Late events are expensive in code-space because their remaining tails force short code lengths. The effective value-to-cost ratio is

$$\pi_n 2^{\lfloor cR_n \rfloor},$$

not merely the prior mass π_n .

Table 30: Prefix-free first-block horizons under the Kraft-tail condition $\sum_{n \leq K} 2^{-\lfloor cR_n \rfloor} \leq 1$. Variable-length coding helps finite shadows, but late events become code-space expensive because their remaining tail capacity is short.

c	geometric	$a = 1$	$a = 2$	log-slow
10	3	4	2	9
30	4	11	3	67
100	6	28	6	3,098
200	7	49	9	97,804

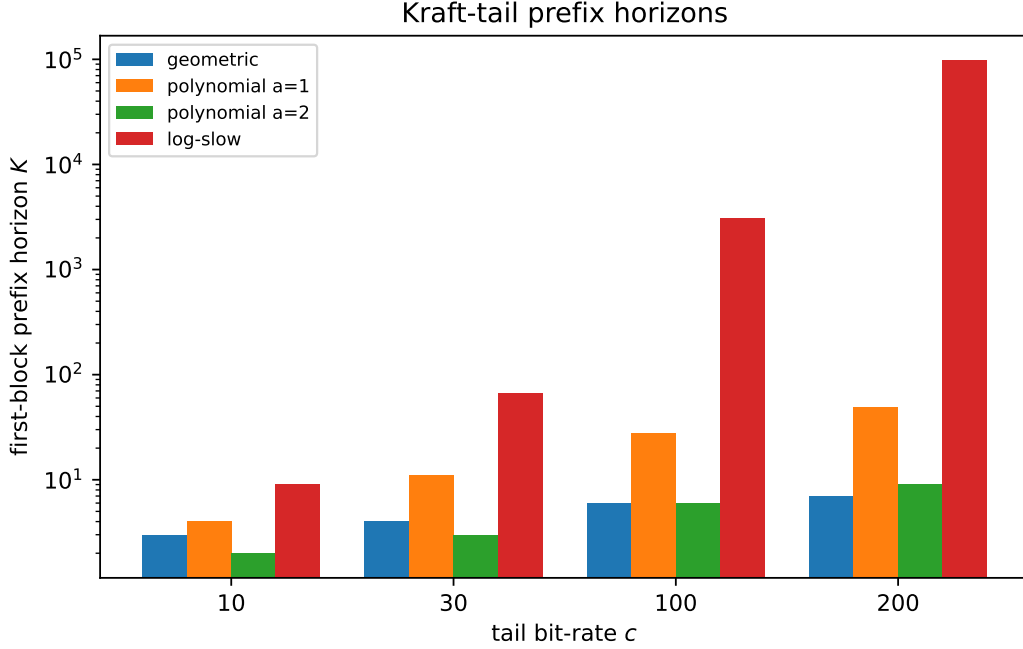


Figure 27: Prefix-free first-block horizons under the Kraft-tail budget. Log-slow schedules can support much larger finite prefix families at the same bit-rate, but again this comes from retaining large remaining tails rather than from completing the endpoint rapidly.

17.5 Preloaded counters do not eliminate late cost

A second loophole is to maintain the address in a counter before the halt and only freeze it when the event occurs. This avoids writing the whole address after the event, but it requires one robust update per computational step. If a distinguishable update costs minimum action Q_{upd} and the update actuator has finite peak power P , then the preloaded-address update horizon obeys

$$N_{\text{pre}}(P) \leq \#\{n : \tau_n \geq Q_{\text{upd}}/P\}.$$

This is the same counting structure as the finite-power latch horizon. Preloading moves the cost from post-event address writing to pre-event address maintenance.

Gray coding is the finite-shadow optimum for worst-case counter updates: it changes one bit per increment. But this optimization does not remove the need for one robust update per step.

Table 31: Counter-preloading stress test. Gray coding minimizes the worst update to one bit flip, but any exact preloaded address still requires one robust update per computational step. Thus preloading shifts address cost into the finite-power update horizon rather than eliminating it.

updates N	binary total flips	Gray total flips	binary worst	Gray worst
2^8	511	256	9	1
2^{16}	131,071	65,536	17	1
2^{24}	33,554,431	16,777,216	25	1
2^{32}	8,589,934,591	4,294,967,296	33	1

17.6 Exceptional findings from Version 0.8

1. Auditable hypercomputation has a causal bit-flow budget. A final witness address must be causally present in the endpoint. It can be stored before the halt, transmitted after the halt, or split between both. The invariant is

$$\boxed{\text{address entropy} \leq \text{pre-event memory} + \text{post-event tail capacity}.}$$

This is a stronger diagnostic than asking whether the final answer bit is stable.

2. Geometric Zeno acceleration is certificate-poor. The standard halving schedule is attractive for ideal boolean halting detection, but it is very poor for late address return. Under the generous model used here, a trillion-bit-rate post-event channel gives only 37 no-preload certified addresses. This suggests that boolean hypercomputation and proof-carrying hypercomputation are separated by a real resource gap.

3. Slow tails reveal a certificate-clock seesaw. The log-slow schedule can support enormous finite certificate shadows because its remaining tail shrinks only as $1/\ln n$. But this is exactly why it is a poor Zeno clock: reaching a small endpoint tail requires astronomically many steps. Certificate capacity and endpoint compression pull the schedule in opposite directions.

4. Prefix-free evidence is a Kraft knapsack. When certificates must be self-delimiting, the active finite shadow is not merely a prefix of the time axis. It is a knapsack whose item cost is $2^{-\lfloor cR_n \rfloor}$. This opens a concrete optimization problem for distributional finite-shadow machines.

17.7 Applications suggested by causal certificate flow

Proof-carrying theorem search. A finite-shadow theorem prover can split endpoint evidence into a preloaded proof-family identifier and a post-event local proof address. The sizing law is

$$\ell_{\text{family}} + \ell_{\text{local}} \leq S + cR_n.$$

This gives a practical way to decide how much endpoint memory must be reserved for a checkable result rather than a bare success flag.

Counterexample-returning verification. Bounded model checking and falsification usually require a concrete counterexample, not only a boolean alarm. The causal-flow law supplies an audit rule: the endpoint must have enough preloaded state and tail capacity to identify the counterexample class and its local address.

Auditable accelerators. Accelerators used in scientific discovery, theorem proving, or safety-critical search should distinguish between “found something” and “returned evidence.” The latter has a larger resource bill. This distinction can prevent overestimating the practical value of a finite-shadow device.

Endpoint architecture. For finite horizons, Gray-coded counters and checkpointed address registers are useful engineering tools. They minimize update bursts, but they should be accounted for as finite-power update resources, not treated as free bookkeeping.

17.8 Targets for further study

Finite-blocklength tail channels. The inequalities above use capacity as a generous upper bound. A sharper endpoint law should include dispersion and error probability. A candidate finite-tail correction is

$$\ell_n \lesssim cR_n - \sqrt{VR_n} Q^{-1}(\alpha) + O(\log R_n),$$

where V is an effective channel dispersion. This correction should make very short endpoint tails even less certificate-capable.

Prefix waterlines with priors. Combining Version 0.6’s Bayesian waterline with the Kraft-tail knapsack should yield a prefix-waterline rule. The active set should be controlled by the ratio $\pi_n 2^{\lfloor cR_n \rfloor}$, with late low-capacity events excluded even when their prior is not negligible.

Joint memory-update thermodynamics. Preloaded address memory requires continual updates. The next model should combine counter update action, error correction, memory retention, and heat dissipation into a single certified-horizon equation.

Quantum endpoint certificates. The quantum version should replace Shannon capacity by accessible information or Holevo information and should include measurement error. The expected invariant remains finite: finite Hilbert-space dimension, finite energy, and nonzero decoding error imply finite auditable address shadow.

Empirical halt-time priors. For practical finite-shadow accelerators, measure halt-time and certificate-length distributions on theorem-proving, SAT, model-checking, and program-analysis workloads. Those empirical priors can be inserted directly into the Kraft-tail knapsack.

18 Version 0.9 appendage: spectral endpoint shadows and finite-bandwidth readout

This appendage studies a remaining analog loophole. A proposed Zeno endpoint might not write a discrete latch or transmit a digital witness; it might disturb a continuous field, imprint a phase, send a pulse, or leave a waveform near the endpoint. Such a proposal can look stronger than the finite latch and certificate-channel models because the final answer is stored in a continuous signal. The test below shows that finite bandwidth turns this analog escape route into another finite shadow.

The relevant mathematical background is the time-bandwidth theory of Slepian, Pollak, and Landau, in which signals essentially limited to time T and bandwidth W have about $2WT$ effective degrees of freedom, together with the finite-bandwidth communication lineage of Nyquist and Shannon [25, 26, 27, 28, 21]. Finite-blocklength information theory then adds finite-time error penalties [17]. The diagnostic principle is

$$\boxed{\text{finite bandwidth} + \text{finite event energy} \implies \text{finite robust event-address horizon.}}$$

Thus a continuous analog endpoint is not automatically a hypercomputational endpoint. It is a finite-resolution channel unless it imports unbounded bandwidth, unbounded energy, noise-free real precision, or a singular final observation rule.

18.1 Bandlimited event signatures

Let an event at external time u produce an endpoint signature $s_u \in L^2(\mathbb{R})$. The generous model used here assumes a shifted signal

$$s_u(t) = s(t - u),$$

with Fourier support contained in $[-W, W]$ and energy $\|s\|_2^2 \leq E$. The L^2 Bernstein inequality for bandlimited functions gives

$$\|s'\|_2 \leq 2\pi W \|s\|_2.$$

Therefore

$$\|s_u - s_v\|_2 \leq |u - v| \sup_{\theta} \|s'(\cdot - \theta)\|_2 \leq 2\pi W \sqrt{E} |u - v|.$$

Theorem 18.1 (Bandlimited Lipschitz collapse). *Let t_n be event times in a finite-time Zeno schedule, with adjacent gaps $\tau_n = t_{n+1} - t_n \rightarrow 0$. Suppose event n is represented by a shifted bandlimited signature s_{t_n} , with bandwidth at most $W < \infty$ and energy at most $E < \infty$. If a robust decoder must distinguish adjacent event addresses under L^2 -noise radius $\eta > 0$, then every protected adjacent pair must satisfy*

$$\tau_n > \frac{\eta}{\pi W \sqrt{E}}.$$

Therefore only finitely many adjacent event addresses can be robustly distinguished at fixed W, E, η .

Proof. Robust decoding of two alternatives under noise radius η requires their signatures to be separated by more than 2η . The Lipschitz bound gives

$$\|s_{t_{n+1}} - s_{t_n}\|_2 \leq 2\pi W \sqrt{E} \tau_n.$$

Thus $\tau_n > \eta/(\pi W \sqrt{E})$. Since $\tau_n \rightarrow 0$, the inequality can hold for only finitely many n . $\square$

This theorem is local. It does not rely on a particular code or on a global channel capacity theorem. It says that when event times cluster at the Zeno endpoint, finite bandwidth makes the corresponding waveforms cluster as well.

18.2 Spectral horizon and utilization

Define the finite-bandwidth event-address horizon

$$N_{\text{BW}}(W, E, \eta) = \# \left\{ n : \tau_n \geq \frac{\eta}{\pi W \sqrt{E}} \right\}.$$

For a fixed schedule this is a shadow depth, not a hypercomputational infinity.

Theorem 18.2 (Zeno spectral-utilization theorem). *Let (τ_n) be any positive summable schedule with $\sum_n \tau_n = T < \infty$. For any fixed $c > 0$, let*

$$D(\delta) = \#\{n : \tau_n \geq \delta\}, \quad N_W = D(c/W).$$

Then

$$\frac{N_W}{W} \rightarrow 0 \quad \text{as } W \rightarrow \infty.$$

Equivalently, every genuine Zeno schedule uses a vanishing fraction of the ordinary $O(WT)$ time-bandwidth event slots available to a finite uniform schedule.

Proof. For an ℓ^1 sequence, its distribution function satisfies $\delta D(\delta) \rightarrow 0$ as $\delta \downarrow 0$; otherwise the layer-cake integral

$$\sum_n \tau_n = \int_0^\infty D(\delta) d\delta$$

would diverge. Taking $\delta = c/W$ gives

$$\frac{N_W}{W} = \frac{D(c/W)}{W} = \frac{1}{c}(c/W)D(c/W) \rightarrow 0.$$

□

For the canonical schedules,

$$\tau_n = 2^{-(n+1)} \implies N_{\text{BW}} = O(\log W),$$

$$R_n \asymp n^{-a} \implies \tau_n \asymp an^{-(a+1)} \implies N_{\text{BW}} = O\left(W^{1/(a+1)}\right),$$

and for the barely summable log-slow schedule,

$$\tau_n \asymp \frac{c}{n \log^2 n} \implies N_{\text{BW}} \sim \frac{cW}{\log^2 W}$$

up to slowly varying factors. Log-slow timing is nearly linear in finite-bandwidth horizon, but still has vanishing utilization and very slow endpoint convergence.

Table 32: Finite-bandwidth distinguishability horizon with margin $\eta = 0.5$, unit event energy, and $T = 1$.

schedule	W	N_{BW}	$N_{\text{BW}}/(2W)$
geometric	10^3	12	6.000e-03
geometric	10^6	22	1.100e-05
geometric	10^9	32	1.600e-08
geometric	10^{12}	42	2.100e-11
power a=1	10^3	78	3.900e-02
power a=1	10^6	2,506	1.253e-03
power a=1	10^9	79,266	3.963e-05
power a=1	10^{12}	2,506,627	1.253e-06
power a=2	10^3	22	1.100e-02
power a=2	10^6	231	1.155e-04
power a=2	10^9	2,324	1.162e-06
power a=2	10^{12}	23,248	1.162e-08
log-slow	10^3	125	6.250e-02
log-slow	10^6	28,335	1.417e-02
log-slow	10^9	$10^{7.05}$	5.646e-03
log-slow	10^{12}	$10^{9.77}$	2.943e-03

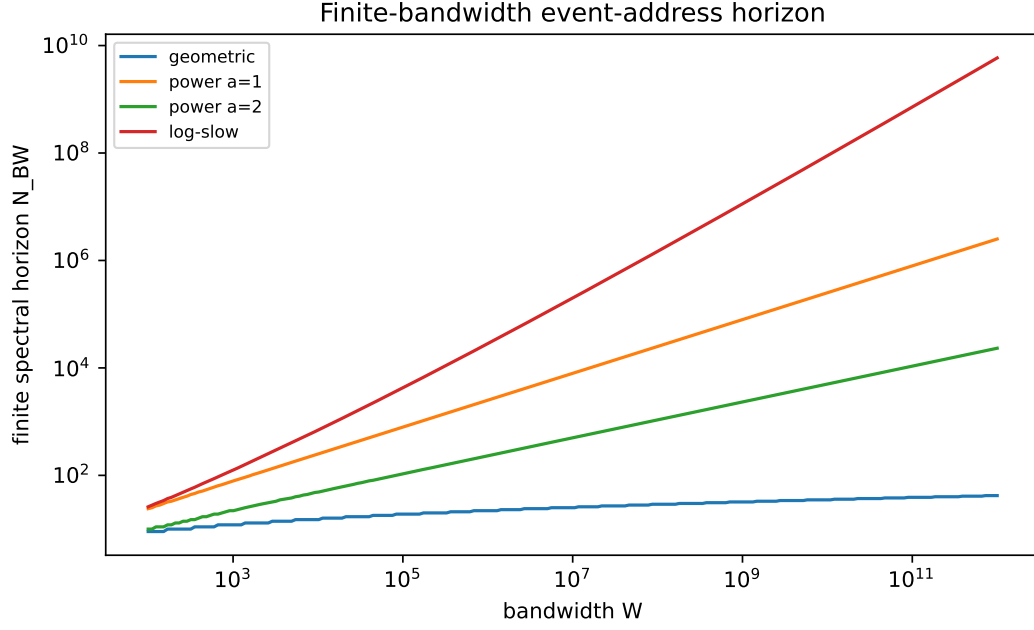


Figure 28: Finite-bandwidth event-address horizon for four schedules.

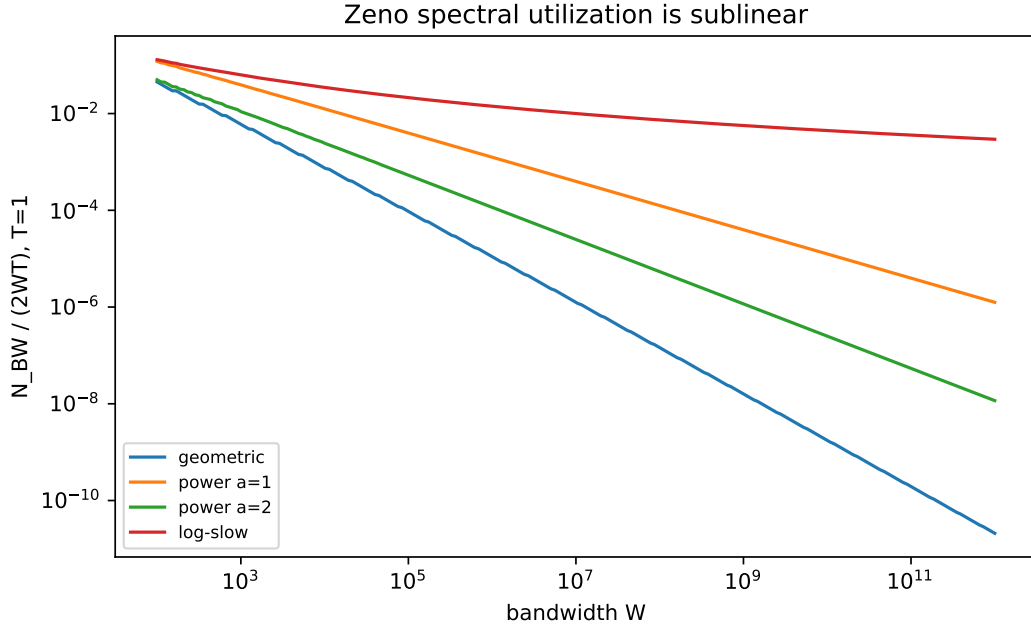


Figure 29: Spectral utilization $N_{BW}/(2WT)$, with $T = 1$. Every summable infinite schedule has vanishing utilization as bandwidth increases.

18.3 Bandwidth-energy complementarity

The collapse theorem can be inverted. If bandwidth is fixed and the design insists on protecting a late adjacent pair with gap τ_n , then event energy must grow:

$$E_n^{\text{BW}} \geq \left(\frac{\eta}{\pi W \tau_n} \right)^2.$$

This differs from the horizon-energy law $E_n \tau_n \geq Q$. The earlier law says that fast distinguishable transitions cost like $1/\tau_n$. The spectral law says that finite-bandwidth address signatures cost like $1/\tau_n^2$. For a polynomial tail $R_n \asymp n^{-a}$,

$$E_n^{\text{BW}} = \Omega \left(R_n^{-(2+2/a)} \right).$$

Thus witness-address waveforms can be more expensive near the endpoint than mere internal state-transition timing.

Table 33: Energy required to keep adjacent event signatures separated at bandwidth $W = 10^6$ and margin $\eta = 0.5$.

schedule	N	$\log_{10} E_{\text{BW}}$
geometric	10	-7.58
geometric	100	46.61
geometric	10^3	588.46
geometric	10^6	602046.39
power a=1	10	-9.51
power a=1	100	-5.59
power a=1	10^3	-1.60
power a=1	10^6	10.40
power a=2	10	-8.08
power a=2	100	-2.19
power a=2	10^3	3.80
power a=2	10^6	21.80
log-slow	10	-9.35
log-slow	100	-6.28
log-slow	10^3	-3.59
log-slow	10^6	3.61

Table 34: Endpoint action-energy and spectral-address energy exponents.

schedule	action endpoint law	spectral address law
geometric	$E = \Omega(R^{-1})$	$E_{\text{BW}} = \Omega(R^{-2})$
power tail $R_n \asymp n^{-a}$	$E = \Omega(R^{-(1+1/a)})$	$E_{\text{BW}} = \Omega(R^{-(2+2/a)})$
log-slow	mild in slot time; extreme in tail time	$E_{\text{BW}} \asymp n^2 \log^4 n$; exponential in $1/R$

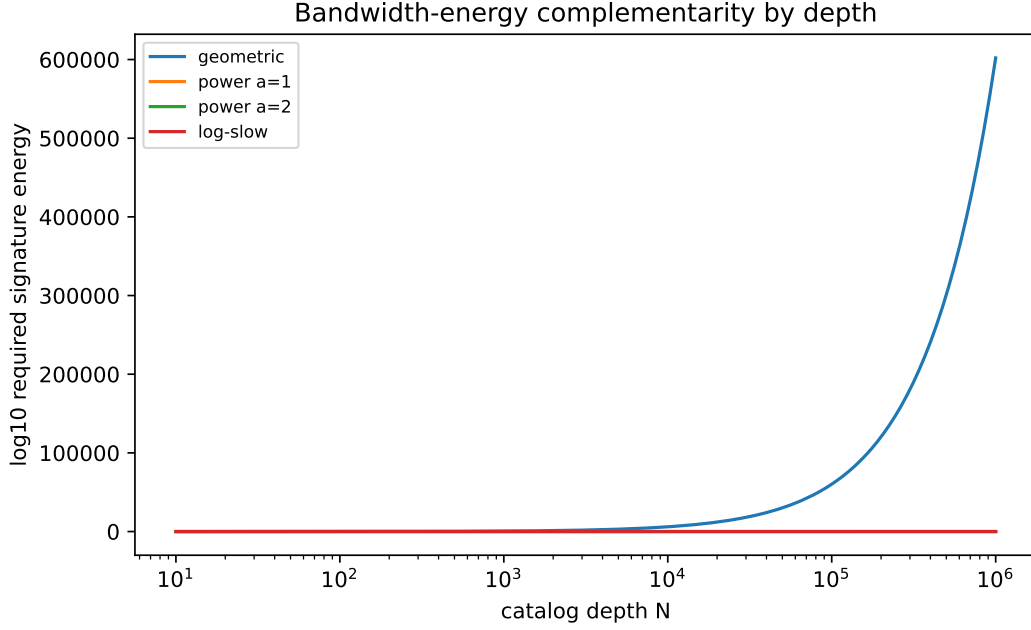


Figure 30: Bandwidth-energy complementarity at $W = 10^6$ and margin $\eta = 0.5$.

18.4 Gram-spectrum experiment: low-pass event catalogs

The numerical stress test models each possible event address by an ideal unit-energy low-pass pulse shifted to event time t_n . For an ideal rectangular spectrum, the normalized inner product is

$$G_{ij} = \langle s_{t_i}, s_{t_j} \rangle = \text{sinc}(2W(t_i - t_j)).$$

The spectrum of G indicates how many nearly independent waveform directions remain in the event catalog.

Table 35: Gram-spectrum stress test for ideal low-pass event signatures at $W=200$.

schedule	N	rank 10^{-3}	rank 10^{-6}	eff. rank	$\log_{10} \kappa$	last coh.
geometric	32	12	13	2.14	19.34	1.000000
geometric	64	11	13	1.42	19.73	1.000000
geometric	128	11	12	1.18	20.07	1.000000
power a=1	32	29	31	25.29	7.13	0.753206
power a=1	64	36	38	19.03	18.90	0.983889
power a=1	128	39	41	10.95	19.43	0.999004
power a=2	32	15	17	4.97	19.10	0.998922
power a=2	64	15	17	2.51	19.59	0.999984
power a=2	128	15	17	1.63	20.00	1.000000
log-slow	32	30	32	26.39	5.92	0.645154
log-slow	64	39	41	25.64	18.72	0.952439
log-slow	128	46	48	20.35	19.13	0.993501

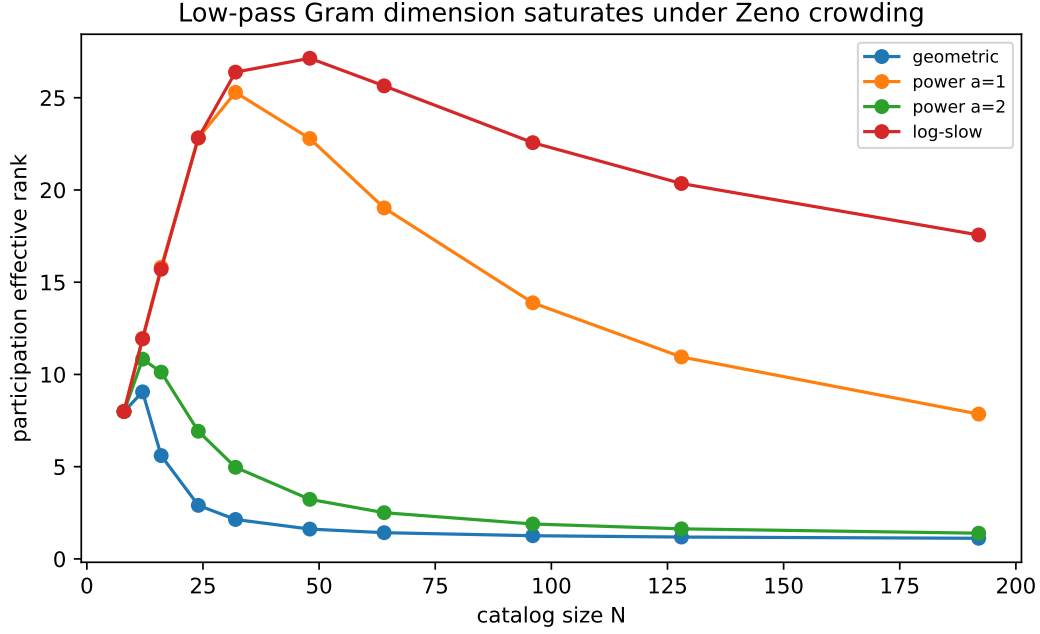


Figure 31: Participation effective rank of the low-pass Gram matrix at $W = 200$.

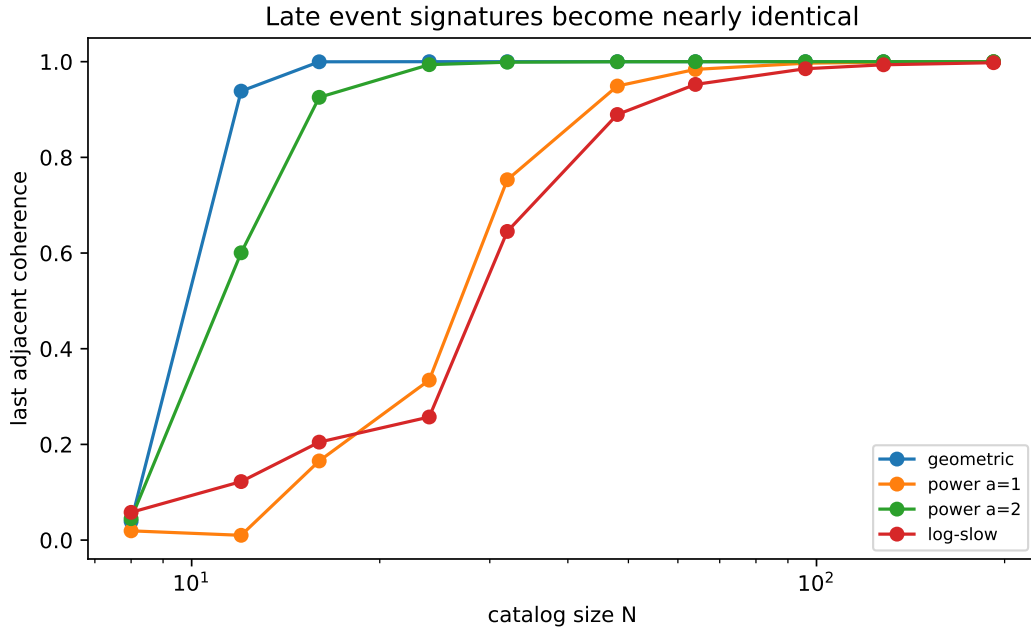


Figure 32: Adjacent low-pass coherence for the last pair in a finite event catalog. Values near one mean that neighboring late event signatures are almost the same waveform.

For geometric timing at $W = 200$, the 10^{-6} -rank is already about 13 for $N = 32$, and it remains about 13 as the catalog grows to $N = 128$. Extra events mostly add near-duplicates. Power-law and log-slow schedules retain more spectral dimensions for the same N , but their last-pair coherence

still approaches one as N grows.

18.5 Spectral de-Zenoization

Once the finite spectral horizon N is accepted, acceleration again becomes a poor design. For a finite catalog of N event slots in total time T , the schedule that maximizes the minimum adjacent gap is uniform:

$$\max_{\substack{\tau_0, \dots, \tau_{N-1} > 0 \\ \sum_{n < N} \tau_n \leq T}} \min_{n < N} \tau_n = \frac{T}{N}.$$

Therefore the minimum bandwidth needed to separate N adjacent event addresses with energy E and margin η satisfies

$$W \geq \frac{\eta N}{\pi T \sqrt{E}}.$$

This is the spectral version of de-Zenoization: once robustness turns a formal infinity into a finite operational shadow, the optimal finite implementation is a resource-matched finite schedule, not a Zeno tail.

18.6 Exceptional findings, applications, and targets

Analog waveform readout has a finite shadow. Finite-bandwidth, finite-energy waveforms are Lipschitz in event time. Accumulating event times therefore produce accumulating endpoint signatures.

Zeno schedules underuse spectral degrees of freedom. The law $N_W/W \rightarrow 0$ holds for every summable infinite schedule. It is not an artifact of geometric halving.

Log-slow timing is a recurring extremal object. The barely summable schedule $\tau_n \asymp 1/(n \log^2 n)$ keeps finite-power, causal-certificate, and finite-bandwidth horizons unusually large, at the price of pathologically slow endpoint convergence.

Event-address energy is harsher than transition energy. Address-preserving finite-bandwidth waveforms require a quadratic $1/\tau_n^2$ energy law. This is a stronger endpoint burden than the earlier action law.

Applications. The formula N_{BW} gives an audit for accelerators that claim to return a proof address, counterexample, or event time through an analog endpoint. It also gives a principled cutoff for Zeno-like chattering in hybrid systems, finite-shadow theorem proving, and Malament-Hogarth receiver analyses.

Targets. The next mathematical target is to replace the generous sinc model by a prolate spheroidal finite-window receiver and obtain a sharper $2WT + O(\log WT)$ dimension cap. A quantum version should use energy-constrained bosonic channel capacity and Holevo-accessible information to produce a bandwidth-energy-Holevo shadow for witness-carrying endpoints.

19 Version 0.10 appendage: quantum endpoint shadows and entropy-limited witnesses

This appendage tests a stronger escape route than classical waveform readout. A Zeno endpoint might try to store the late-event address in a quantum state, use coherent phase kicks, exploit nonorthogonal codes with a collective measurement, or place the final answer in a high-dimensional quantum memory. The question is whether quantum readout can avoid the finite-shadow laws derived for scalar, statistical, latch, causal, and bandlimited endpoints.

The answer developed here is negative for any endpoint with finite energy spread, finite accessible information, or finite entropy. Quantum theory can improve constants and coding strategies, but it does not turn a finite endpoint into an infinite witness-return channel. The relevant background is the Helstrom theory of state discrimination, the Holevo bound on accessible classical information, Mandelstam–Tamm type quantum speed limits, Bekenstein and holographic entropy bounds, and finite-dimensional packing bounds such as Welch’s inequality [29, 30, 31, 32, 33, 34].

The diagnostic principle is

$$\boxed{\text{finite quantum speed} + \text{finite entropy} \implies \text{finite auditable endpoint shadow.}}$$

A quantum endpoint can return a boolean bit with small entropy, but an auditable hypercomputer must often return, preserve, or certify a witness address. Witnesses are where the quantum endpoint pays its resource debt.

19.1 Quantum time-shift event signatures

Let an event at external time u produce a quantum endpoint state

$$\rho_u = U(u)\rho U(u)^*, \quad U(u) = \exp(-iHu/\hbar),$$

or more generally a path whose time-averaged energy uncertainty is bounded by ΔE . Let

$$A(\rho, \sigma) = \arccos F(\rho, \sigma)^{1/2}$$

be the Bures angle. The Mandelstam–Tamm speed-limit form gives

$$A(\rho_u, \rho_v) \leq \frac{\Delta E}{\hbar} |u - v|.$$

For binary discrimination of two equiprobable states, the Helstrom formula gives

$$P_e^*(\rho, \sigma) = \frac{1 - D_{\text{tr}}(\rho, \sigma)}{2}, \quad D_{\text{tr}}(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1.$$

The Fuchs–van de Graaf relation implies $D_{\text{tr}} \leq \sin A$. Therefore achieving binary error at most $\alpha < 1/2$ requires

$$\sin A \geq 1 - 2\alpha, \quad A \geq a_\alpha := \arcsin(1 - 2\alpha).$$

Combining the inequalities yields a quantum separation threshold.

Theorem 19.1 (Quantum time-shift collapse). *Let t_n be event times in a finite-time Zeno schedule with adjacent gaps $\tau_n = t_{n+1} - t_n \rightarrow 0$. Suppose adjacent event addresses are encoded as quantum*

states whose Bures speed is bounded by $\Delta E/\hbar$. If every protected adjacent pair must be distinguished with equal-prior binary error at most $\alpha < 1/2$, then

$$\tau_n \geq \frac{\hbar a_\alpha}{\Delta E}, \quad a_\alpha = \arcsin(1 - 2\alpha).$$

Consequently, finite ΔE protects only finitely many adjacent event addresses in any genuine Zeno schedule.

Proof. The Helstrom condition $P_e^* \leq \alpha$ requires $D_{\text{tr}} \geq 1 - 2\alpha$. Since $D_{\text{tr}} \leq \sin A$, the Bures angle must obey $A \geq \arcsin(1 - 2\alpha)$. The speed limit gives $A \leq \Delta E \tau_n / \hbar$ for adjacent events, hence the stated lower bound on τ_n . Since $\tau_n \rightarrow 0$, only finitely many gaps can satisfy this bound at fixed ΔE . $\square$

Define the quantum speed-limit shadow

$$N_{\text{QSL}}(\Delta E, \alpha) = \# \left\{ n : \tau_n \geq \frac{\hbar a_\alpha}{\Delta E} \right\}.$$

This is the quantum analogue of the finite-bandwidth horizon N_{BW} .

Corollary 19.2 (Quantum Zeno utilization). *For every positive summable schedule $\sum_n \tau_n = T < \infty$ and fixed $\alpha < 1/2$,*

$$\frac{N_{\text{QSL}}(\Delta E, \alpha)}{\Delta E/\hbar} \rightarrow 0 \quad (\Delta E/\hbar \rightarrow \infty).$$

Thus a true Zeno schedule uses a vanishing fraction of the event-address slots that an ordinary finite schedule would obtain from the same quantum speed scale.

Proof. It is the same distribution-function argument as the spectral utilization theorem. Let $D(\delta) = \#\{n : \tau_n \geq \delta\}$. For any ℓ^1 schedule, $\delta D(\delta) \rightarrow 0$ as $\delta \downarrow 0$. Put $\delta = \hbar a_\alpha / \Delta E$. $\square$

Table 36: Quantum time-shift distinguishability horizon at binary error target $\alpha = 0.05$ and $\hbar = 1$.

schedule	$\Delta E/\hbar$	N_{QSL}	$N_{\text{QSL}}/(\Delta E/\hbar)$
geometric	10^3	9	9.000e-03
geometric	10^6	19	1.900e-05
geometric	10^9	29	2.900e-08
geometric	10^{12}	39	3.900e-11
power a=1	10^3	29	2.900e-02
power a=1	10^6	944	9.440e-04
power a=1	10^9	29,883	2.988e-05
power a=1	10^{12}	945,007	9.450e-07
power a=2	10^3	11	1.100e-02
power a=2	10^6	120	1.200e-04
power a=2	10^9	1,212	1.212e-06
power a=2	10^{12}	12,132	1.213e-08
log-slow	10^3	33	3.300e-02
log-slow	10^6	5,666	5.666e-03
log-slow	10^9	2,009,562	2.010e-03
log-slow	10^{12}	$10^{8.99}$	9.869e-04

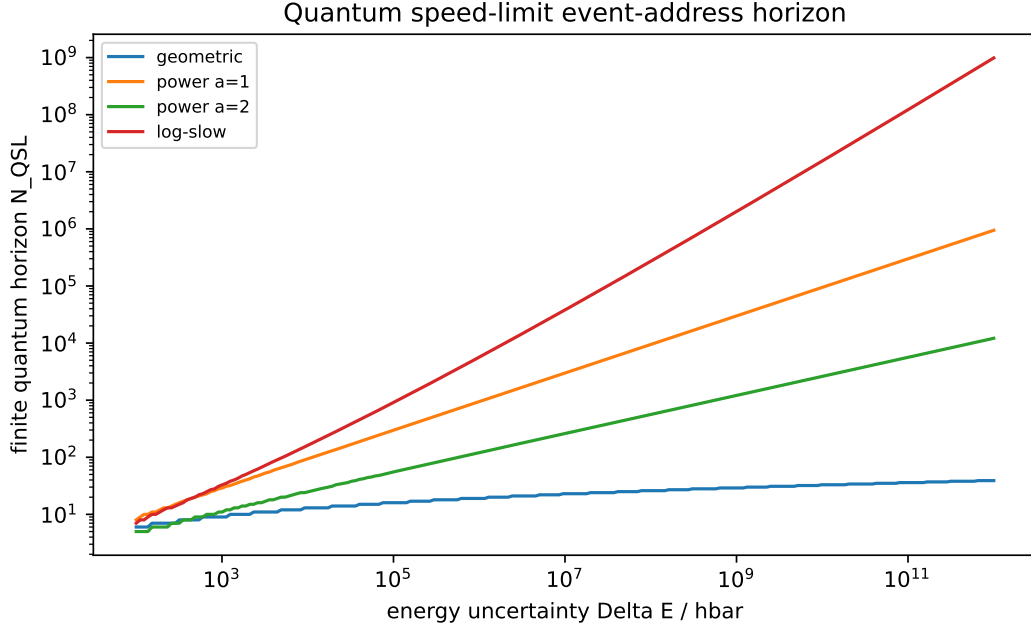


Figure 33: Quantum speed-limit event-address horizon at error target $\alpha = 0.05$.

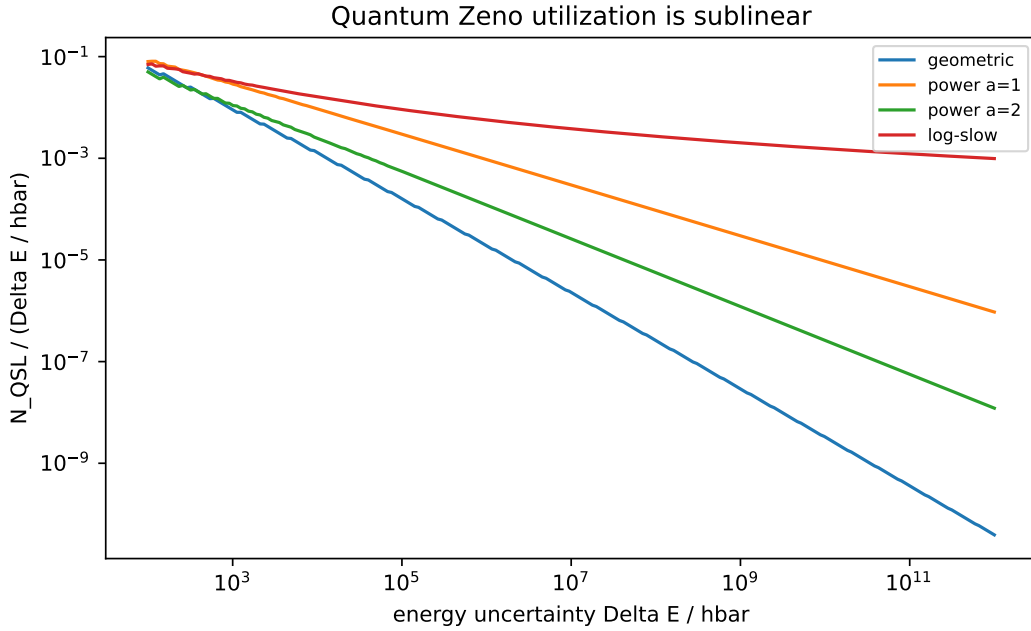


Figure 34: Quantum utilization $N_{\text{QSL}}/(\Delta E/\hbar)$. Every summable infinite schedule is sublinear in the available quantum speed scale.

The inversion of [Theorem 19.1](#) is a quantum event-address energy law:

$$\Delta E_n \geq \frac{\hbar a_\alpha}{\tau_n}.$$

A late quantum phase kick or state shift cannot remain uniformly address-distinguishable with bounded energy variance. For a polynomial tail $R_n \asymp n^{-a}$, this gives

$$\Delta E_n = \Omega \left(R_n^{-(1+1/a)} \right),$$

the same endpoint exponent as the generic action law but now derived from quantum state discrimination rather than from a postulated logical-action cost.

19.2 Holevo-Fano address shadow

Quantum measurement cannot extract arbitrary classical information from a finite quantum state. Let an endpoint encode an address $X \in \{1, \dots, M\}$, uniformly distributed for the audit problem, into states ρ_x . A measurement returns $\hat{X}$ with average error probability $P_e \leq \alpha$. Fano's inequality gives

$$I(X; \hat{X}) \geq \log_2 M - h_2(\alpha) - \alpha \log_2(M - 1),$$

where h_2 is binary entropy. Holevo's theorem gives

$$I(X; \hat{X}) \leq \chi = S \left(\sum_x M^{-1} \rho_x \right) - M^{-1} \sum_x S(\rho_x) \leq S_{\max},$$

where $S_{\max}$ is the endpoint entropy budget in bits. Since $\log_2(M - 1) \leq \log_2 M$, every auditable quantum address endpoint satisfies

$$\log_2 M \leq \frac{S_{\max} + h_2(\alpha)}{1 - \alpha}.$$

Theorem 19.3 (Holevo-Fano endpoint-address shadow). *A quantum endpoint with accessible entropy budget $S_{\max} < \infty$ cannot return one of infinitely many possible event addresses with uniformly bounded average error $\alpha < 1$. Its auditable address catalog obeys*

$$M \leq M_Q(S_{\max}, \alpha) := 2^{(S_{\max} + h_2(\alpha))/(1 - \alpha)}.$$

This theorem is independent of the measurement architecture. The measurement may be collective, nonprojective, adaptive, or optimized over all POVMs; the Holevo bound has already optimized over what can be classically accessed from the quantum encoding.

Table 37: Holevo-Fano address bound for a finite entropy endpoint at target average error $\alpha = 0.05$.

$S_{\max}$ bits	α	$\log_2 M_{\max}$	$\log_{10} M_{\max}$
1	0.05	1.35	0.41
4	0.05	4.51	1.36
8	0.05	8.72	2.63
16	0.05	17.14	5.16
32	0.05	33.99	10.23
64	0.05	67.67	20.37

19.3 Physical entropy caps: Bekenstein and holographic shadows

If the endpoint is a finite physical region, entropy bounds turn [Theorem 19.3](#) into a spacetime resource law. A Bekenstein-type cap gives, in bits,

$$S_{\text{Bek}} \leq \frac{2\pi ER}{\hbar c \ln 2},$$

where E is the region energy and R is an enclosing radius. A covariant or holographic area cap gives the schematic light-sheet bound

$$S_{\text{hol}} \leq \frac{A}{4\ell_P^2 \ln 2}.$$

Substitution yields

$$\log_2 M \leq \frac{1}{1-\alpha} \left(\frac{2\pi ER}{\hbar c \ln 2} + h_2(\alpha) \right)$$

and

$$\log_2 M \leq \frac{1}{1-\alpha} \left(\frac{A}{4\ell_P^2 \ln 2} + h_2(\alpha) \right).$$

These are not proposed as final quantum-gravity theorems. They are audit equations: if a hypercomputational endpoint is claimed to return an unbounded witness address from a finite receiver region, the analysis must identify which entropy assumption is being rejected or which singular resource is being imported.

Table 38: Illustrative physical entropy caps converted to auditable-address horizons at $\alpha = 0.05$. Natural units are used for compactness.

bound	scale	S_{max} bits	$\log_2 M_{\text{max}}$	$\log_{10} M_{\text{max}}$
Bekenstein	$ER/(\hbar c) = 0.1$	0.91	1.26	0.38
Bekenstein	$ER/(\hbar c) = 1$	9.06	9.84	2.96
Bekenstein	$ER/(\hbar c) = 10$	90.65	95.72	28.81
Bekenstein	$ER/(\hbar c) = 100$	906.47	954.48	287.33
area	$A/\ell_P^2 = 10$	3.61	4.10	1.23
area	$A/\ell_P^2 = 100$	36.07	38.27	11.52
area	$A/\ell_P^2 = 1000$	360.67	379.96	114.38
area	$A/\ell_P^2 = 10000$	3606.74	3796.87	1142.97

The numerical scale is instructive. Entropy caps can be enormous; the point is not that physical endpoints are small. The point is that finite entropy gives finite witness capacity. Hypercomputation requires an infinity, not merely an astronomical catalog.

19.4 Finite-dimensional state crowding

A finite Hilbert space is the cleanest model. If the endpoint memory has dimension d , then $S_{\text{max}} \leq \log_2 d$, so [Theorem 19.3](#) implies

$$\log_2 M \leq \frac{\log_2 d + h_2(\alpha)}{1-\alpha}.$$

Equivalently, for fixed d , the average address error must eventually rise as M/d grows.

There is also a geometric crowding view. For pure address states $\psi_1, \dots, \psi_M$ in $\mathbb{C}^d$, Welch's bound gives

$$\max_{i \neq j} |\langle \psi_i, \psi_j \rangle|^2 \geq \frac{M - d}{d(M - 1)} \quad (M > d).$$

So a finite-dimensional endpoint catalog cannot keep all address states arbitrarily close to orthogonal. The seeded random-codebook experiment below compares the Welch lower bound with observed maximum pairwise fidelities and the Fano-Holevo error floor.

Table 39: Finite-dimensional quantum address crowding. Fano uses accessible information at most $\log_2 d$ bits; the last column is a seeded random-pair stress sample, not an exhaustive optimum.

d	M	Fano $P_e \geq$	Welch $\mu^2 \geq$	sample max μ^2
8	32	0.242	9.677e-02	6.428e-01
16	64	0.210	4.762e-02	4.656e-01
64	256	0.168	1.176e-02	1.331e-01
64	1,024	0.311	1.466e-02	1.786e-01
256	1,024	0.141	2.933e-03	4.041e-02
256	4,096	0.264	3.663e-03	4.520e-02

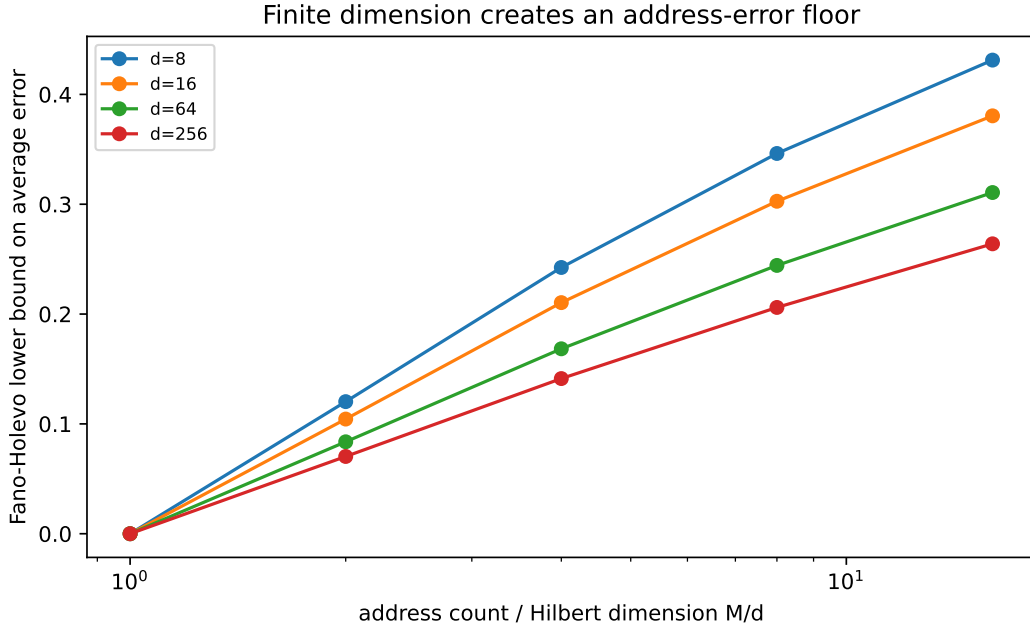


Figure 35: Fano-Holevo lower bound on average address error as the number of possible addresses exceeds the Hilbert dimension.

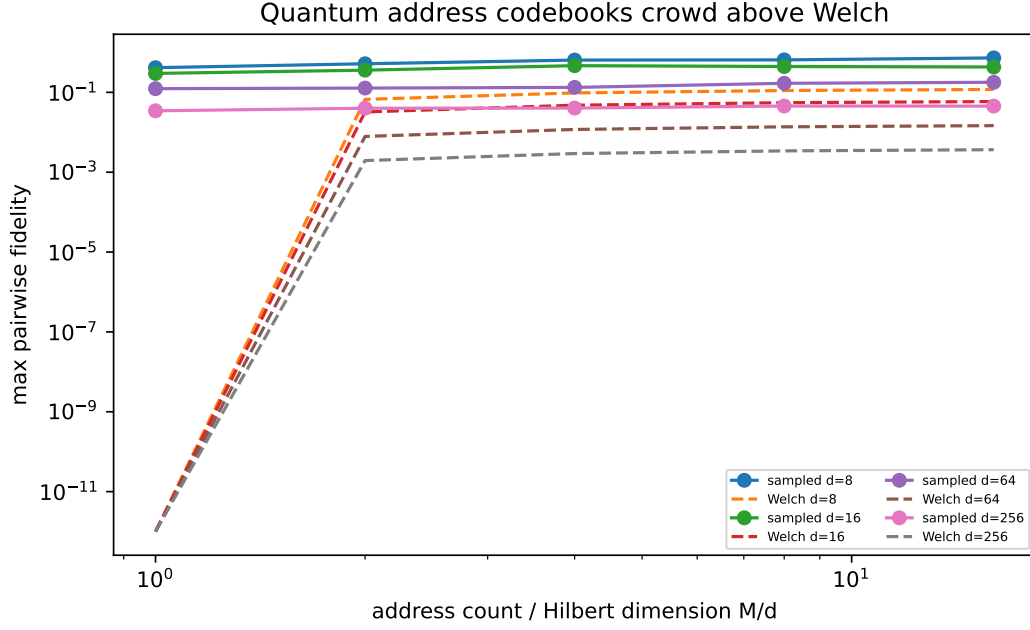


Figure 36: Sampled maximum pairwise fidelity of random complex quantum address codebooks, compared with the Welch lower bound. Random codebooks crowd much faster than the best possible spherical-code lower bound.

The experiment separates two limitations. Fano-Holevo bounds the average recoverable address information, while Welch-style geometry detects near-collisions between particular state pairs. A hypercomputational endpoint needs both global capacity and local separation; finite quantum memory supplies neither at infinite scale.

19.5 Quantum shadow equation

The robust-shadow equation can now be extended with quantum terms. For a witness-returning or address-auditable endpoint,

$$N_{\text{rob}}^Q = \min\{N_{\text{clk}}, N_{\text{read}}, N_{\text{stat}}, N_{\text{pow}}, N_{\text{rel}}, N_{\text{BW}}, N_{\text{QSL}}, N_{\text{Holevo}}, N_{\text{ent}}, N_{\text{dim}}\}.$$

Here

$$N_{\text{QSL}} = \#\{n : \tau_n \geq \hbar a_\alpha / \Delta E\},$$

while N_{Holevo} , N_{ent} , and N_{dim} are the largest event-address catalogs allowed by the Holevo-Fano, physical entropy, and Hilbert-dimension bounds. The minimum is the operational shadow. A formal Zeno tail can be infinite while every term in this physical quantum audit is finite.

This also clarifies a common ambiguity. A boolean endpoint asks for one bit. A certified endpoint asks for an address, a proof, a counterexample, a reproducible physical mark, or a causal certificate. Quantum mechanics does not erase this distinction. A single qubit can answer a yes/no question, but it cannot reveal an unbounded classical witness address with bounded error.

19.6 Exceptional findings, applications, and targets

Quantum readout inherits the finite-shadow pattern. The same structure found for scalar signals, latches, causal certificates, and bandlimited waveforms reappears in quantum theory: finite

endpoint resources imply a finite robust catalog.

Quantum speed and quantum information split the burden. The Mandelstam–Tamm/Helstrom argument limits how densely event-time signatures can be placed. The Holevo-Fano argument limits how many address labels can be recovered even when the states are otherwise optimally encoded.

Entropy bounds are finite-infinity separators. Bekenstein or holographic caps may allow unimaginably many addressable states, but they still do not allow countably infinite witness return from a finite region.

Applications. The formulas above can audit quantum analog hypercomputers, phase-kick Zeno devices, proof-carrying theorem search, counterexample-returning verification, and Malament-Hogarth receiver designs. They also suggest practical design rules for finite shadow accelerators: use quantum resources for a high-value finite witness catalog, then apply the Bayesian waterline and de-Zenoization results from earlier sections.

Targets for further study. The next quantum target is an energy-constrained bosonic endpoint theorem: combine finite-blocklength quantum channel coding, photon-number cost, loss, thermal noise, and Holevo information into a single formula for $N_{\text{boson}}(E, K, N_B, \alpha)$. A second target is a quantum Fisher version of the event-time shadow, replacing a global ΔE bound with a local metric tensor on endpoint states. A third target is a concrete Malament-Hogarth entropy audit: compute the light-sheet area, redshifted channel capacity, and quantum speed-limit budget for an explicit receiver geometry.

20 Synthesis: a resource-normalized Church-Turing principle

The pieces above suggest the following draft principle.

Conjecture 20.1 (Resource-normalized Church-Turing principle). *A physical process with computable finite descriptions of its dynamics, bounded total action, bounded accessible energy, finite statistical signal budget, bounded quantifier-shadow volume, finite Bayesian endpoint budget with a computable prior, finite endpoint channel capacity, finite endpoint bandwidth, bounded quantum speed/entropy, bounded online residual-state width, nonzero non-summable error floor, and robust effectively finite observation cannot compute a non-Turing total predicate with positive uniform reliability.*

This principle is deliberately conditional. It does not deny that non-Turing behavior can be placed into a mathematical model. Instead, it claims that any such behavior must be paid for by

violating at least one of the following constraints:

effective finite readout
 bounded endpoint energy/action
 finite statistical signal budget
 bounded quantifier-shadow volume
 finite witness-address capacity
 finite bandwidth and quantum entropy
 bounded online aggregation width
 summably controlled error
 computable physical input
 robust observation at limit traces.

Thus a proposed hypercomputer should be audited by fourteen questions:

1. Which infinite trace is generated?
2. Which finite readout functional converts the trace to an answer?
3. Is the readout continuous, effectively continuous, or discontinuous?
4. What finite-margin gap separates negative traces from arbitrarily late positive traces?
5. What endpoint signal budget maintains that gap under readout uncertainty?
6. If a persistent latch is used, what is its finite-power write horizon $N_{\text{lat}}(P; m)$?
7. If an iterated observer or bounded jump is claimed, what is the quantifier shadow volume $V = \prod_i N_i$?
8. If an endpoint returns a witness or certificate, what channel capacity identifies its address?
9. If a continuous waveform is used, what finite-bandwidth horizon N_{BW} remains?
10. If a quantum endpoint is used, what are N_{QSL} and the Holevo-Fano entropy address bound?
11. If primitive leaves are compressed internally, what is the online residual width $W_\pi(F)$?
12. Does noisy readout pay the look-elsewhere debt over the possible pivotal leaves?
13. What happens to E_n , $E_n\tau_n$, and the total action near the endpoint?
14. Does the infinite product of correctness probabilities remain positive after correction?

21 Open problems and next directions

21.1 Formal originality checks

The aperture variable $\rho_n = \tau_n/R_n$ is mathematically elementary, and related tail-ratio methods may exist in summability theory, renewal processes, or asymptotic analysis. A useful next step is a literature search for tail-ratio parameterizations of supertasks and accelerated computation. If the exact packaging is absent, the aperture calculus may be a publishable unifying notation.

21.2 Readout degree taxonomy

The inequality $f_{\mathcal{A}} \leq_W O$ should be developed into a detailed taxonomy:

$$O_{\exists}, \quad O_{\infty}, \quad \text{lim}, \quad \text{lim sup}, \quad \text{closed choice}, \quad \text{jump}(O),$$

with each observer matched to known Weihrauch degrees and arithmetical or analytical hierarchies.

21.3 Endpoint thermodynamics

The horizon-energy law should be refined using specific physical models:

1. reversible classical gates;
2. quantum orthogonalization;
3. driven open quantum systems;
4. signal emission to a finite observer;
5. gravitational redshift and blueshift in Malament-Hogarth-like geometries.

The core question is whether a stronger law can be proved:

$$\text{non-Turing readout at finite time} \implies \int_0^T E(t) dt = \infty$$

under broad physical assumptions.

21.4 Reliability with correction

The current reliability theorem assumes that any uncorrected error corrupts the answer. A stronger theory should model error correction. A likely form is:

$$\sum_n p_n^{\text{logical}} < \infty$$

where p_n^{logical} is the post-correction logical error probability. This would connect Zeno computation to fault-tolerance thresholds under increasing clock rates and shrinking time windows.

21.5 Robust negative certificates

The main weakness of halting-style Zeno computation is the nonhalting trace 0^ω . A future theorem should characterize all predicates whose negative instances require non-robust infinite absence certificates. This may produce a robust/non-robust split inside the arithmetical hierarchy.

21.6 Finite-margin event horizons

The signal-budget law should be generalized beyond scalar positive weights. Important variants include signed signals, adaptive signals, finite-dimensional vector signals, quantum measurements, and stochastic channels. The central conjecture is that every finite-resource, finite-noise version has a finite robust event horizon, with the horizon controlled by an appropriate metric entropy or information-capacity bound.

22 Conclusion

The working theory can be compressed into four equations:

$$f(x) = O(G(x)), \quad G \text{ computable}, \quad \deg(f) \leq \deg(O)$$

$$\tau_n = \rho_n R_n, \quad R_n = T \exp\left(-\sum_{k < n} -\log(1 - \rho_k)\right)$$

$$E_n \geq \frac{Q}{\rho_n R_n}$$

$$P_{\text{clean}} = \prod_n (1 - p_n) > 0 \iff \sum_n p_n < \infty.$$

Version 0.2 adds two further equations:

$$N_{\text{robust}} \leq \frac{B}{2\eta}$$

for scalar finite-budget endpoint event detection, and

$$E_n^{\text{rel}} \geq \frac{Q + c \log(n+2)}{\rho_n R_n}$$

for summably reliable Zeno schedules under exponential logical-error suppression. Version 0.3 adds

$$N_{\text{stat}} = \left\lfloor \frac{B}{4\sigma^2 z_{1-\alpha}^2} \right\rfloor$$

for noisy Gaussian endpoint observation and

$$N_{\text{rob}} = \min\{N_{\text{clk}}, N_{\text{read}}, N_{\text{stat}}, N_{\text{pow}}, N_{\text{rel}}\}$$

for the robust finite shadow. Version 0.4 adds

$$N_{\text{lat}}(P; m) = \#\{n : \tau_n \geq Q_m/P\}$$

and the asymptotic utilization law

$$\frac{Q_m N_{\text{lat}}(P; m)}{PT} \rightarrow 0$$

for every true infinite finite-time schedule. Version 0.5 adds

$$B_{Q, \text{margin}} \geq 4\eta^2 \prod_i N_i$$

for deterministic finite-margin bounded quantifier shadows and

$$B_{Q, \text{stat}} \gtrsim 2\sigma^2 V \log V, \quad V = \prod_i N_i$$

for orthogonal Gaussian endpoint channels with unknown pivotal location. Version 0.6 adds

$$b_n^* = \kappa^{-1}[\log(\pi_n/\theta)]_+, \quad \sum_n [\log(\pi_n/\theta)]_+ = \kappa B$$

for Bayesian finite-shadow allocation and the active-horizon laws

$$K_B \sim \sqrt{2\kappa B/s}, \quad K_B \sim \left(\frac{\kappa(p+1)B}{ap} \right)^{1/(p+1)}, \quad K_B \sim \kappa B/\beta$$

for geometric, stretched-exponential, and power-law priors respectively. Version 0.7 adds

$$\log_2 M \leq \frac{C + h_2(\alpha)}{1 - \alpha}$$

for endpoint witness-address capacity and

$$W_\pi(F) = \max_t |\{F(a, \cdot) : a \in \{0, 1\}^t\}|, \quad 2^{f_\pi} \leq W_\pi(F_N^{\exists\forall}) \leq 2^{f_\pi} + 1$$

for online aggregation/frontier shadows. Together they shift attention from “Can infinite steps occur in finite time?” to a more diagnostic question: “What singular interface, resource divergence, signal budget, or reliability assumption converts those steps into a stable finite answer?”

A Appendix: derivation of the polynomial endpoint exponent

Let

$$R_n = T(n+1)^{-a}.$$

Then

$$\rho_n = 1 - \frac{R_{n+1}}{R_n} = 1 - \left(\frac{n+1}{n+2} \right)^a \sim \frac{a}{n+1}.$$

Solving $R_N \approx \varepsilon$ gives

$$N+1 \sim (T/\varepsilon)^{1/a}.$$

Therefore

$$E_\varepsilon = \frac{Q}{\rho_N R_N} \sim \frac{Q}{a(N+1)^{-1}\varepsilon} = \frac{Q}{a} T^{1/a} \varepsilon^{-(1+1/a)}.$$

This proves the endpoint exponent $1 + 1/a$.

B Appendix: source files

The numerical tables and figures in this draft were generated from a deterministic Python script accompanying the source bundle. The script does not call external services. It implements the formulas in [Theorems 4.1, 5.2, 6.1, 9.2, 10.2, 12.1 and 12.5](#) and writes CSV tables for reproducibility. Version 0.2 adds the script `make_v02_experiments.py`, which regenerates the signal-budget and reliability-tax tables and figures. Version 0.3 adds `make_v03_experiments.py`, which regenerates the statistical endpoint, de-Zenoization, robust-shadow, and prior-tail tables and figures. Version 0.4 adds `make_v04_experiments.py`, which regenerates the latch-horizon, Zeno-utilization, power-requirement, and power-clock-seesaw tables and figures. Version 0.5 adds

`make_v05_experiments.py`, which regenerates the quantifier-certificate, look-elsewhere, depth-horizon, and pivotal-influence tables and figures. Version 0.6 adds `make_v06_experiments.py`, which regenerates the Bayesian endpoint waterline, hard-versus-soft coverage, active-depth, residual-risk, and allocation-profile tables and figures. Version 0.7 adds `make_v07_experiments.py`, which regenerates the endpoint-address capacity shadow, semantic compression, residual-width, frontier-memory, and cost-migration tables and figures. Version 0.8 adds `make_v08_causal_experiments.py`. Version 0.9 adds `make_v09_spectral_experiments.py`, which regenerates finite-bandwidth horizon, spectral-utilization, Gram-spectrum, and bandwidth-energy tables and figures. Version 0.10 adds `make_v10_quantum_experiments.py`, which regenerates the quantum speed-limit horizon, entropy-address, physical entropy, Fano-Holevo error, and random quantum crowding tables and figures.

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Version 0.11 Appendage

Stochastic Advice, Random Oracles, and Semantic Alignment Debt

Purpose. This increment tests whether randomness can replace the singular endpoint resource. The answer developed here is that randomness alone is not enough: the random source must be semantically aligned with the target noncomputable predicate.

Central finding. Incomputability and semantic usefulness separate. A fair random real is almost surely incomputable, but it almost surely does not compute a fixed noncomputable target. A nonrecursive biased coin can act as an oracle only because the target was already encoded in the probability law, and extracting late bits carries an exponential sampling debt.

stochastic hypercomputation $\Rightarrow$ noncomputable law

1. Effective stochastic collapse

Let M be a finite-branching probabilistic machine with computable transition probabilities. If M halts almost surely and decides a predicate with a fixed bounded-error gap, then a computable halting-tail modulus collapses the model back to ordinary Turing computability.

$$\Pr[M(x) = A(x)] \geq \frac{1}{2} + \gamma$$

Effective stochastic collapse. Enumerate the random transition tree to depth t . The lower acceptance mass is computable, and the upper acceptance mass is lower acceptance plus live mass. Once the live mass is effectively below the error gap, the interval cannot straddle $1/2$. Thus non-Turing power must come from a noncomputable probability law, a non-effective convergence interface, or a weakened output criterion.

2. Semantic mutual information

Let A_I be a k -bit target block and Z the complete physical observation. Fano-style inequalities imply that reconstructing target bits requires aligned information, not merely entropy:

$$I(A_I; Z) \geq k(1 - h_2(\epsilon))$$

Semantic alignment debt. A stochastic source used as a finite-shadow oracle for k independent target bits with bit-error at most ϵ must carry at least $k(1 - h_2(\epsilon))$ bits of target-aligned mutual information. Entropy independent of the target cannot be converted into reliable oracle answers by computation alone.

capacity C	eps=0.1	eps=0.05	eps=0.01	eps=0.001
16	30	22	17	16
64	120	89	69	64
256	482	358	278	258
1024	1928	1434	1114	1035
4096	7713	5739	4456	4143

Table 1. Fano-style semantic-capacity shadow. Allowing bit errors increases target bits per capacity bit, but exact oracle behavior remains near one aligned bit per target bit.

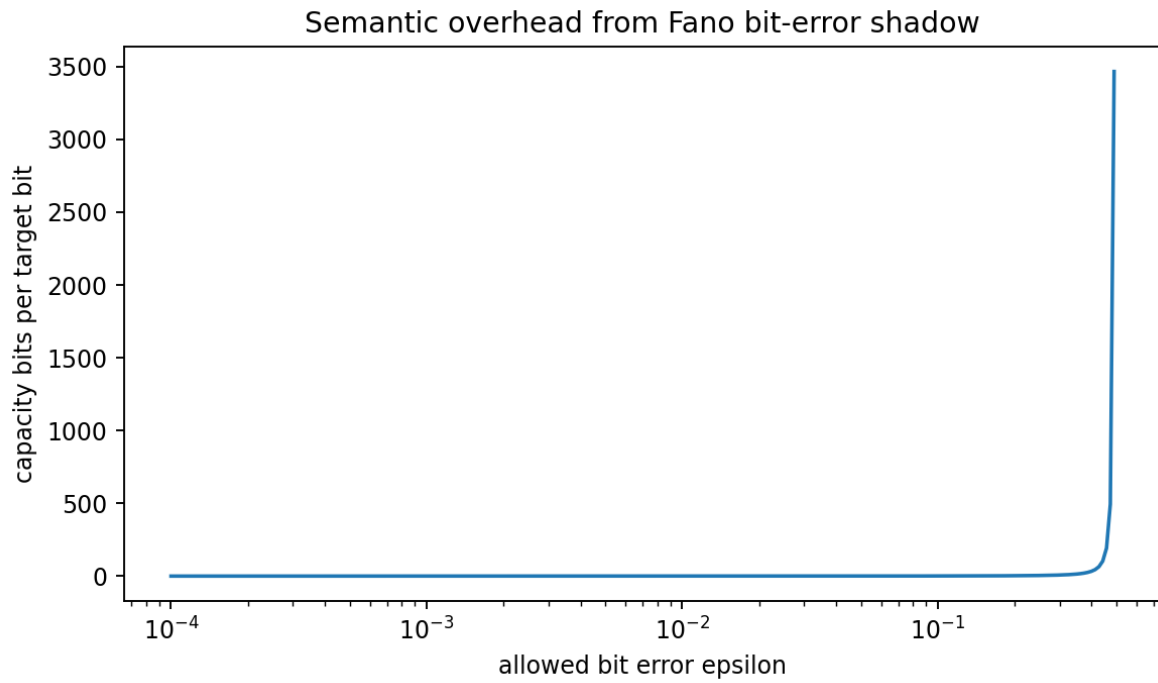


Figure 1. Semantic capacity overhead implied by the bit-error Fano shadow.

3. Blind random advice search

If candidate advice strings are sampled independently of a fixed target, one k -bit prefix match has probability 2^{-k} . With S independent candidates:

$$P_{k,S} = 1 - (1 - 2^{-k})^S, \quad S_q(k) \sim -\log(1 - q)2^k$$

This finite-prefix law mirrors the infinite cone obstruction: most random reals are incomputable, but blind sampling almost never lands in the semantic cone of a desired noncomputable object.

target prefix bits k	$P(\text{one candidate matches})$	$\log_{10}$ trials for 50%	$\log_{10}$ trials for 95%
10	2^{-10}	2.85	3.49
20	2^{-20}	5.86	6.50
40	2^{-40}	11.88	12.52
80	2^{-80}	23.92	24.56
160	2^{-160}	48.01	48.64
320	2^{-320}	96.17	96.81

Table 2. Random advice portfolio size needed to match a fixed target prefix.

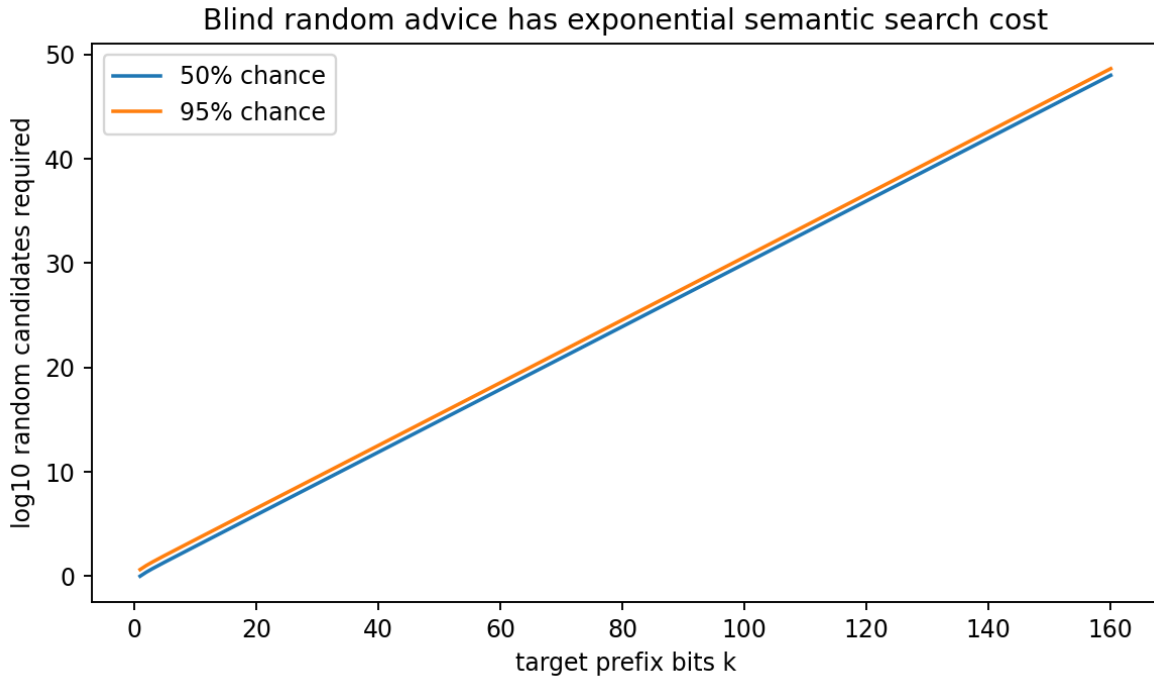


Figure 2. Blind random advice has exponential semantic search cost.

4. Biased coins as compressed oracles

A nonrecursive biased coin differs from an unselected random stream: the target information is placed directly into the stochastic law. The remaining task is metrology. If the n th bit is separated from a decision threshold by guard margin g_n , then Hoeffding-scale extraction requires:

$$m_n = \Theta\left(\frac{\log(1/\delta_n)}{g_n^2}\right), \quad \log_{10} m_n = 2(n+2)\log_{10} 2 + O(\log \log(1/\delta_n))$$

encoded	bit n	gap g_n	log10 samples delta=0.01	delta_n summable	log10 samples summable
5		2 ⁻⁷	4.64	1.689e-04	4.89
10		2 ⁻¹²	7.65	5.024e-05	7.95
20		2 ⁻²²	13.67	1.379e-05	14.02
30		2 ⁻³²	19.69	6.326e-06	20.07
40		2 ⁻⁴²	25.71	3.616e-06	26.11
60		2 ⁻⁶²	37.75	1.634e-06	38.17
80		2 ⁻⁸²	49.79	9.266e-07	50.23

Table 3. Hoeffding sample budgets for reading guarded oracle bits. The sample scale is dominated by g_n^{-2} .

For infinitely many extracted bits, the per-bit error schedule must be summable. This adds only a logarithmic correction to the sample exponent, but it reconnects biased-coin extraction to the reliability-summability law.

bit n	gap g_n	samples m	empirical error	Hoeffding upper
4	1.562e-02	2048	0.0793	0.3679
4	1.562e-02	4096	0.0229	0.1353
4	1.562e-02	8192	0.0018	0.0183
6	3.906e-03	32768	0.0800	0.3679
6	3.906e-03	65536	0.0260	0.1353
6	3.906e-03	131072	0.0020	0.0183
8	9.766e-04	524288	0.0690	0.3679
8	9.766e-04	1048576	0.0210	0.1353
8	9.766e-04	2097152	0.0030	0.0183
10	2.441e-04	8388608	0.0760	0.3679
10	2.441e-04	16777216	0.0250	0.1353
10	2.441e-04	33554432	0.0025	0.0183

Table 4. Monte Carlo check for extracting a guarded biased-coin bit using a sample-mean threshold.

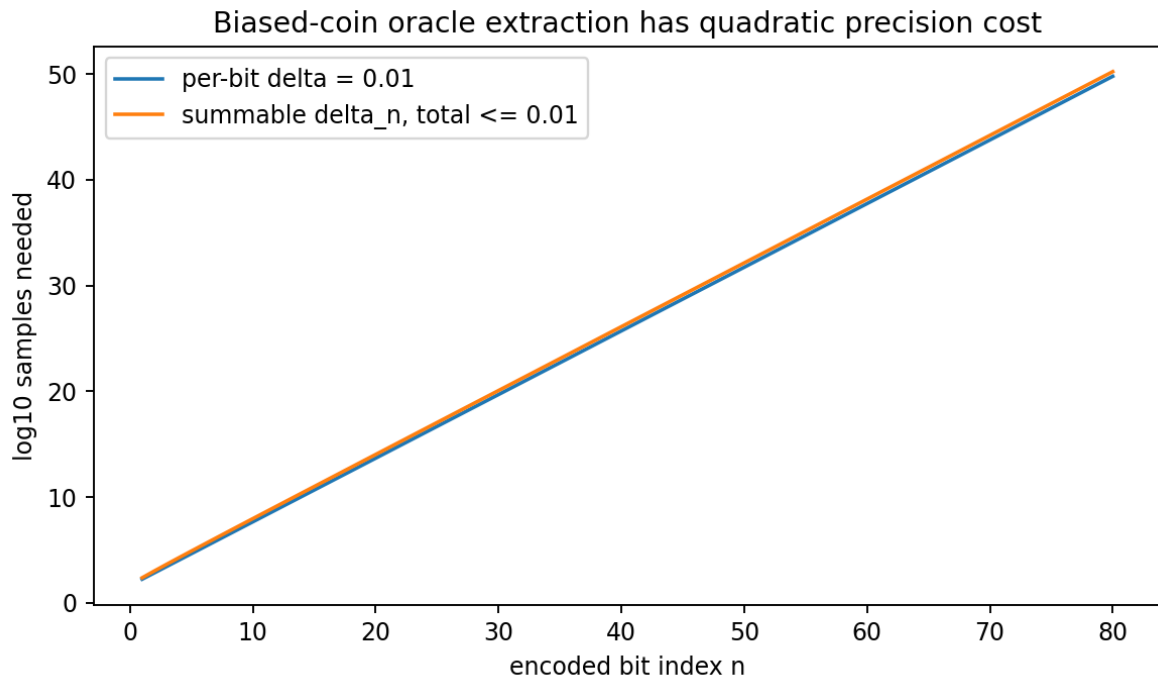


Figure 3. Later encoded bits require exponentially many samples because threshold margins shrink exponentially.

5. Exceptional findings

Incomputability is not enough. A random source may be noncomputable without carrying usable information about the halting predicate, a formal theory, or any specified oracle set.

Biased-coin hypercomputation is a precision oracle. A nonrecursive bias works because the oracle is already encoded in the probability law. Randomness is the measurement mechanism, not the origin of the target information.

Random advice has a finite-prefix exponential wall. Blind advice has success cost $\Theta(2^k)$ for k target bits. In the infinite limit, useful oracle cones have measure zero for fixed noncomputable targets.

Reliability summability reappears. Stochastic oracle extraction is not exempt from the infinite-product law; extracting infinitely many bits with nonzero joint success probability requires a summable error schedule.

6. Applications and targets

Quantum-randomness audits. A hypercomputation claim based on random quantum output must specify semantic alignment with the target predicate, not only algorithmic randomness of the source.

Nonrecursive physical constants. Any oracle encoded into a real-valued physical constant should report a finite sampling law for extracting the n th useful bit with summable error.

Theorem-proving portfolios. Randomized theorem search should be modeled as a prior-weighted finite-shadow process, not as blind access to noncomputable advice.

Verification and fuzzing. Randomness expands exploration, but only target correlations turn exploration into reliable discovery.

Target: semantic Weihrauch resources. Define a resource preorder separating raw entropy from mutual information about a target representation.

Target: stochastic endpoint waterlines. Combine the Bayesian endpoint waterline with biased-coin extraction, allocating samples only to oracle bits whose posterior value exceeds a waterline.

Target: cone-measure finite analogues. Develop finite versions of upper-cone obstructions using bounded decoders, limited query depth, and prefix measure.

Target: random-parameter metrology. Replace Hoeffding toy bounds with quantum estimation and Cramer-Rao bounds, then feed those bounds into aperture, reliability, and causal-screen laws.

References for this appendage

The stochastic-collapse framing follows classical probabilistic Turing-machine results; the alignment discussion uses standard Fano information bounds; the random-oracle warning is consistent with measure-zero upper-cone phenomena; and biased-coin extraction is treated here as a metrology problem rather than as free oracle access.

Version 0.11 Supplement

Analog Oracle Constants and Metrological Finite Shadows

Purpose. This supplement studies exact real constants, noncomputable initial conditions, analog weights, and calibration parameters as another proposed source of hypercomputational power.

Central finding. A real-oracle machine compresses infinitely many target bits into vanishing amplitude gaps. Finite metrology converts that infinite object into a finite advice string.

$$\text{Zeno gap } \tau_n \rightarrow 0 \iff \text{oracle metric gap } \Delta_N \rightarrow 0$$

1. Prefix packing

A real number can mathematically encode a noncomputable set through its digit expansion. The physical question is how many of those digits can be extracted with finite precision. A robust ternary code is:

$$\theta_A = \sum_{n=0}^{\infty} 2A(n)3^{-(n+1)} \in [0, 1]$$

For arbitrary scalar encodings, packing already forces exponential shrinkage. If 2^N prefix representatives lie in $[0, L]$, the minimum separation obeys:

$$\Delta_N \leq \frac{L}{2^N - 1}$$

Thus no bounded scalar parameter can provide uniformly separated prefixes of unbounded length. A different base changes constants; it does not remove the shrinking-gap law.

2. Fisher extraction law

With total Fisher information J and Gaussian local confidence multiplier z , robust extraction of N scalar-coded bits requires the estimation radius to be below half the prefix gap:

$$J_{\min}(N) \geq \frac{4z_1^2 - \alpha}{L^2} 4^N, \quad N_{\text{Fish}}(J) \leq \frac{1}{2} \log_2 J + O(1)$$

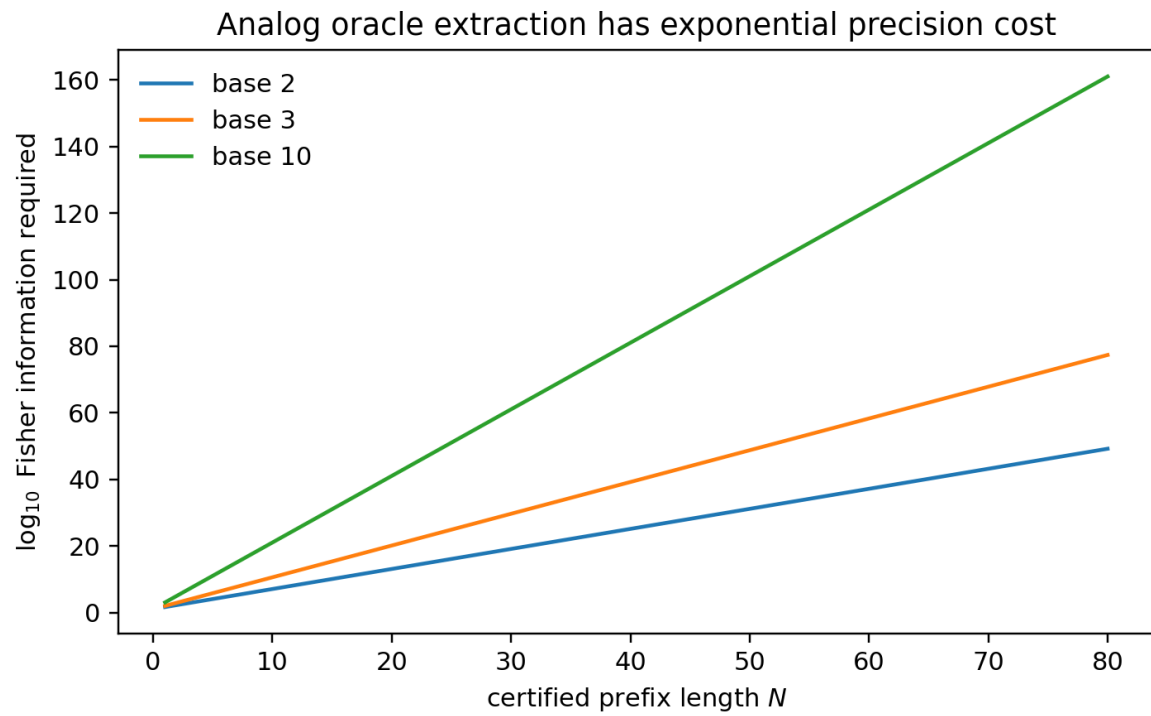


Figure S1. Scalar analog-oracle extraction has exponential Fisher-information cost.

prefix N	scalar code	$\log_{10} J_{\min}$	$J_{\min}$
5	binary packing	4.04	1.11e4
5	ternary Cantor	5.81	6.39e5
10	binary packing	7.05	1.13e7
10	ternary Cantor	10.58	3.77e10
20	binary packing	13.08	1.19e13
20	ternary Cantor	20.12	1.32e20
40	binary packing	25.12	1.31e25
40	ternary Cantor	39.20	1e39.2
80	binary packing	49.20	1e49.2
80	ternary Cantor	77.37	1e77.4

Table S1. Representative Fisher-information costs at $\alpha=0.05$.

3. Dimension and capacity shadows

Finite dimension helps only linearly. Packing 2^N prefixes into a bounded d -dimensional parameter region gives:

$$N_{d, \text{Fish}}(J) \leq \frac{d}{2} \log_2 J + O(d)$$

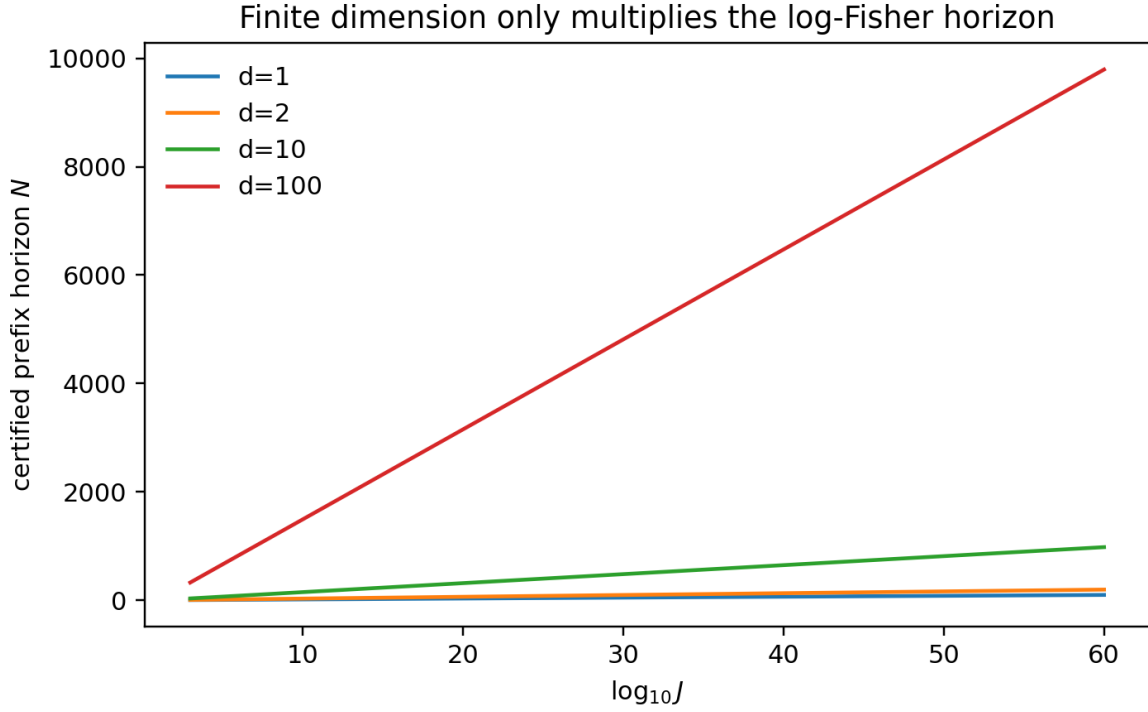


Figure S2. Finite dimension multiplies the accessible prefix but preserves logarithmic Fisher scaling.

J budget	d=1	d=2	d=10	d=100
1e6	8	16	82	824
1e12	18	36	182	1821
1e24	38	76	381	3814
1e60	97	195	979	9793

Table S2. Approximate prefix horizons using N about $(d/2) \log_2(J/(4z^2))$.

A measurement apparatus also returns a finite record. If C bits of mutual information are available about independent oracle bits at bit error α , Fano accounting requires:

$$C \geq N(1 - h_2(\alpha))$$

C bits	alpha	h2(alpha)	bits recoverable
10	0.01	0.081	10
10	0.05	0.286	14
10	0.10	0.469	18
100	0.01	0.081	108
100	0.05	0.286	140

C bits	alpha	h2(alpha)	bits recoverable
100	0.10	0.469	188
1e+03	0.01	0.081	1087
1e+03	0.05	0.286	1401
1e+03	0.10	0.469	1883
1.00e6	0.01	0.081	1087894
1.00e6	0.05	0.286	1401339
1.00e6	0.10	0.469	1883223

Table S3. Capacity gives a linear finite-prefix bound; scalar geometry adds exponential metrological cost.

4. Drift and the calibration sweet spot

If measurement averaging reduces variance as a/t but the parameter drifts with variance bt , then longer averaging eventually becomes harmful:

$$V(t) = \frac{a}{t} + bt, \quad t^* = \sqrt{a/b}, \quad V_{\min} = 2\sqrt{ab}$$

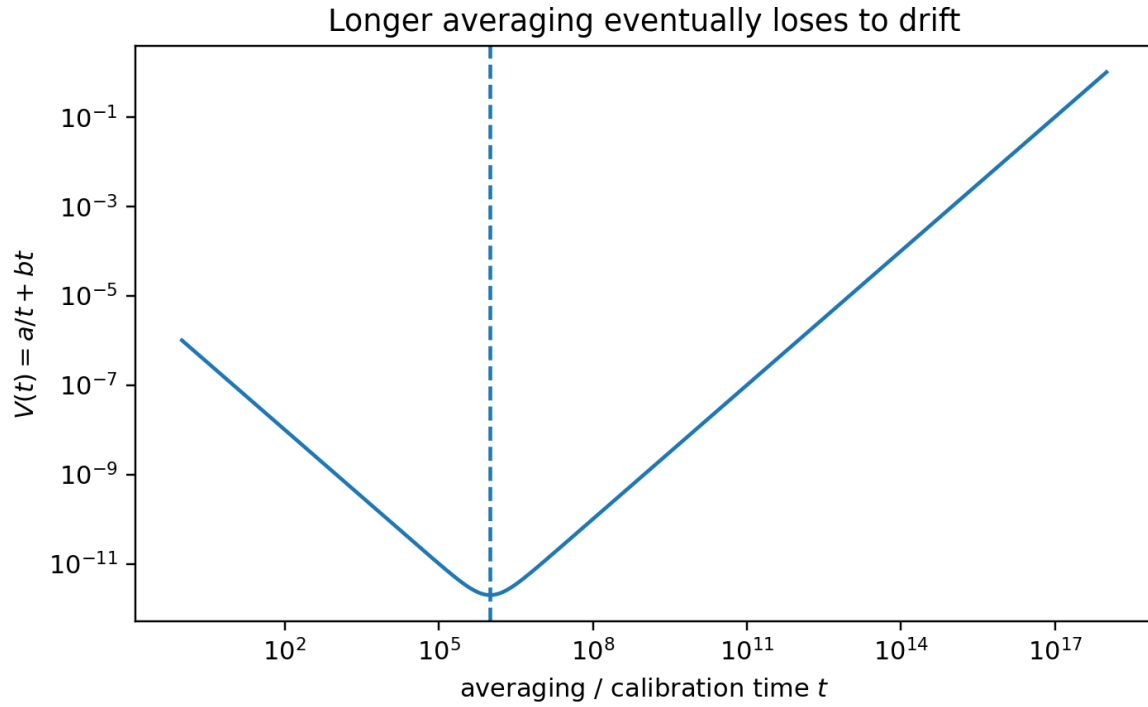


Figure S3. Calibration sweet spot: averaging helps until drift dominates.

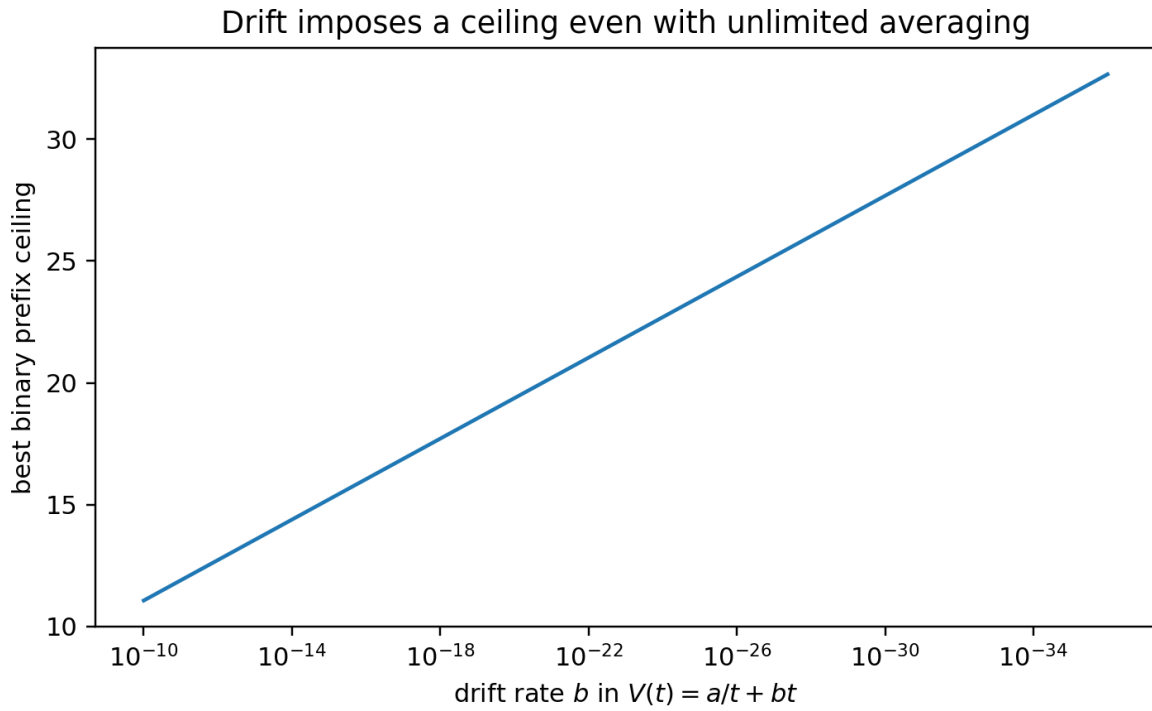


Figure S4. Suppressing drift by many orders of magnitude increases the extractable prefix only logarithmically.

drift b	t^*	$V_{\min}$	binary prefix ceiling
1e-12	1e+03	2.00e-9	12
1e-18	1.00e6	2.00e-12	17
1e-24	1.00e9	2.00e-15	22
1e-30	1.00e12	2.00e-18	27
1e-36	1.00e15	2.00e-21	32

Table S4. Drift ceilings for $V(t)=1e-6/t + bt$ at $\alpha=0.05$.

5. Error cliff and finite-advice factorization

For the ternary-Cantor code under a Gaussian local model:

$$P_{\text{err}}(N; J) \approx \text{erfc}\left(\frac{3^{-N}\sqrt{J}}{2\sqrt{2}}\right)$$

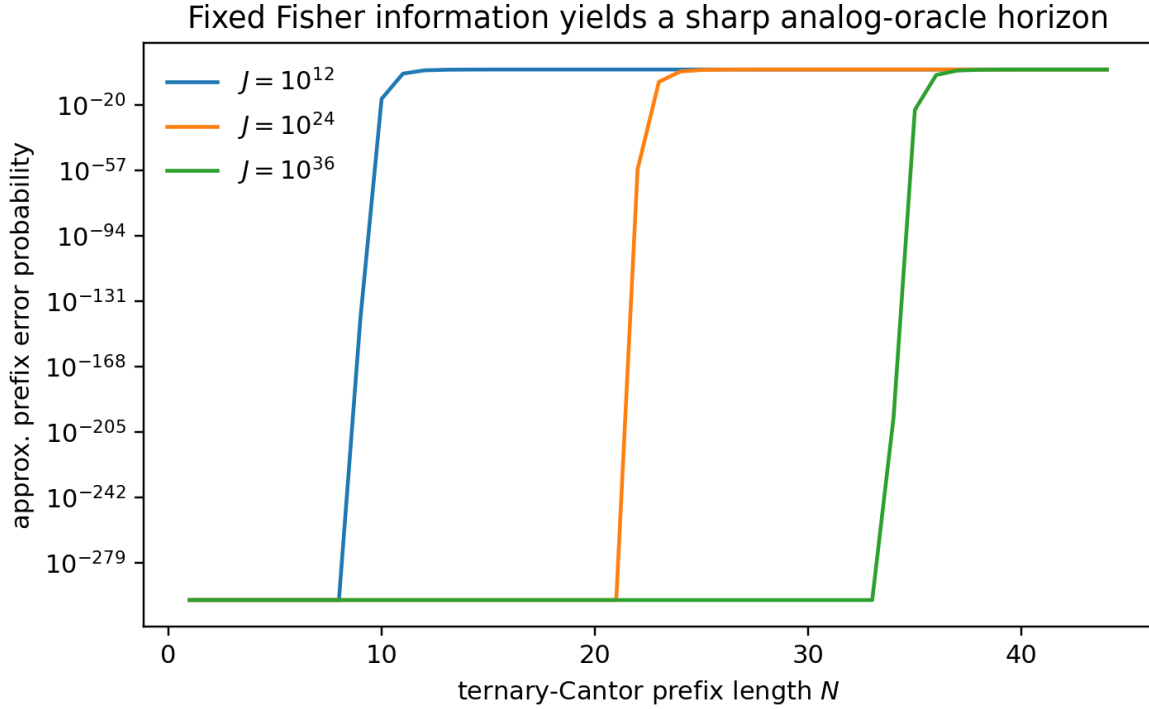


Figure S5. Fixed Fisher information yields a sharp analog-oracle horizon.

Finite-advice shadow principle. If only N robust bits of a real parameter are extractable, the auditable computation factors through an ordinary machine with an N -bit advice string.

$$f_{\text{robust}}(x) = M(x, \theta_{1:N}), \quad N < \infty$$

The robust content is not computation with an exact real; it is computation with finite advice whose length is bounded by the smallest active metrological shadow:

$$N_{\text{oracle}} = \min\{N_{\text{Fisher}}, N_{\text{cap}}, N_{\text{dim}}, N_{\text{drift}}, N_{\text{entropy}}\}$$

6. Applications and targets

Exact-real machine audits. BSS-style or real-RAM claims can be translated into range, dimension, Fisher, comparator-margin, and returned-capacity budgets.

Analog neural networks and reservoirs. Arbitrary real weights become finite advice once weight preparation, drift, and readout precision are specified.

Physical constants as oracles. A proposal that encodes answers in a coupling constant must supply an extraction protocol $V(t)$, not merely an exact mathematical value.

Target. Replace the Cramer-Rao heuristic with minimax Le Cam bounds and a quantum Fisher/Bures version.

Target. Combine drift with the horizon-energy law to estimate thermodynamic price per extracted oracle bit.

Exceptional synthesis: Zeno machines concentrate index information into time gaps; analog-oracle machines concentrate it into metric gaps. Finite robustness converts both into finite shadows.

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Version 0.12 Appendage

Chronometric Shadows and Time-Warp Cost

Purpose. This increment audits a resource not counted by endpoint readout: the clock or phase reference that orders and triggers the shrinking computational slots of an accelerating machine.

Central finding. A Zeno implementation can hide an oracle inside the time reparametrization. Infinite internal time requires infinite phase advance. If the clockwork cost is eventually linear in speed, the time-warp cost diverges before endpoint readout is considered.

$$\int_0^T v(t) dt = \infty, \quad C_\psi = \int_0^T \psi(v(t)) dt, \quad \psi(v) \geq cv \Rightarrow C_\psi = \infty$$

1. Time-warp convexity collapse

Let s be internal computational time, t external time, and $v(t)=ds/dt$. A finite-time supertask requires $\int v(t)dt$ to diverge on a finite interval. If the cost density $\psi(v)$ is eventually at least linear, the divergent part of the speed integral lies in the region where $\psi(v) \geq cv$, and the clockwork cost diverges. Therefore a hypercomputational time warp requires sublinear asymptotic clock cost, a singular controller, or an uncounted resource.

For a discrete schedule with slot lengths τ_n , a power-law clockwork model $\psi(v)=v^p$ gives $C_p(N)=\sum \tau_n^{1-p}$. The critical exponent p_* is the boundary between finite and divergent infinite clock cost.

schedule	asymptotic τ_n	critical p^*	$p=.25$	$p=.50$	$p=.75$	$p=1$	$p=1.25$
geometric	$\Theta(\lambda^n)$	1	finite	finite	finite	diverges	diverges
power $a=1$	$\Theta(n^{-2})$	1/2	finite	diverges	diverges	diverges	diverges
power $a=2$	$\Theta(n^{-3})$	2/3	finite	finite	diverges	diverges	diverges
log-slow	$\Theta(1/(n \log^2 n))$	0	diverges	diverges	diverges	diverges	diverges

Table 1. Time-warp power-cost phase diagram. The log-slow schedule is maximally fragile: every positive speed-cost exponent makes the infinite tail cost diverge.

N	geometric	power $a=1$	power $a=2$	log-slow
10	29.27	23.58	31.27	23.84
20	194.8	63.41	98.3	61.12
50	3.64e+4	242.1	467	214
100	2.11e+8	675.9	1545	550.7
200	7.08e+15	1899	5155	1409
500	2.67e+38	7475	2.55e+4	4832

Table 2. Superlinear clockwork cost samples for $p=1.25$.

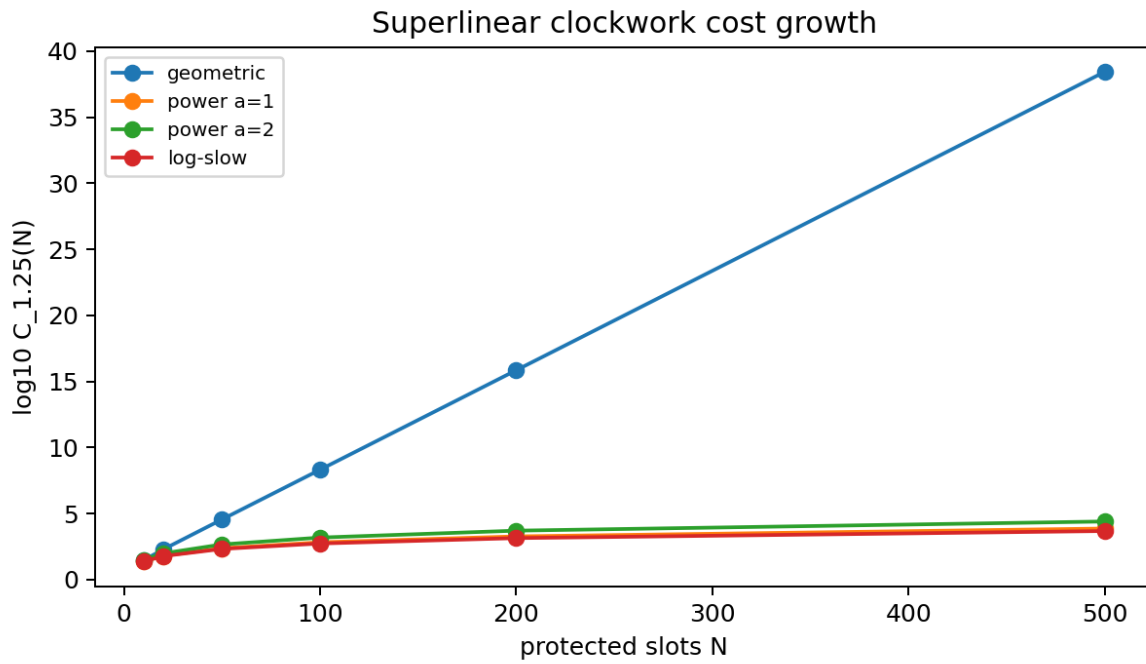


Figure 1. Superlinear time-warp cost growth. Endpoint-friendly acceleration can be clock-hostile.

2. Chronometric reliability law

Let J_n be independent Gaussian timing jitter with standard deviation σ_n . If a slot has guard $g \tau_n$, a gross timing failure occurs when $|J_n| \geq g \tau_n$. With $r_n = g \tau_n / \sigma_n$, the infinite product of local survival probabilities is positive exactly when gross failure probabilities are summable.

$$P_N^{\text{time}} = \prod_{n < N} [1 - 2\Phi(-r_n)], \quad r_n = \frac{g \tau_n}{\sigma_n}$$

$$r_n^2 \geq 2(1 + \delta) \log n \Rightarrow P_\infty^{\text{time}} > 0, \quad r_n^2 \leq 2(1 - \delta) \log n \Rightarrow P_\infty^{\text{time}} = 0$$

A fixed five- or six-sigma guard can pass long finite tests while still having zero infinite survival. The relative phase margin must grow slowly but without bound.

guard ratio model	N=1,000	N=100,000	N=1,000,000	infinite limit
constant 5-sigma	0.9994	0.9443	0.5637	0
constant 6-sigma	1	0.9998	0.998	0
$\sqrt{2 \cdot 0.8 \log(n+1000)}$	0.5247	0.002873	2.07e-5	0
$\sqrt{2 \cdot 1.2 \log(n+1000)}$	0.9709	0.8813	0.8588	positive
$\sqrt{2 \cdot 2.0 \log(n+1000)}$	0.9999	0.9999	0.9999	positive

Table 3. Timing survival under independent Gaussian jitter. The threshold is the logarithmic guard exponent $\beta=1$.

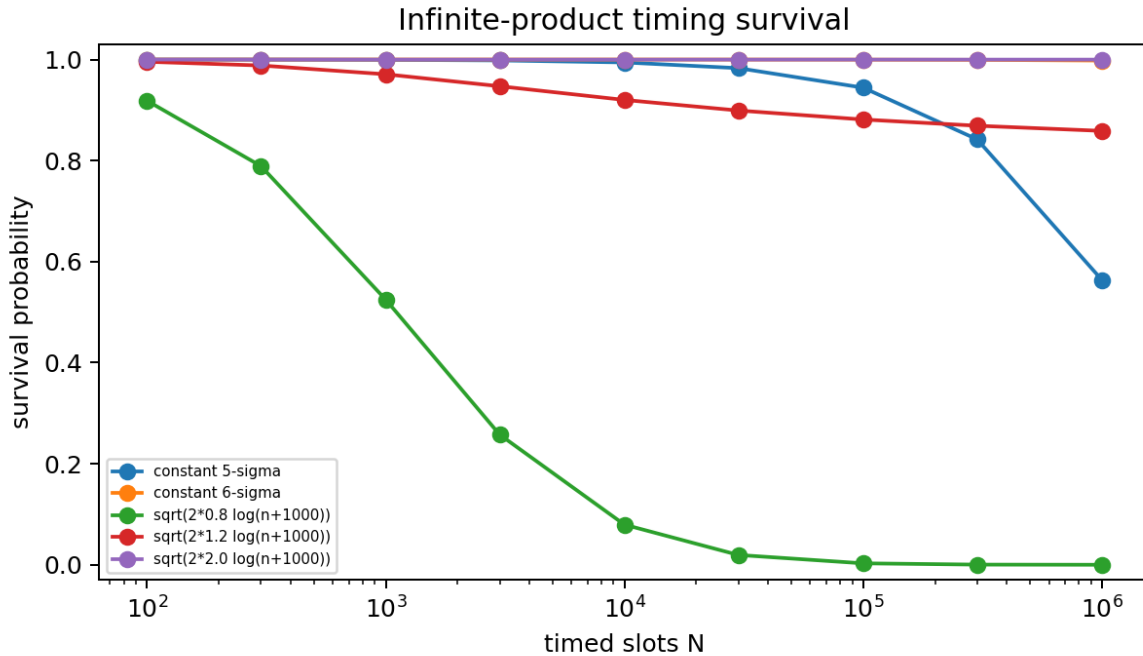


Figure 2. Constant high-sigma timing guards have zero infinite survival; logarithmic guards can survive.

3. Timing-Fisher shadow and drift counterfinding

Represent timing precision by Fisher information $I_n = \sigma_n^{-2}$. The logarithmic guard law implies a cumulative precision budget. Finite timing-Fisher information induces a finite phase shadow independent of endpoint signal, latch power, bandwidth, or quantum state dimension.

$$N_{\text{phase}}(J) = \max \left\{ N : \sum_{n < N} \frac{2 \log(n+2)}{g^2 \tau_n^2} \leq J \right\}$$

schedule	J=1e+6	J=1e+12	J=1e+18	J=1e+24
geometric	8	18	28	38
power a=1	14	216	3,164	47,294
power a=2	8	60	420	2,913
log-slow	16	469	20,597	1,151,557

Table 4. Finite timing-Fisher horizons.

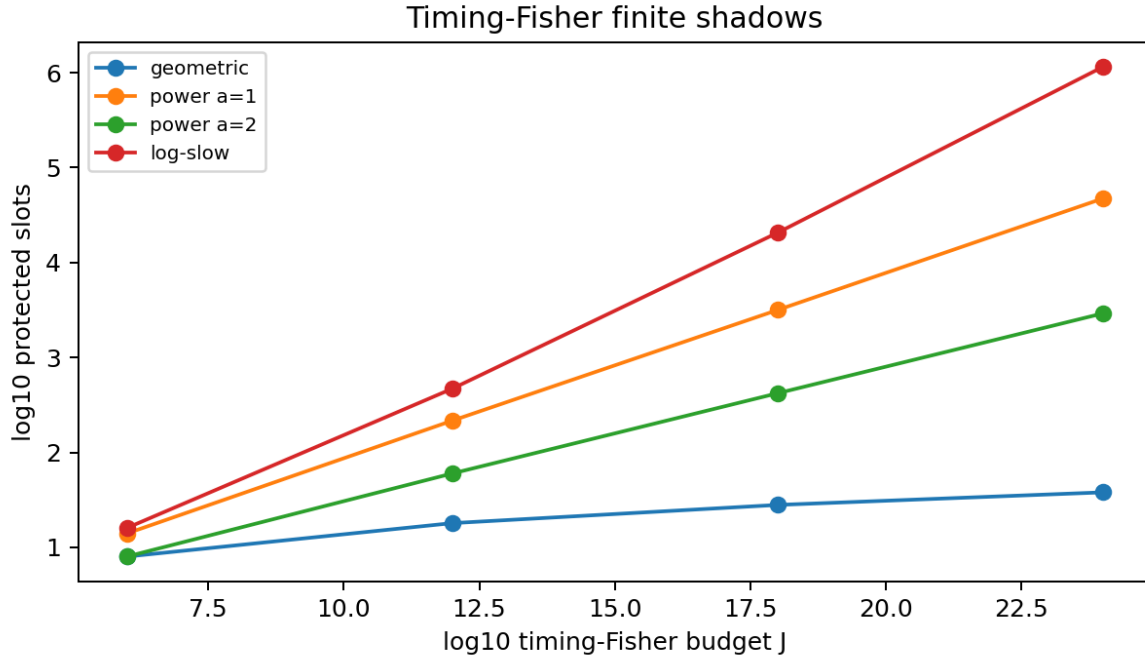


Figure 3. Timing-Fisher finite shadows for representative schedules.

Counterfunding: endpoint-time drift is not necessarily the main obstruction. If $\sigma_n = \gamma \tau_n$ and jitters are independent, then $\text{Var}(\sum J_n) = \gamma^2 \sum \tau_n^2 \leq \gamma^2 T^2$. Endpoint time can remain tight while phase correctness collapses from infinitely many local guard failures.

schedule	sum τ_n^2	RMS factor $\sqrt{\text{sum } \tau_n^2}$
geometric	0.3333	0.5774
power a=1	0.2899	0.5384
power a=2	0.5849	0.7648
log-slow	0.1628	0.4034

Table 5. Relative independent jitter has finite cumulative endpoint variance for the tested schedules.

4. Applications and targets

Clock audits for Zeno hardware. A proposed accelerator should state the resource that orders and triggers operations, not only the endpoint readout mechanism.

Hybrid-system Zeno control. The phase shadow distinguishes mathematically infinite switching from finite precision actuation.

Malament-Hogarth synchronization. Receiver models should count clock alignment, phase reference, and causal-window discrimination as independent resources.

Clock waterlines. Combine prior-weighted endpoint waterlines with timing-Fisher allocation to protect only high-influence event windows.

Thermodynamic uncertainty refinements. Replace the Gaussian jitter toy model with TUR-based dissipation/precision bounds for stochastic clocks.

Quantum reference frames. Couple temporal precision, localization, energy spread, and endpoint distinguishability in one clock-channel shadow.

Tauberian schedule classification. Use p_* and related critical exponents to classify schedules independently of their endpoint semantics.

Reproducibility note

The tables and figures in this chronometric appendage are regenerated by `make_v12_chronometric_experiments.py`. The stress models are deliberately simple diagnostic envelopes; the invariant claims are the divergence, summability, and finite-shadow laws.

Version 0.12 Appendage

Reversible Zeno Computation and Entropy Logistics

Purpose. This increment tests whether logical reversibility can evade the endpoint, latch, bandwidth, quantum, stochastic, and metrological finite-shadow laws.

Central finding. Reversibility changes the currency but not the accounting. A reversible interior avoids direct erasure only by retaining the information needed to reverse the trajectory, exporting it to an entropy sink, or moving it into final measurement. Finite retained garbage, finite entropy export, finite checkpoint memory, and summable reliability imply finite horizons.

$$\mathcal{I}_N = \sum_{n < N} \ell_n, \quad \ell_n = \log_2 \max_y |F_n^{-1}(y)|$$

1. Information-loss volume and garbage-heat shadow

For primitive irreversible maps F_n , the information-loss volume $\mathcal{I}_N$ counts the missing preimage bits. A reversible physical implementation cannot destroy these indices; it must store them as garbage, export them as entropy, or place them in a measurement record.

$$\sum_{n < N} \ell_n \leq G + H, \quad N_{gh}(G, H) = \max\{N : \sum_{n < N} \ell_n \leq G + H\}$$

retained garbage G bits	entropy-export H bits	loss ell bits/step	N_gh=floor((G+H)/ell)
1,000	0	1	1,000
1.000e6	0	1	1.000e6
1,000	1.000e6	1	1.001e6
1.000e9	1.000e9	1	2.000e9
1.000e6	0	10	1.000e5
1.000e9	1.000e9	10	2.000e8
1.000e12	1.000e12	100	2.000e10

Table 1. Garbage-heat finite shadows for a constant information-loss rate.

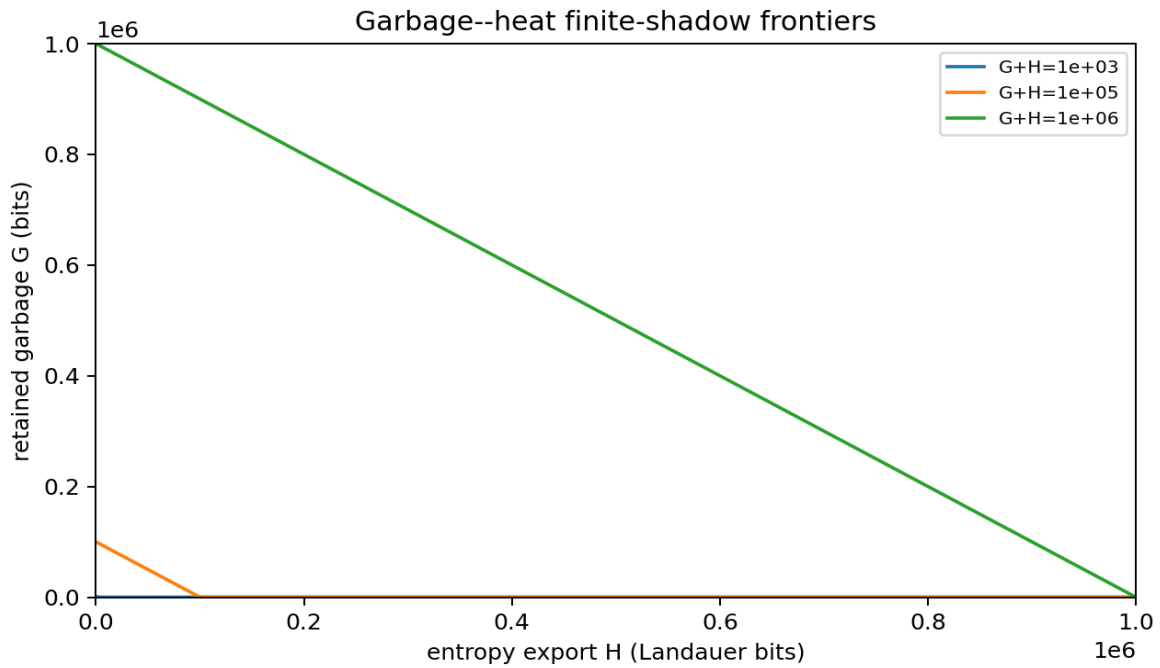


Figure 1. Garbage-heat frontiers: heat export can substitute for retained garbage, but finite budgets still imply finite primitive depth.

2. Bennett checkpoint horizons

Checkpointed reversible simulation can make finite computations extraordinarily cheap in retained history. In audit form, take G approximately $\gamma S \log_2 N$, where S is work space and γ absorbs checkpoint constants. The horizon can be astronomical, but it remains finite for finite G .

$$N_{\text{Ben}} \lesssim 2^{G/(\gamma S)}$$

work space S bits	available reversible memory	gamma	log10 N_Ben	N_Ben
1,000	1.000e6	4	75.3	$10^{75.3}$
1,000	1.000e9	4	7.53e+04	$10^{7.53e+04}$
1,000	1.000e12	4	7.53e+07	$10^{7.53e+07}$
1.000e6	1.000e6	4	0.0753	1
1.000e6	1.000e9	4	75.3	$10^{75.3}$
1.000e6	1.000e12	4	7.53e+04	$10^{7.53e+04}$

Table 2. Bennett-style reversible memory horizons.

3. Finite-time erasure stress and de-Zenoization

If garbage is erased rather than retained or uncomputed, finite-time Landauer corrections add a short-time penalty. A generic diagnostic model is $W_n \geq kBT \ln 2 b_n + \Lambda b_n^2 / \tau_n$. For a finite active horizon, Cauchy-Schwarz gives an ordinary schedule optimizer:

$$\tau_n^* = T \frac{b_n}{\sum_{k < N} b_k}, \quad \min \sum_{n < N} \frac{b_n^2}{\tau_n} = \frac{\left(\sum_{n < N} b_n \right)^2}{T}$$

For equal erasure burdens, the optimizer is uniform, not Zeno-like. Infinite acceleration is a bad finite-time cleanup policy once the robust horizon has collapsed to finite N .

schedule	N=10	N=100	N=10000	N=1e+06
geometric	3.31	30.4	3.01e+03	3.01e+05
power a=1	2.64	5.54	11.5	17.5
power a=2	3.26	7.11	15.1	23.1
log-slow	2.64	5.11	9.74	14.1

Table 3. Finite-time erasure stress $\log_{10} \sum 1/\tau_n$ in the toy excess-work model.

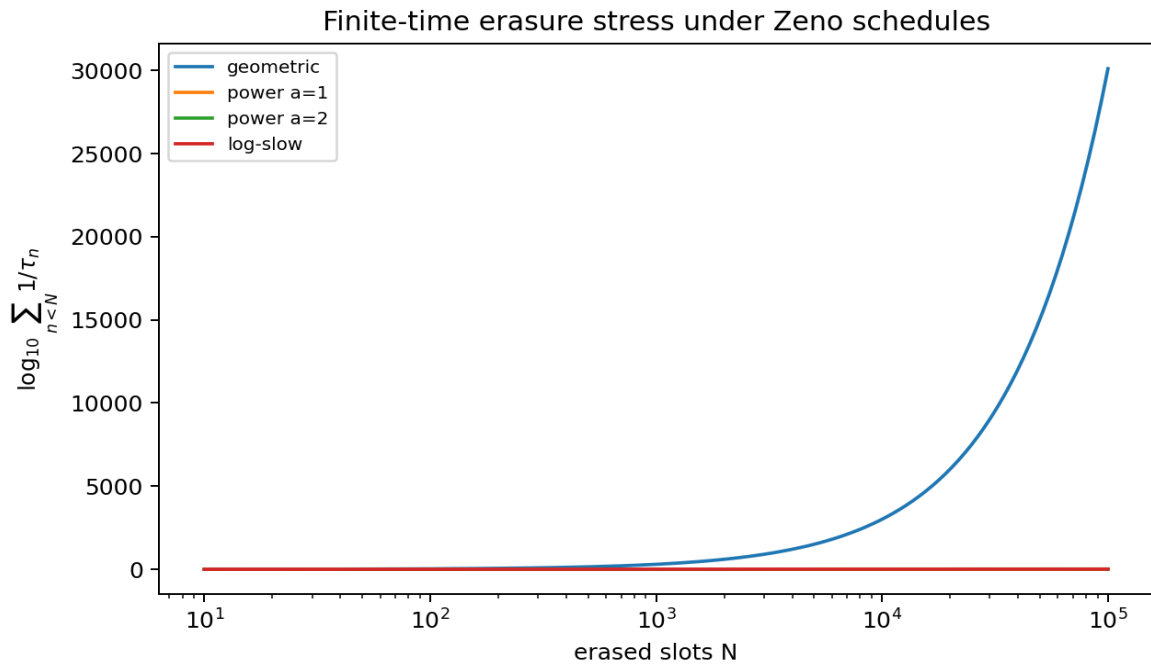


Figure 2. Geometric schedules explode under finite-time cleanup; slower tails delay but do not remove the divergence.

4. Reliability entropy and the Lambert shadow

Reversible gates are not automatically reliable gates. If logical error in slot n is $p_n = \exp(-\alpha c_n)$, then making the total error over N slots at most δ requires correction entropy of order $N \log(N/\delta)$. Solving this budget gives a Lambert-W finite shadow.

$$C_N^* = \frac{N}{\alpha} \log\left(\frac{N}{\delta}\right), \quad N_{\text{relH}} = \left\lfloor \frac{\alpha H}{W(\alpha H/\delta)} \right\rfloor$$

entropy budget H bits	delta	N_relH	log10 N_relH	H/N_relH bits/slot
1,000	1e-06	56	1.75	17.8
1.000e6	1e-06	4.092e4	4.61	24.4
1.000e9	1e-06	3.215e7	7.51	31.1
1.000e12	1e-06	2.645e10	10.4	37.8
1.000e18	1e-06	1.948e16	16.3	51.3

Table 4. Lambert reliability horizons for $\alpha=1$ and $\delta=10^{-6}$.

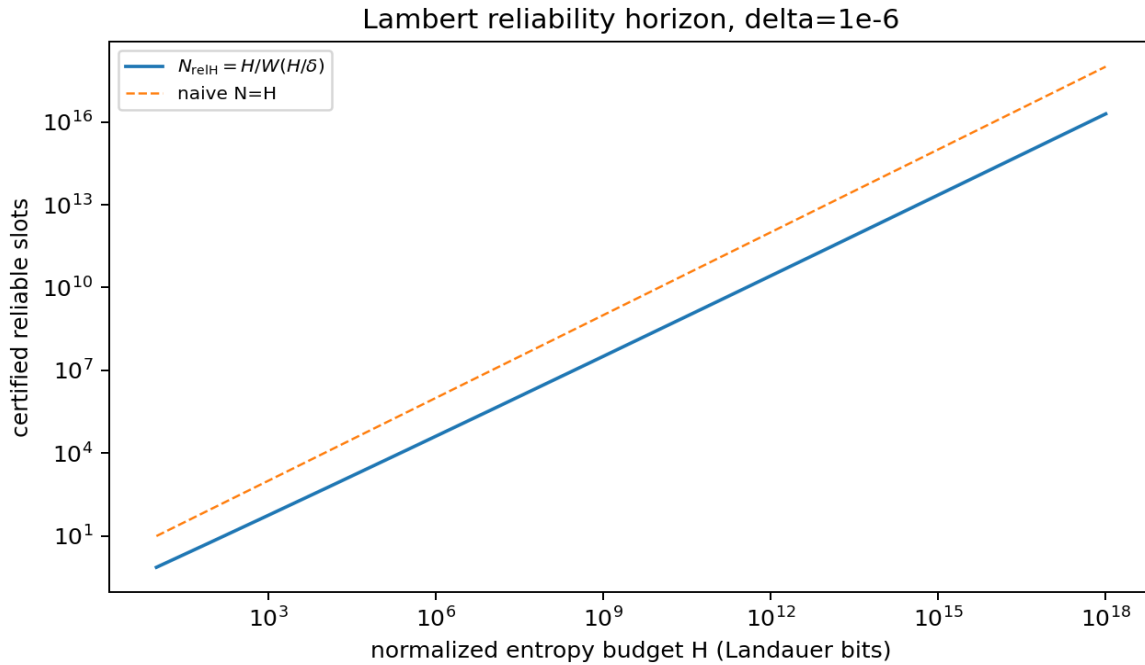


Figure 3. Reliability entropy produces an $H/W(H)$ horizon, not a naive linear horizon.

5. Composite reversible finite shadow

The reversible audit adds new finite horizons to the earlier robust-shadow minimum. The active constraint can be primitive information-loss volume, checkpoint memory, reliability entropy, or one of the previous endpoint limits.

$$N_{\text{rev}} = \min\{N_{\text{gh}}, N_{\text{Ben}}, N_{\text{relH}}, N_{\text{read}}, N_{\text{lat}}, N_{\text{BW}}, N_{\text{QSL}}, N_{\text{oracle}}, N_{\text{phase}}\}$$

case	G	H	S	N_gh	N_Ben	N_relH	active finite shadow
small lab	1.000e6	1.000e6	1,000	2.000e+06	$10^{*75.3}$	40,925	40,925
large device	1.000e12	1.000e12	1.000e6	$10^{*12.3}$	$10^{*7.53e+04}$	$10^{*10.4}$	$10^{*10.4}$

case	G	H	S	N_gh	N_Ben	N_reIH	active finite shadow
memory rich / cold	1.000e15	1.000e6	1.000e6	10 ¹⁵	10 ^{7.53e+07}	40,925	40,925
heat rich / sparse memory	1.000e6	1.000e15	1.000e6	10 ¹⁵	1	10 ^{13.4}	1

Table 5. Composite reversible finite shadows.

6. Exceptional findings

Reversibility changes the currency, not the accounting. It avoids direct heat only by keeping, exporting, or measuring the information that an irreversible primitive would have discarded.

Bennett checkpointing is powerful but finite. Finite reversible memory can make the shadow enormous, but not infinite.

Reliability reintroduces entropy. A reversible supertask with a positive error floor fails; summable reliability needs a correction budget growing like $N \log N$ in the exponential-suppression model.

Finite-time cleanup de-Zenozes the implementation. Once the active shadow is finite, the erasure-stress optimizer is burden-proportional, and uniform for equal burdens.

The reusable invariant is information-loss volume. It exposes where apparently heat-free implementations hide entropy.

7. Applications and targets

Reversible hypercomputer audits. Publish a ledger of lost preimage bits, retained garbage, entropy export, checkpoint memory, correction entropy, and final measurement compression.

Proof search and theorem proving. A reversible proof search can preserve its trajectory, but a proof-producing endpoint must still return, verify, compress, or erase enough information to expose the proof.

Thermodynamic compiler scheduling. For finite-shadow accelerators, allocate cleanup time in proportion to burden rather than preserving a Zeno tail.

Fault-tolerant reversible devices. Test whether concrete reversible and quantum error-correction schemes alter the $N \log N$ correction law or only its constants.

Reversible Weihrauch shadows. Add garbage, entropy, checkpoint, and cleanup coordinates to observer degrees.

Pebbling lower bounds for alternating observers. Combine reversible pebbling with frontier-width laws to locate the memory cost of fair alternating branches.

Measurement-compression theorem. Treat final measurement as a many-to-few map from a large reversible state space to a small classical answer and derive readout and entropy-screen laws from it.

Reproducibility note

The tables and figures in this reversible appendage are regenerated by `make_v12_reversible_entropy_experiments.py`. The models are diagnostic envelopes; the intended invariants are the finite shadows and optimization laws.

G Version 0.13 appendage: chronology fixed points, selector cost, and spectral finite shadows

This appendage tests a limit interface that is not primarily a Zeno endpoint. In a Deutschian closed-timelike-curve model, the chronology-violating register must satisfy a self-consistency condition. In the quantum case this has the form

$$\rho_{\text{CTC}} = \text{Tr}_{\text{CR}}(U(\rho_{\text{CTC}} \otimes \rho_{\text{CR}})U^\dagger),$$

and in the finite classical stochastic case it becomes the stationary-distribution equation

$$\pi = \pi P.$$

Deutsch introduced the fixed-point condition as a way to avoid paradoxes in the presence of closed timelike lines [1]. Aaronson and Watrous showed that polynomial-size Deutschian CTC computers have exactly PSPACE power in both the classical and quantum cases [2]. In the unbounded computability setting, Aaronson, Bavarian, Cubitt, Grewal, Gueltrini, O'Donnell, and Raat identify the corresponding power with Δ_2^0 , equivalently Turing reducibility to the halting problem, and frame the problem as approximate fixed points of computable Markov chains and quantum channels [3].

The blindspot pursued here is operational. A fixed point can be postulated, but a finite physical process must either select one among many fixed points or stabilize a unique fixed point with finite precision. Multiplicity creates a selection interface. Uniqueness creates a condition-number interface. The fixed-point analogue of the limit-interface conservation law is therefore

$\begin{aligned} \text{CTC computational power} \leq & \text{ordinary evolution} + \text{fixed-point selection} \\ & + \text{inverse spectral gap} + \text{global reset/entropy cost.} \end{aligned}$

The singular resource has moved from an infinite clock horizon to the set-valued map

$$P \mapsto \{\pi : \pi P = \pi\},$$

where the missing operation is not iteration but exact self-consistent selection.

G.1 Selection-free output and the fixed-point polytope

Let P be a finite Markov kernel on a state set S , and let $o : S \rightarrow \{0, 1\}$ be the observable answer bit. Define the stationary-output set

$$I(P, o) = \left\{ \sum_{s \in S} \pi(s) o(s) : \pi P = \pi \right\}.$$

Since the stationary set is a convex polytope and the output is linear, $I(P, o)$ is a closed interval. A CTC decision is *selection-free with margin* γ if

$$I(P, o) \subseteq [0, 1/2 - \gamma] \quad \text{or} \quad I(P, o) \subseteq [1/2 + \gamma, 1].$$

If this condition fails, the architecture has not yet produced an operational answer. It has produced a set of possible answers plus an implicit selector.

Proposition G.1 (Fixed-point output interval). *For a finite Markov kernel P and output o , all possible Deutschian stationary outputs are exactly the interval $I(P, o) = [m(P, o), M(P, o)]$, where m and M are obtained by two linear programs over the stationary polytope. For a deterministic map, the extreme values are attained by stationary distributions uniform on recurrent cycles.*

Proof. The equations $\pi P = \pi$, $\pi \geq 0$, and $\sum_s \pi(s) = 1$ define a compact convex polytope. The output expectation is a linear functional on that polytope, so its image is a compact interval and the extrema occur at extreme stationary distributions. For a finite deterministic map, each recurrent class is a directed cycle, and the invariant distribution on that class is uniform on the cycle; convex combinations of these cycle measures give all stationary distributions. $\square$

Candidate law G.2 (Selector-oracle law). *If a CTC architecture permits arbitrary selection among fixed points whose observable outputs differ, then the selector is an oracle-like resource. For any set $A \subseteq \mathbb{N}$, there is a computable family of two-output CTC instances for which a selector choosing the $A(n)$ -labelled fixed point computes $A(n)$.*

The construction is intentionally simple. Give the n th instance two absorbing recurrent states labelled 0 and 1. Both are stationary. A selector $\mathcal{S}_A$ that chooses the $A(n)$ state decides A . Multiplicity alone does no computation; the computational content lies in the selection rule. Deutsch’s maximum-entropy prescription is one possible rule, but whenever maximum entropy returns a mixture across contradictory answer classes, the readout becomes a near-threshold probability rather than a robust decision.

G.2 Unique fixed points and the inverse-gap interface

One can try to avoid selection by adding leakage, damping, noise, or thermalization so that the fixed point becomes unique. This replaces multiplicity by a spectral condition number. For an irreducible, aperiodic finite Markov chain, write

$$g(P) = 1 - \max\{|\lambda| : \lambda \in \text{spec}(P), \lambda \neq 1\}$$

for the absolute spectral gap. Markov-chain perturbation and mixing theory relate g^{-1} to relaxation and stationary-distribution sensitivity [4, 5]. In the present framework, g^{-1} is the amount of chronology resolution hidden inside the fixed-point postulate.

Candidate law G.3 (Chronology condition-number law). *A unique fixed-point CTC with output margin γ and model perturbation scale δ can be robust only when*

$$g(P) \gtrsim \frac{\delta}{\gamma}.$$

Equivalently, a family with $g(P_n) \rightarrow 0$ imports an inverse-gap singularity unless the output is invariant under the corresponding nearly stationary modes.

The constant depends on the norm and perturbation model, but the interface is not optional: stationary distributions become ill-conditioned when the spectral gap closes. It is useful to name the logarithmic condition number

$$\Xi(P) = -\log g(P)$$

the *chronology aperture depth*. A Zeno schedule accumulates aperture depth by making the remaining time R_n vanish. A fixed-point CTC accumulates aperture depth by making the self-consistency equation harder to stabilize.

G.3 Local chronology cycles and the square-root precision shadow

A minimal model of a chronology register carrying a long computational history is a lazy nearest-neighbor walk on an L -cycle:

$$P_L = (1 - \mu)I + \frac{\mu}{2}(S + S^{-1}), \quad 0 < \mu \leq 1,$$

where S is the cyclic shift. Its spectral gap is

$$g_L = \mu \left(1 - \cos \frac{2\pi}{L}\right) \sim \frac{2\pi^2\mu}{L^2}.$$

Consequently, an ordinary relaxation process needs on the order of

$$T_{\text{relax}}(\epsilon) \gtrsim \frac{1}{g_L} \log \frac{1}{\epsilon}$$

steps to suppress the slowest nonstationary mode. If a CTC fixed point represents an exponentially long PSPACE history, this relaxation scale is exponential in the represented history length. A self-consistency postulate may bypass that time, but then the bypass itself is the computational resource.

Theorem G.4 (Local-cycle finite-shadow law). *Assume that a local chronology cycle of length L has gap $g_L \sim 2\pi^2\mu/L^2$, output margin γ , and stationary-distribution sensitivity bounded at scale δ/g_L . Then robust certification requires*

$$L \lesssim \sqrt{\frac{2\pi^2\mu\gamma}{\delta}}.$$

Thus the robust history length grows only as $\delta^{-1/2}$, and the number of robust history-address bits grows only as

$$\log_2 L_{\text{rob}} \leq \frac{1}{2} \log_2 \frac{1}{\delta} + O(1).$$

Proof. The condition-number law requires, up to model-dependent constants, $\delta/g_L \lesssim \gamma$. Substituting $g_L \sim 2\pi^2\mu/L^2$ and solving for L gives the stated bound. Taking a binary logarithm gives the history-bit form. $\square$

Table 1: Local chronology-cycle gaps. The lazy walk uses $\mu = 1/2$. The relaxation estimate is $\log(10^6)/g_L$, and $\delta_{\text{crit}} = \gamma g_L$ uses margin $\gamma = 0.1$.

L	g_L	relaxation steps	δ_{crit}
8	0.146	94.3	0.0146
16	0.0381	363	0.00381
32	0.00961	$1.44e + 03$	9.61×10^{-4}
64	0.00241	$5.74e + 03$	2.41×10^{-4}
128	6.02×10^{-4}	2.29×10^4	6.02×10^{-5}
256	1.51×10^{-4}	9.17×10^4	1.51×10^{-5}
512	3.76×10^{-5}	3.67×10^5	3.76×10^{-6}
1024	9.41×10^{-6}	1.47×10^6	9.41×10^{-7}
4096	5.88×10^{-7}	2.35×10^7	5.88×10^{-8}
16384	3.68×10^{-8}	3.76×10^8	3.68×10^{-9}

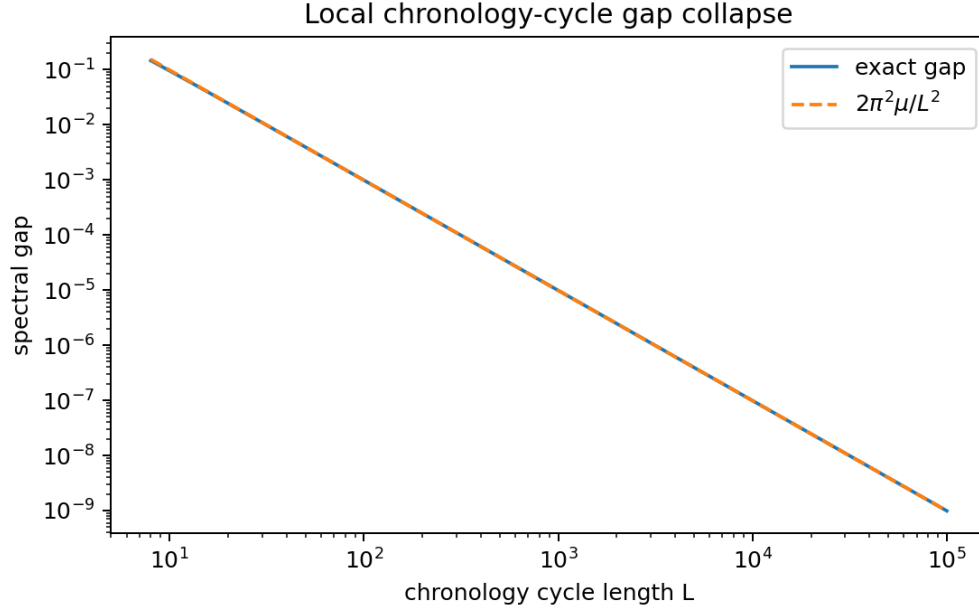


Figure 1: Spectral gap of the local chronology cycle. The L^{-2} law means that long self-consistent histories are stationary but increasingly ill-conditioned.

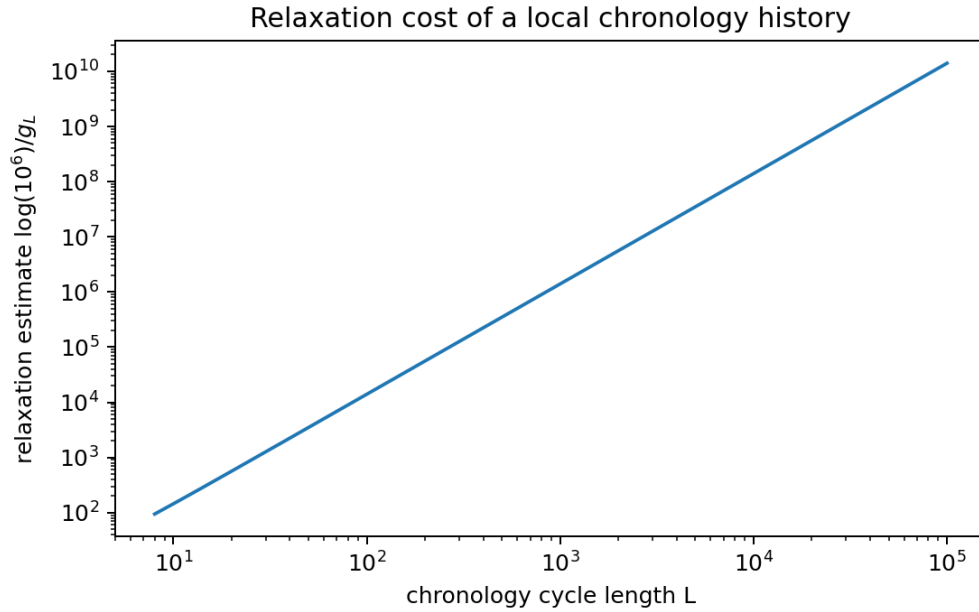


Figure 2: Relaxation estimate $\log(10^6)/g_L$ for suppressing the slowest mode to 10^{-6} . Fixed-point postulation avoids this cost by assumption; finite physical preparation does not.

Table 2: Robust history-address depth allowed by local-cycle perturbation precision. Parameters: $\mu = 1/2$, output margin $\gamma = 0.1$.

δ	L_{rob}	history bits
10^{-6}	993	9.96
10^{-9}	3.14×10^4	14.94
10^{-12}	9.93×10^5	19.92
10^{-15}	3.14×10^7	24.90
10^{-18}	9.93×10^8	29.89
10^{-24}	9.93×10^{11}	39.85
10^{-30}	9.93×10^{14}	49.82
10^{-40}	9.93×10^{19}	66.43
10^{-60}	9.93×10^{29}	99.65
10^{-100}	9.93×10^{49}	166.09

The precision table is a severe finite-shadow result. With $\mu = 1/2$ and margin $\gamma = 0.1$, perturbation control at $\delta = 10^{-30}$ protects only about fifty binary history-address bits in the local-cycle model. This does not refute the mathematical CTC complexity model; it identifies a hidden physical condition number that the ideal model treats as free.

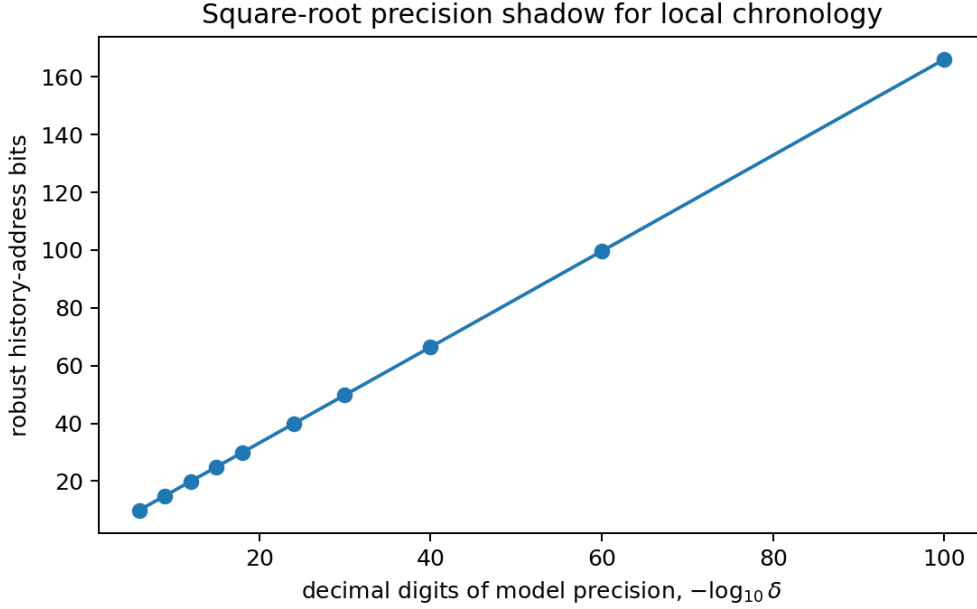


Figure 3: Robust history bits as a function of model precision. Each additional two decimal digits of perturbation control buy only about one additional history bit.

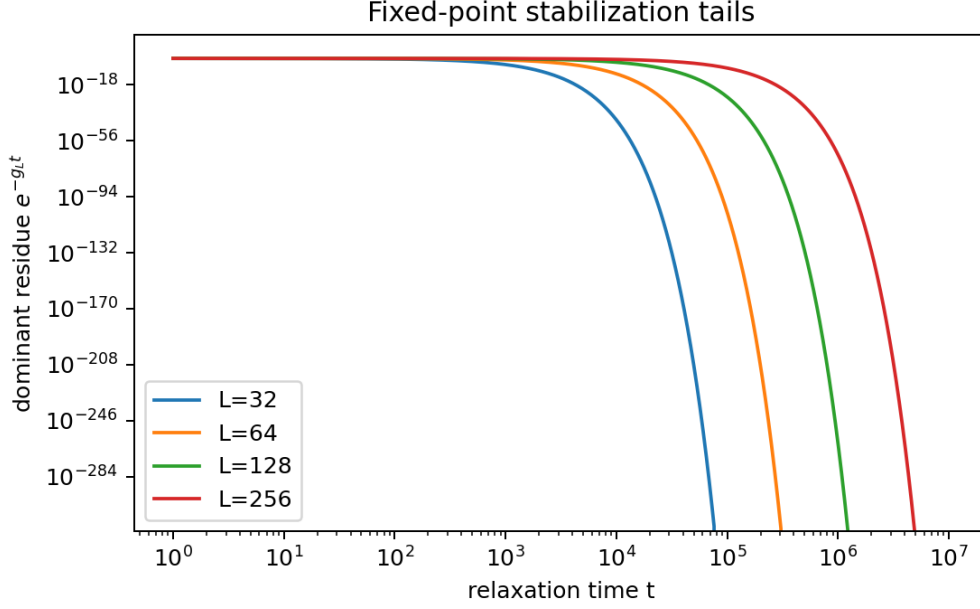


Figure 4: Dominant-mode residue $e^{-g_L t}$ for local chronology cycles. Increasing the history length by a factor of four increases the relaxation scale by roughly a factor of sixteen.

G.4 Multiplicity is typical in random deterministic fixed-point machines

The previous laws are worst-case diagnostics. A separate question is whether ambiguous stationary-output intervals are rare. To test this, random deterministic maps $f : S \rightarrow S$ were sampled, binary output labels were assigned independently, and the stationary-output interval was computed from the cycle averages of f . A deterministic map with several recurrent cycles has a stationary polytope whose vertices are those cycle measures. If the minimum and maximum cycle averages straddle $1/2$, then a yes/no answer requires a selector.

Table 3: Random deterministic-map stationary-output intervals with random binary labels. A selection-free margin uses $\gamma = 0.1$. The crossing fraction estimates how often the fixed-point polytope straddles the decision threshold.

$ S $	samples	mean cycles	mean width	crosses $1/2$	selection-free
16	500	2.16	0.427	0.590	0.400
32	500	2.46	0.477	0.664	0.300
64	500	2.84	0.549	0.724	0.220
128	500	3.09	0.546	0.748	0.198
256	500	3.35	0.557	0.780	0.124
512	500	3.78	0.591	0.840	0.084
1024	500	4.13	0.593	0.870	0.046
2048	500	4.44	0.622	0.874	0.042

The experiment suggests that fixed-point selection is not an exotic corner case. As the state space grows from 16 to 2048 states, the fraction of random instances whose stationary-output interval crosses the decision threshold rises from about 0.59 to about 0.87, while the fraction with a 0.1 selection-free margin falls from about 0.40 to about 0.04. The candidate empirical law is:

Generic recurrent fixed-point machines develop selector demand before they develop a large robust decision margin.

This is not a theorem about all ensembles, but it gives a useful warning sign. A proposed fixed-point computer should report not only one preferred fixed point but the stationary-output interval or an independently justified selector.

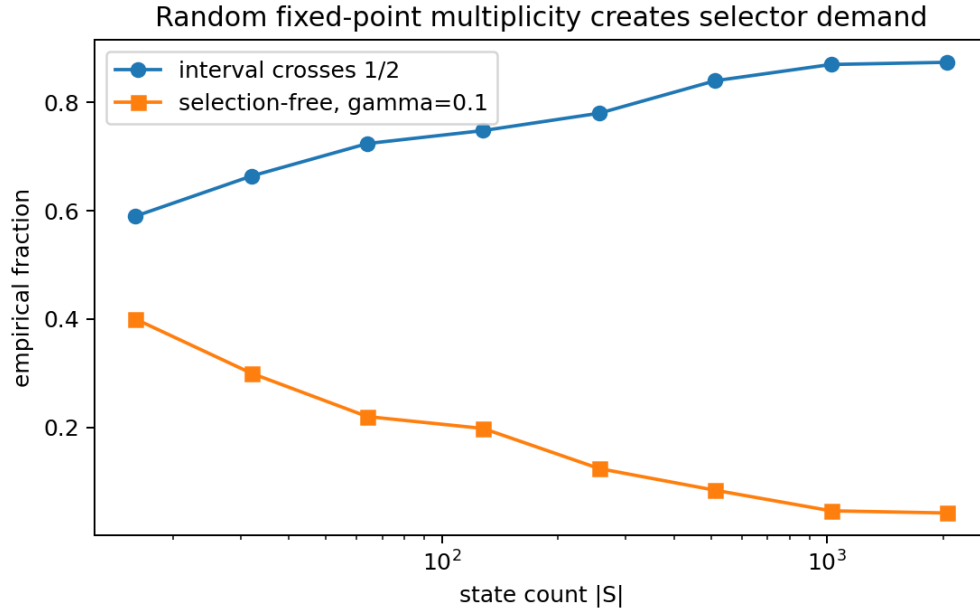


Figure 5: Empirical selector demand for random deterministic maps with random binary outputs. Threshold-crossing intervals become common, while selection-free margins become rare.

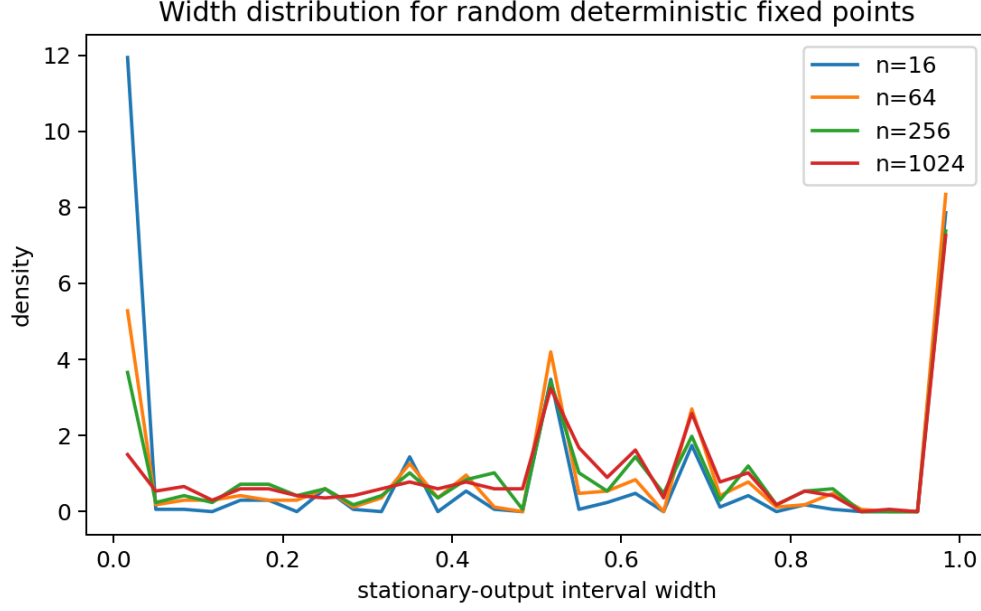


Figure 6: Distribution of stationary-output interval widths. Larger state spaces tend to offer more distinct recurrent averages, widening the selector burden.

G.5 Leaky uniqueness does not rescue spurious fixed points

The simplest model of fixed-point ambiguity is a two-state system with two nearly absorbing answer states. To force uniqueness, add leakage:

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}.$$

The unique stationary probability of answer 1 is

$$\pi_1 = \frac{a}{a+b}, \quad g = a+b.$$

If a and b are both small, the fixed point is unique but fragile. A perturbation $|\Delta a|, |\Delta b| \leq \delta$ changes π_1 by at most

$$|\Delta \pi_1| \leq \frac{\delta}{g-2\delta} \quad (2\delta < g).$$

Thus the limit $a, b \rightarrow 0$ does not converge to a robust physical selector. It converges to an infinite condition number.

Table 4: Two-state leaky selector with $\pi_1 = 0.9$. Shrinking $g = a + b$ preserves the formal answer while increasing sensitivity and relaxation cost.

$g = a + b$	a	b	$1/g$	δ_{crit}	relax
0.1	0.09	0.01	10	0.00833	138
0.01	0.009	1.00×10^{-3}	100	8.33×10^{-4}	$1.38e + 03$
0.001	9.00×10^{-4}	1.00×10^{-4}	$1e + 03$	8.33×10^{-5}	1.38×10^4
1.00×10^{-4}	9.00×10^{-5}	1.00×10^{-5}	1.00×10^4	8.33×10^{-6}	1.38×10^5
1.00×10^{-6}	9.00×10^{-7}	1.00×10^{-7}	1.00×10^6	8.33×10^{-8}	1.38×10^7
1.00×10^{-8}	9.00×10^{-9}	1.00×10^{-9}	1.00×10^8	8.33×10^{-10}	1.38×10^9
1.00×10^{-10}	9.00×10^{-11}	1.00×10^{-11}	1.00×10^{10}	8.33×10^{-12}	1.38×10^{11}
1.00×10^{-12}	9.00×10^{-13}	1.00×10^{-13}	1.00×10^{12}	8.33×10^{-14}	1.38×10^{13}

The table fixes $\pi_1 = 0.9$ while decreasing $g = a + b$. The answer remains formally the same, but the tolerated absolute perturbation and the relaxation time scale linearly with g and $1/g$. A tiny symmetry-breaking leak is not a free choice rule; it is an analog constant whose usable precision must be audited.

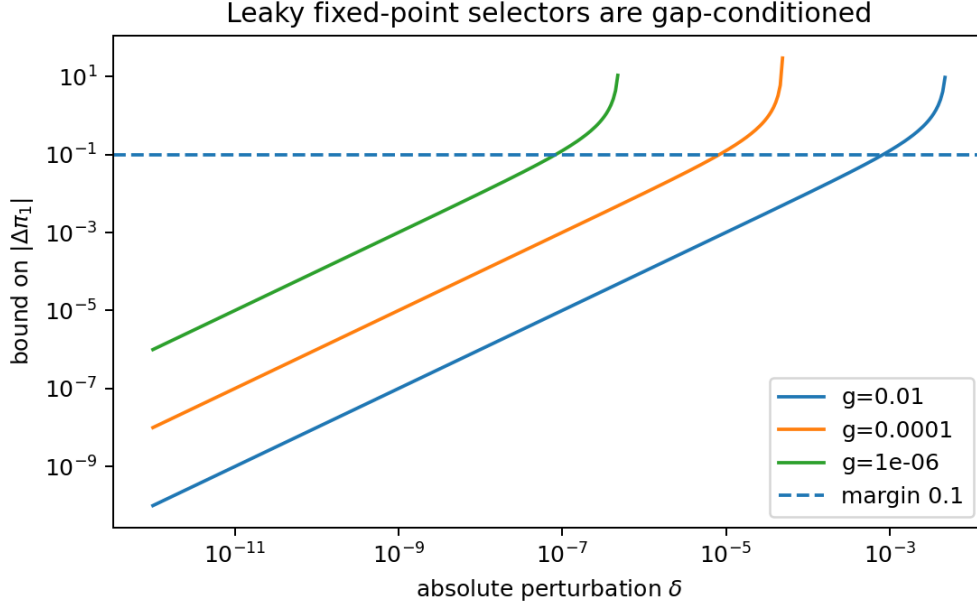


Figure 7: Two-state leaky selector sensitivity. The relevant small parameter is not the absolute perturbation but the perturbation divided by the spectral gap.

G.6 Gap injection shifts cost to global reset capacity

A natural countermeasure is to add a reset or teleportation move:

$$P_r = (1 - r)P + r\mathbf{1}q^T.$$

For a doubly stochastic local cycle and uniform q , the nonconstant eigenvalues are multiplied by $1 - r$, so the new gap is

$$g(P_r) = r + (1 - r)g(P).$$

This can keep the gap bounded away from zero even when L is large. But the reset is no longer local. A uniform reset over L possible chronology states requires a global address source with $\log_2 L$ bits of entropy or control per reset event. More structured reset distributions may reduce entropy, but then they import information about which recurrent regions matter.

Candidate law G.5 (Gap-injection conservation). *A closing chronology gap can be repaired by global reset, teleportation, or damping only by adding a new resource: reset entropy, global addressing, nonlocal coupling, or a prior over important fixed-point regions. Increasing g without changing the computational answer moves cost from inverse-gap stabilization into reset-channel complexity.*

Table 5: Gap injection by a global reset floor $r = 10^{-3}$. The gap no longer collapses, but a uniform reset over L chronology states imports $\log_2 L$ address/control bits.

L	local gap	gap with reset	reset bits
16	0.0381	0.039	4
64	0.00241	0.00341	6
256	1.51×10^{-4}	0.00115	8
1024	9.41×10^{-6}	0.00101	10
4096	5.88×10^{-7}	0.001	12
16384	3.68×10^{-8}	0.001	14
65536	2.30×10^{-9}	0.001	16

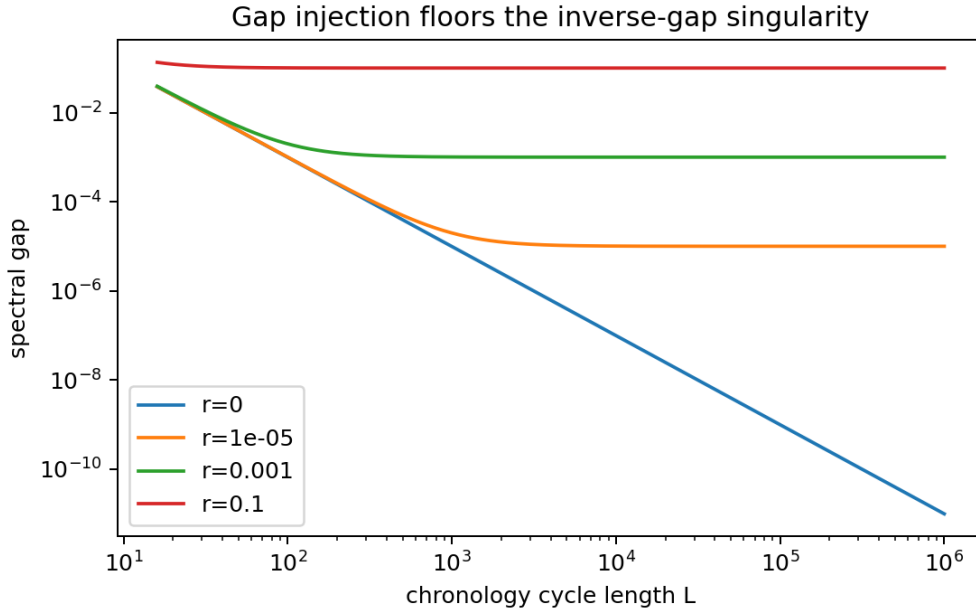


Figure 8: A global reset floor prevents the spectral gap from closing, but it does so by adding a nonlocal reset channel. The gap has been bought, not obtained for free.

G.7 Chronology robust-shadow equation

The preceding components combine into a fixed-point analogue of the robust-shadow equation:

$$N_{\text{CTC}} = \min\{N_{\text{sel}}, N_{\text{gap}}, N_{\text{mix}}, N_{\text{pert}}, N_{\text{reset}}, N_{\text{entropy}}\}.$$

Here N_{sel} is the amount of reliable information supplied by a selector over multiple fixed points; N_{gap} is the depth allowed by spectral gap; N_{mix} is the depth allowed by finite relaxation time; N_{pert} is the depth allowed by model precision; N_{reset} is the depth allowed by global reset/address capacity; and N_{entropy} is the depth allowed by finite state or receiver entropy. A proposed CTC computer is operationally meaningful only to the smallest of these depths unless the model postulates the missing term as exact.

Table 6: Illustrative chronology robust-shadow ledgers. The active history-depth is the smallest among perturbation precision, finite relaxation time, reset/address entropy, and fixed-point selector information.

case	δ	N_{pert}	N_{mix}	N_{reset}	N_{sel}	active
high precision, no reset	10^{-24}	39	19	∞	∞	19 mix
reset-limited	10^{-30}	49	49	20	∞	20 reset
mixing-limited	10^{-18}	29	13	∞	∞	13 mix
selector-limited	10^{-40}	66	66	80	12	12 selector
balanced finite	10^{-24}	39	26	40	50	26 mix

The equation separates three claims that are often conflated:

$$\begin{aligned}
\text{existential fixed point} & : \exists \pi \ (\pi P = \pi), \\
\text{selected fixed point} & : \pi = \mathcal{S}(P), \\
\text{stable fixed point} & : \pi \text{ is unique and robust under finite perturbations.}
\end{aligned}$$

The first claim is mathematical. The second requires a selector. The third requires a spectral, metrological, and entropy audit.

G.8 Exceptional findings from the chronology increment

Fixed-point selection is a limit interface. The CTC analogue of exact endpoint readout is exact fixed-point choice. If contradictory fixed points are allowed and no selection-free output condition is imposed, the selector can carry arbitrary noncomputable information.

Uniqueness can hide a singularity. A unique stationary distribution is not automatically robust. When the gap closes, the stationary distribution becomes an ill-conditioned function of the transition law. Small leaks resolve paradoxes algebraically but create an analog precision debt.

Local chronology histories have a square-root precision law. For the natural local-cycle model, robust history length scales as $\delta^{-1/2}$ and history-address depth as $\frac{1}{2} \log_2(1/\delta)$. This is the fixed-point counterpart of the Fisher, bandwidth, causal-screen, and timing finite-shadow laws developed earlier.

Random recurrent maps expose typical selector demand. The random-map experiment indicates that ambiguous stationary-output intervals become common as state count grows. This motivates a new reporting standard: any fixed-point accelerator should publish its stationary-output interval, not merely the output of a chosen simulator or selector.

Gap repair is resource migration. Global reset can make the gap large, but the reset channel must address or randomize over a large chronology state space. The cost moves from stabilization time to entropy, address capacity, nonlocal control, or prior design.

G.9 Applications

The fixed-point shadow laws apply beyond speculative time travel. Any engineered recurrent system that claims a computational advantage by settling into a self-consistent state can be audited by the same quantities: stationary-output interval, selector rule, spectral gap, perturbation margin, and reset entropy. This includes analog constraint solvers, recurrent neural computation, physical annealers, feedback model-checking circuits, theorem-proving portfolios that use equilibrium selection as a search primitive, and self-consistent program synthesis loops.

A constructive use is also visible. CTC-inspired finite accelerators may be valuable when the fixed-point polytope is selection-free and the gap is certified. For example, bounded model-checking instances with small recurrent frontiers can be compiled into stable stationary systems whose finite robust shadows return certificates or counterexamples. The audit equations identify when this is engineering rather than hidden hypercomputation.

G.10 Targets for further study

1. **Quantum-channel version.** Replace Markov spectral gap by the peripheral spectrum, Dobrushin coefficient, or Birkhoff contraction coefficient of a completely positive trace-preserving map. Prove a quantum chronology condition-number law in trace distance.
2. **Selector-degree taxonomy.** Classify fixed-point selectors using Weihrauch closed choice, cluster-point, and limit degrees. Determine which selector principles correspond to PSPACE-like, Δ_2^0 -like, and higher resources under finite descriptions.
3. **Spurious fixed-point certification.** Develop algorithms that certify selection-free output intervals for large implicit Markov chains without enumerating the stationary polytope.
4. **Gap-versus-entropy lower bounds.** Prove tradeoffs showing that any reset mechanism that raises the gap of an L -state local chronology history above a constant must import $\Omega(\log L)$ address entropy, nonlocal control, or problem-specific prior information.
5. **Finite-blocklength fixed-point solvers.** Combine Markov perturbation theory with finite-time statistical estimation to obtain nonasymptotic confidence intervals for stationary-output decisions.
6. **Causal-loop thermodynamics.** Quantify the heat, work, and entropy production needed to maintain a large gap in a chronology register whose natural local dynamics would have $g \sim L^{-2}$.

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H Version 0.14 appendage: adiabatic spectral shadows and ground-state limit interfaces

This appendage tests another escape route: perhaps the limiting operation can be placed not in an accelerating clock, an endpoint latch, an analog constant, or a chronology fixed point, but in the preparation and readout of a ground state of an infinite-dimensional Hamiltonian. Kieu’s proposed quantum adiabatic approach to Hilbert’s tenth problem explicitly uses a countably infinite Fock basis and attempts to identify the answer from the final ground state of a problem Hamiltonian [2, 1]. The adiabatic model is also a central ordinary model of quantum computation, where the running time is controlled by spectral gaps along the interpolation path [4, 5]. Smith’s critique of Kieu’s proposal emphasized decoy-state and convergence-time obstructions, while also noting that certain infinite-dimensional ground-energy tasks are themselves uncomputable [3]. Separately, Cubitt, Perez-Garcia, and Wolf proved that the spectral gap problem is undecidable for families of quantum spin systems [6].

The blindspot pursued here is operational. A ground-state oracle is not merely a Hamiltonian. It is a compound interface consisting of state preparation, gap protection, thermal suppression, finite-resolution energy measurement, and finite support in the physical degrees of freedom. The candidate conservation law is

$\begin{aligned} \text{adiabatic hypercomputational power} &\leq \text{ordinary unitary evolution} + \text{ground-state selection} \\ &\quad + \text{inverse spectral gap} + \text{zero-temperature readout} \\ &\quad + \text{infinite Fock support} + \text{exact energy measurement.} \end{aligned}$

Thus the limiting resource moves from an endpoint in time to an endpoint in spectrum. The new diagnostic quantity is not the remaining Zeno time R_n , but the smallest operational gap that protects the computationally relevant ground state from nearby alternatives.

H.1 Spectral-gap finite-shadow law

Let $H_N(s)$, $s \in [0, 1]$, be a finite truncation or finite active sector of an adiabatic proposal. Let

$$\Delta_N = \min_{s \in [0, 1]} (E_1^{(N)}(s) - E_0^{(N)}(s)), \quad \Lambda_N = \max_{s \in [0, 1]} \|\dot{H}_N(s)\|.$$

A simple stress model for adiabatic preparation is

$$T_N \geq K \frac{\Lambda_N}{\Delta_N^2},$$

where K absorbs the target adiabatic error and path regularity. More refined adiabatic bounds have additional derivative terms and sometimes stronger gap powers, but this form is sufficient as a conservative audit variable.

Candidate law H.1 (Spectral-gap finite-shadow law). *Given finite adiabatic time T , the certified active depth is*

$N_{\text{ad}}(T) = \max \left\{ N : K \frac{\Lambda_N}{\Delta_N^2} \leq T \right\}.$

If $\Delta_N \rightarrow 0$ or $\Lambda_N \rightarrow \infty$ along the sequence of sectors needed to encode an unbounded problem, then finite T certifies only a finite spectral shadow unless an additional singular preparation resource is supplied.

The gap plays the same role in spectral proposals that the tail time R_n plays in Zeno machines. A narrow gap is an energy-domain aperture. The aperture may be traversed in formal mathematics, but finite preparation time resolves only a finite prefix of the aperture tower.

gap model	$T = 10^6$	$T = 10^{12}$	$T = 10^{24}$
$\Delta_N = (N + 1)^{-1}$	999	1.00×10^6	1.00×10^{12}
$\Delta_N = (N + 1)^{-2}$, $\Lambda_N = N + 1$	14	250	6.31×10^4
$\Delta_N = e^{-0.05N}$	138	276	552
$\Delta_N = 1/\log(N + 3)$	$>10^{18}$	$>10^{18}$	$>10^{18}$
$\Delta_N \approx (\log 2)/(N + 1)^2$	25	831	8.33×10^5

Table 1: Adiabatic finite-shadow horizons for the stress law $T_N \geq \Lambda_N/\Delta_N^2$. Even enormous adiabatic time gives only logarithmic depth for exponential gaps and fourth-root depth for random spectral crowding.

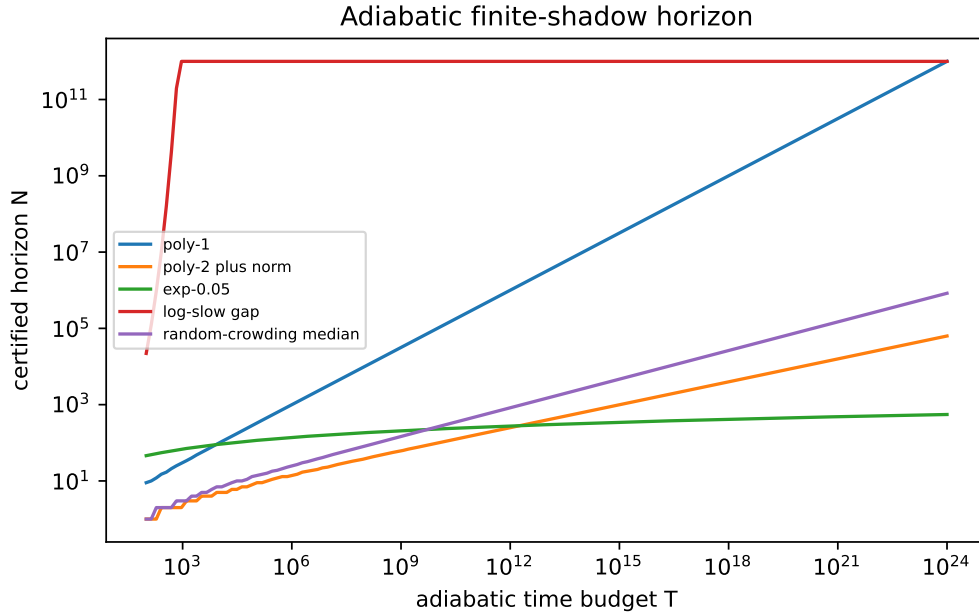


Figure 1: Adiabatic finite-shadow horizons under the stress law $T_N \geq \Lambda_N/\Delta_N^2$. Exponential gaps give logarithmic depth in the time budget. Random spectral crowding behaves like an effective $\Delta \sim M^{-2}$ law and therefore gives a fourth-root horizon.

The log-slow gap row is intentionally included as a counterexample to naive pessimism. If $\Delta_N = 1/\log(N + 3)$, then the adiabatic time horizon is enormous: N can grow like $\exp(\sqrt{T})$. This does not produce hypercomputation, because every finite T still gives a finite N , and the other shadows below can dominate. The finding is useful because it shows that not all spectral obstructions are equally severe; the active bottleneck must be measured rather than assumed.

H.2 Finite-temperature ground-state law

Ground-state readout also requires protection against thermal occupation of excited states. Suppose the relevant first excited band has effective degeneracy d_N and gap Δ_N . For a Gibbs-like endpoint

at inverse temperature β , the excited-to-ground statistical weight obeys the elementary comparison

$$\frac{p_{\text{exc}}}{p_0} \lesssim d_N e^{-\beta \Delta_N}.$$

To achieve ground-state probability at least $1 - \alpha$, it is therefore sufficient to demand

$$d_N e^{-\beta \Delta_N} \leq \frac{\alpha}{1 - \alpha},$$

or

$$\boxed{\beta \Delta_N \geq \log\left(\frac{d_N(1 - \alpha)}{\alpha}\right)}.$$

This gives a second horizon,

$$\boxed{N_{\text{temp}}(\beta, \alpha) = \max\left\{N : \beta \Delta_N \geq \log\left(\frac{d_N(1 - \alpha)}{\alpha}\right)\right\}}.$$

Even when the adiabatic path can in principle be followed, a finite-temperature endpoint cannot certify an arbitrarily small gap. Degeneracy adds a logarithmic look-elsewhere debt.

Energy measurement adds a third floor. If the measurement uncertainty is η , then neighboring alternatives separated by less than about 2η cannot be robustly classified:

$$\boxed{N_{\text{meas}}(\eta) = \max\{N : \Delta_N \geq 2\eta\}}.$$

Combining the three gives the spectral robust-shadow equation

$$\boxed{N_{\text{spec}} = \min\{N_{\text{ad}}, N_{\text{temp}}, N_{\text{meas}}, N_{\text{support}}, N_{\text{entropy}}\}}.$$

gap model	N_{ad}	N_{temp}	N_{meas}	N_{spec}	active bound
$\Delta_N = (N + 1)^{-1}$	1.00×10^6	7.09×10^4	5.00×10^5	7.09×10^4	thermal
$\Delta_N = (N + 1)^{-2}, \Lambda_N = N + 1$	250	336	706	250	adiabatic
$\Delta_N = e^{-0.05N}$	276	233	262	233	thermal
$\Delta_N = 1/\log(N + 3)$	$>10^{18}$	$>10^{18}$	$>10^{18}$	$>10^{18}$	adiabatic
$\Delta_N \approx (\log 2)/(N + 1)^2$	831	283	587	283	thermal

Table 2: Composite spectral shadow with $T = 10^{12}$, $\beta = 10^6$, $\eta = 10^{-6}$, $\alpha = 0.05$, and degeneracy proxy $d_N = N + 1$. The robust horizon is the smallest certified depth.

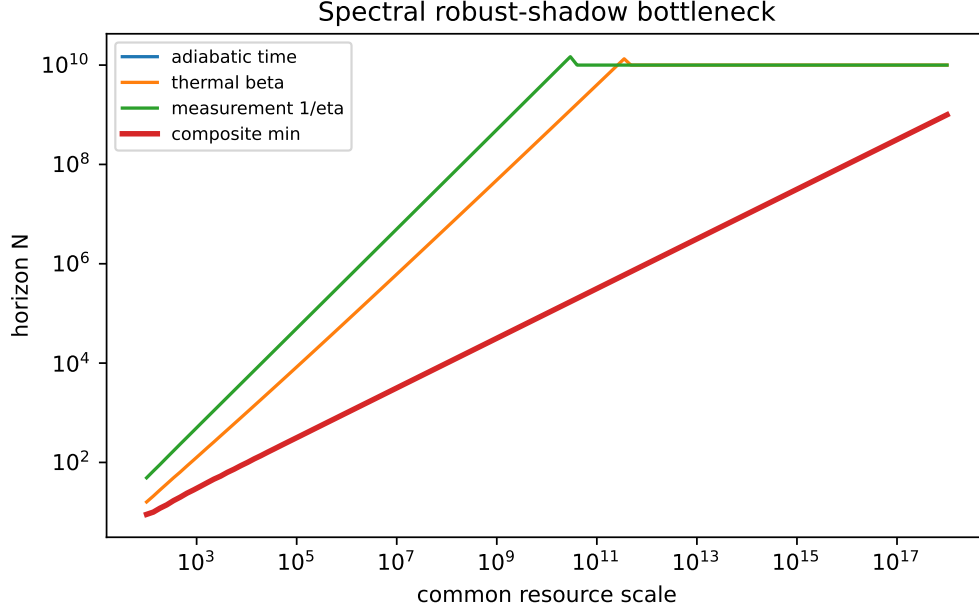


Figure 2: Composite spectral robust shadow for the polynomial gap model $\Delta_N = (N + 1)^{-1}$ when the common resource scale controls adiabatic time, inverse temperature, and measurement precision. The useful depth is the lower envelope, not the most favorable individual curve.

Remark H.2 (Undecidable gaps are not gap detectors). The undecidability of the spectral gap is a mathematical obstruction to an algorithm that classifies arbitrary Hamiltonian families. It is not by itself a physical oracle. A finite experiment observes finite sectors, finite times, finite temperatures, and finite resolution. It therefore implements a thresholded question such as “is the gap above the certified floor?” rather than the exact dichotomy “gapped or gapless in the thermodynamic limit.” The exact dichotomy remains a limit interface.

H.3 Spectral catalog capacity and the Lambert gap tax

A different spectral loophole is to encode many possible witnesses or answers as distinct energy levels inside a bounded window of width Ω . If M distinguishable addresses must fit in the window, the pigeonhole principle gives

$$\Delta_{\min} \leq \frac{\Omega}{M - 1}.$$

Thermal robustness with unknown address also requires roughly

$$\beta \Delta_{\min} \geq \log\left(\frac{M - 1}{\alpha}\right).$$

Combining the two yields

$$(M - 1) \log\left(\frac{M - 1}{\alpha}\right) \leq \beta \Omega.$$

Inverting with the Lambert W function gives the finite catalog law

$$M - 1 \leq \frac{\beta \Omega}{W(\beta \Omega / \alpha)}.$$

This is the spectral analogue of the endpoint-address capacity shadow. The logarithmic penalty appears because the identity of the protected level is not known in advance.

$\beta\Omega$	max catalog size M	Lambert approximation
1000	128	127.5
1.00×10^6	7.06×10^4	7.06×10^4
1.00×10^9	4.83×10^7	4.83×10^7
1.00×10^{12}	3.66×10^{10}	3.66×10^{10}

Table 3: Spectral catalog capacity from the joint constraints $\Delta \leq \Omega/(M-1)$ and $\beta\Delta \geq \log((M-1)/\alpha)$. The inverted law is $M-1 \lesssim \beta\Omega/W(\beta\Omega/\alpha)$.

H.4 Random spectral crowding law

The pigeonhole law is worst-case and can be too optimistic for unstructured spectra. If M candidate levels are placed randomly in a unit window, the minimum spacing is typically of order M^{-2} , not M^{-1} . A quick derivation is that the expected number of pairs within distance ϵ is approximately $M(M-1)\epsilon$. Hence

$$\Pr\{\Delta_{\min} > \epsilon\} \approx e^{-M(M-1)\epsilon},$$

so the median is approximately

$$\Delta_{\min,1/2} \approx \frac{\log 2}{M^2}.$$

This produces a sharper generic stress law:

$$T_{\text{ad}} \sim \Delta_{\min}^{-2} \sim M^4, \quad \beta_{\text{temp}} \sim \Delta_{\min}^{-1} \log M \sim M^2 \log M.$$

M	median min gap	10 pct.	90 pct.	$(\log 2)/M^2$	$1/M$
16	0.002794	4.16×10^{-4}	0.009025	0.002708	0.0625
32	6.94×10^{-4}	1.17×10^{-4}	0.002252	6.77×10^{-4}	0.03125
64	1.78×10^{-4}	2.69×10^{-5}	5.76×10^{-4}	1.69×10^{-4}	0.01562
128	4.42×10^{-5}	6.38×10^{-6}	1.42×10^{-4}	4.23×10^{-5}	0.007812
256	1.05×10^{-5}	1.57×10^{-6}	3.54×10^{-5}	1.06×10^{-5}	0.003906
512	2.61×10^{-6}	4.13×10^{-7}	9.03×10^{-6}	2.64×10^{-6}	0.001953
1024	6.62×10^{-7}	9.62×10^{-8}	2.33×10^{-6}	6.61×10^{-7}	9.77×10^{-4}

Table 4: Random spectral catalogs are much more crowded than the pigeonhole bound: the minimum spacing follows the M^{-2} scale rather than M^{-1} .

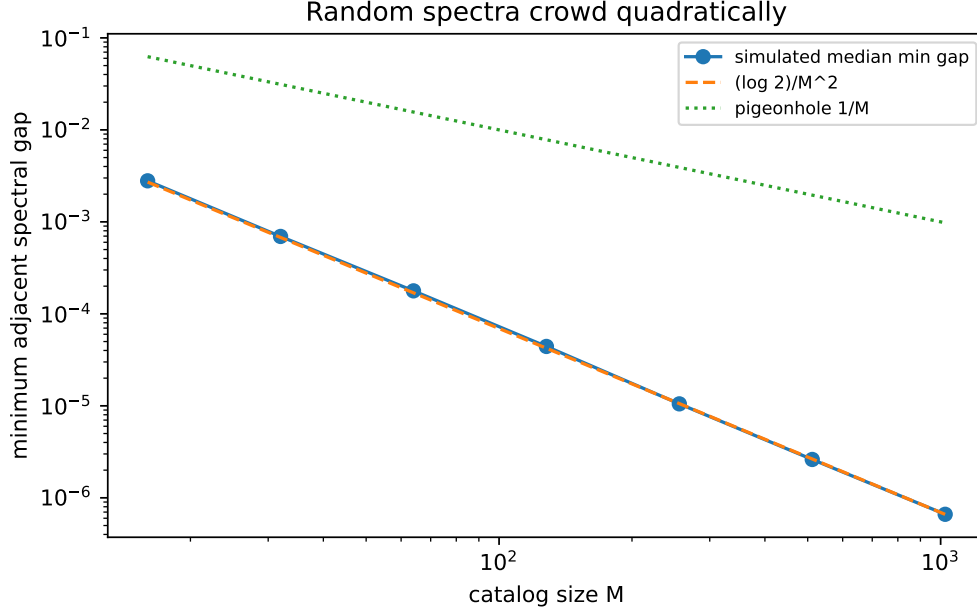


Figure 3: Random spectral catalog experiment with 2000 trials per catalog size. The minimum gap follows the M^{-2} median law rather than the M^{-1} pigeonhole scale. Unstructured spectral encodings are therefore quadratically more crowded than the weakest deterministic bound suggests.

H.5 Fock-support finite shadow

Infinite-dimensional adiabatic proposals often rely on a countably infinite occupation basis. Finite energy does not remove this basis, but it does limit how much of it can be occupied with high probability. Let $\hat{n}$ be an occupation-number observable and suppose a prepared state satisfies $\mathbb{E}[\hat{n}] \leq \bar{n}$. Then Markov's inequality gives

$$\Pr\{\hat{n} > N\} \leq \frac{\bar{n}}{N + 1}.$$

Thus any finite mean occupation has a finite high-confidence support cutoff

$$N_{\text{support}}(\delta) \leq \left\lceil \frac{\bar{n}}{\delta} \right\rceil - 1$$

in the conservative Markov sense. Coherent states are much more concentrated; if $\hat{n} \sim \text{Poisson}(\lambda)$, then the cutoff grows approximately as

$$N_{\delta} \approx \lambda + (2\lambda \log(1/\delta))^{1/2} + \frac{2}{3} \log(1/\delta).$$

The cutoff may be large, but it is finite for finite mean energy and fixed confidence.

coherent mean λ	Poisson N_δ	tail at N_δ	Markov bound	saddle estimate
10	28	7.6×10^{-7}	1.00×10^7	35.83
100	151	8.0×10^{-7}	1.00×10^8	161.8
1000	1154	9.1×10^{-7}	1.00×10^9	1175
1.00×10^4	1.05×10^4	9.7×10^{-7}	1.00×10^{10}	1.05×10^4

Table 5: Finite Fock-support shadows for a coherent-state occupation model. N_δ is the smallest cutoff for which $\Pr\{X > N_\delta\} \leq 10^{-6}$. The actual cutoff is far smaller than the Markov bound but remains finite for finite mean energy.

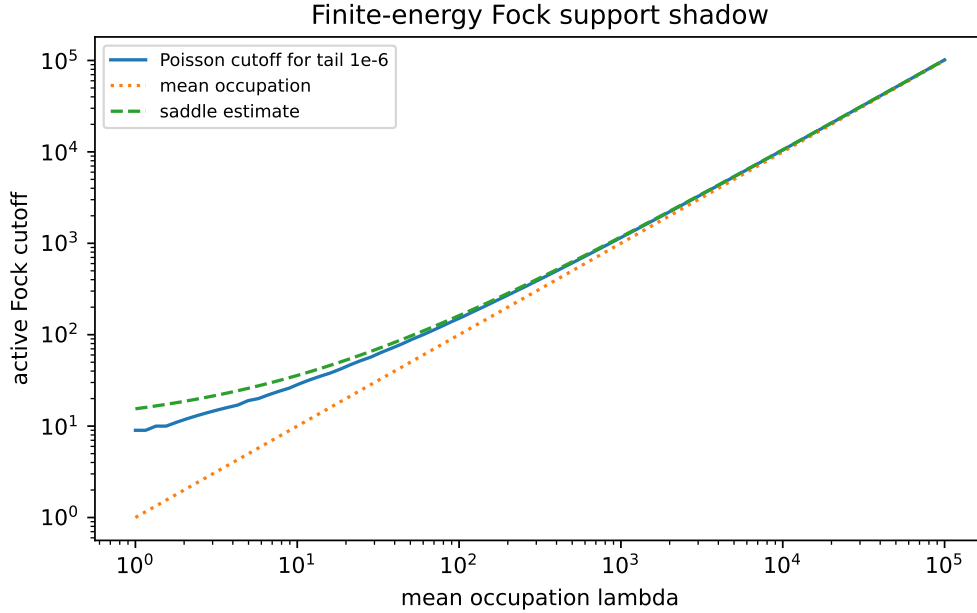


Figure 4: Finite Fock-support shadow for a coherent-state occupation model. The active high-confidence support grows nearly linearly with mean occupation, with a square-root-log correction. Infinite Fock space is not the same as infinite physically occupied support.

H.6 Exceptional findings from Version 0.14

The strongest findings of this appendage are as follows.

1. **Gap is the spectral aperture.** The Zeno tail R_n has a spectral analogue, Δ_N . Hypercomputation claims based on ground states must specify how the relevant gap is protected, measured, and thermally isolated.
2. **Spectral robustness is a minimum over several shadows.** Large adiabatic time is not enough. The certified horizon is

$$N_{\text{spec}} = \min\{N_{\text{ad}}, N_{\text{temp}}, N_{\text{meas}}, N_{\text{support}}, N_{\text{entropy}}\}.$$

A favorable gap law can be defeated by temperature, degeneracy, measurement floor, or finite support.

3. **Unknown spectral addresses pay a Lambert tax.** A bounded energy window and finite inverse temperature allow at most

$$M - 1 \leq \frac{\beta\Omega}{W(\beta\Omega/\alpha)}$$

robustly protected addresses. This is weaker than a linear $\beta\Omega$ count by a slowly varying but unavoidable factor.

4. **Random spectra crowd quadratically.** The simulation confirms the median law $\Delta_{\min} \approx (\log 2)/M^2$. A random spectral encoding therefore converts an address catalog of size M into adiabatic time cost of order M^4 under the inverse-square gap stress law.
5. **Finite energy creates a finite Fock shadow.** A countably infinite basis is not an infinite active search region. With finite mean occupation and fixed miss probability, the occupied sector has a finite certified cutoff.

H.7 Applications and targets generated by Version 0.14

Adiabatic hypercomputation audits. Any proposal based on ground-state readout should report the tuple

$$(\Delta_N, \Lambda_N, \beta, \eta, d_N, \bar{n}, N_{\text{spec}})$$

for its active sector. Without this tuple, the claimed oracle is not operationally specified.

Finite-dimensional spectral accelerators. Although the results do not support actual hypercomputation under finite resources, they do suggest practical finite-shadow accelerators: choose an active spectral catalog, allocate gaps according to prior or verification value, and stop at the composite spectral horizon.

Theorem-proving and counterexample search. For proof search, spectral encodings can be used as bounded witness catalogs. The random crowding law warns that unstructured encodings are poor; designed spectra with nearly uniform spacing are preferable when the endpoint must return a witness address.

Ground-state complexity versus physical extraction. Undecidable or very hard ground-state questions should be separated from physically extractable answers. The target is a theorem of the form: finite time, finite temperature, finite measurement resolution, and finite energy reduce an infinite-dimensional ground-state oracle to an ordinary computation with finite advice and a finite spectral cutoff.

Open mathematical targets. The next targets are:

1. Prove minimax versions of the random spectral crowding law for structured but unknown spectral encodings.
2. Combine the Bayesian endpoint waterline of Version 0.6 with spectral gap allocation, producing a *spectral waterline* that assigns larger gaps to higher-prior witnesses.
3. Replace the elementary thermal bound with finite-blocklength quantum state-discrimination bounds for low-temperature endpoint measurements.

4. Derive quantum Fisher versions of N_{meas} for estimating small ground-state energy differences.
5. Build an adiabatic finite-shadow benchmark suite for truncated Diophantine Hamiltonians, recording gap, decoy-state probability, and witness-support tails as the cutoff grows.

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15 Version 0.15 appendage: chaotic amplification shadows and fractal decision interfaces

This appendage tests another possible escape route. Instead of putting the limit interface in an endpoint clock, a ground-state gap, a fixed-point selector, or a real-valued oracle constant, one may place it in the sensitive dependence of a dynamical system. Moore’s work on generalized shifts and low-dimensional dynamics showed that ordinary dynamical systems can simulate Turing machines and can have undecidable long-term questions [1, 2]. Siegelmann and Sontag’s analog neural-network model showed that real-valued recurrent systems have computational structure beyond finite-state digital models when exact real values are admitted [3]. More recent work on universal analog computation also treats dynamical systems as a natural language for analog universality [4]. The blindspot pursued here is not whether such systems can encode complicated or undecidable behavior. They can. The question is how many semantically aligned bits can be robustly extracted from finite time, finite expansion, finite precision, and finite-noise observations.

The central finding is that chaotic amplification is a finite advice channel unless its integrated expansion, state precision, or observation precision becomes singular. Sensitive dependence is a bit pump: it can move small-scale initial information into macroscopic coordinates, but finite-time smooth dynamics pump only finitely many bits across a finite uncertainty floor.

15.1 Lyapunov advice-packing bound

Let $F_t : X \rightarrow Y$ be the finite-time map implemented by a dynamical system and let $O : Y \rightarrow Z$ be the observation map. Write $H_t = O \circ F_t$. Suppose the relevant one-dimensional advice manifold has diameter D , H_t is L_t -Lipschitz on it, and the final observation has uncertainty radius η . If $M = 2^N$ different N -bit advice prefixes are to be robustly identified, their observed signatures must be separated by at least 2η .

Theorem 15.1 (Lipschitz advice-packing bound). *For scalar advice encoded in an interval of diameter D , any robust extraction of $M = 2^N$ messages through an L_t -Lipschitz endpoint must satisfy*

$$2^N - 1 \leq \frac{L_t D}{2\eta}.$$

Equivalently,

$$N \leq \log_2 \left(1 + \frac{L_t D}{2\eta} \right).$$

Proof. Among M ordered code points in an interval of diameter D , some adjacent pair is separated by at most $D/(M-1)$. The corresponding endpoint signatures are separated by at most $L_t D/(M-1)$. Robust identification under uncertainty radius η requires every decisive pair to be separated by at least 2η . Rearranging gives the claim. $\square$

If $L_t \leq \exp \Lambda(t)$, where

$$\Lambda(t) = \int_0^t \lambda_+(s) ds$$

is an integrated positive expansion rate, then

$$N_{\text{Lip}}(t, \eta) \leq \log_2 \left(1 + \frac{D e^{\Lambda(t)}}{2\eta} \right).$$

Thus a finite-dimensional smooth chaotic amplifier has a finite bit horizon whenever $\Lambda(t) < \infty$ and $\eta > 0$.

Candidate law 15.2 (Lyapunov advice capacity). *The robust advice capacity of a scalar chaotic limit interface is bounded, up to additive resolution constants, by*

$$C_{\text{Lya}}(t, \eta) = \frac{\Lambda(t)}{\ln 2} + \log_2 \frac{D}{2\eta}.$$

Infinite advice extraction in finite external time therefore requires at least one of:

$$\Lambda(T) = \infty \quad \vee \quad \eta = 0 \quad \vee \quad D = \infty \quad \vee \quad \text{non-Lipschitz observation/state preparation.}$$

integrated expansion Λ	$\eta = 10^{-3}$	$\eta = 10^{-6}$	$\eta = 10^{-9}$
10	23	33	43
25	45	54	64
50	81	91	101
100	153	163	173

Table 1: Robust scalar advice horizon from $2^N - 1 \leq e^\Lambda/(2\eta)$. Finite integrated expansion and nonzero readout uncertainty give a finite bit horizon.

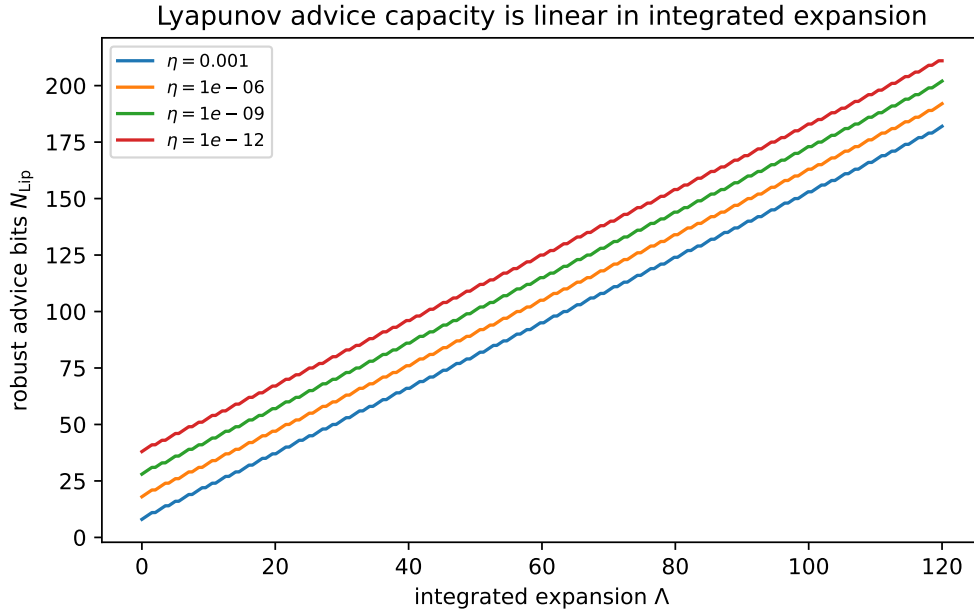


Figure 1: Robust advice depth as a function of integrated expansion. The relationship is nearly linear in $\Lambda/\ln 2$; each additional bit requires another factor of two in finite-time expansion or another factor of two in measurement precision.

This gives a useful reinterpretation of chaotic hypercomputation proposals. Chaos can reveal hidden initial bits, but it does not create them. It spends integrated expansion to move them above the observation floor. The hidden singularity is therefore either exact initial condition preparation, exact observation, or unbounded finite-time expansion.

15.2 Doubling-map digit extraction and the summability threshold

The doubling map

$$D(x) = 2x \pmod{1}$$

is the cleanest model of chaotic digit extraction. If

$$x = 0.b_1b_2b_3\cdots,$$

then, away from dyadic ambiguities,

$$b_{n+1} = \mathbf{1}\{D^n(x) \geq 1/2\}.$$

Thus the dynamics shifts hidden binary advice into the leading observable bit. With exact readout, this appears to expose an arbitrary oracle sequence. With finite readout noise, the oracle becomes a reliability problem.

Suppose the n -th threshold observation has independent uniform additive noise $U_n \sim \text{Unif}[-\eta_n, \eta_n]$ with $0 < \eta_n \leq 1/2$. For a uniformly distributed phase, the error probability is exactly

$$p_n = \frac{\eta_n}{2}.$$

Only phases within distance η_n of the decision threshold can be flipped, and averaging over the triangular flip probability gives the factor $1/2$.

Corollary 15.3 (Chaotic digit-reliability law). *The probability that an infinite sequence of shifted chaotic digits is read without error is*

$$P_{\text{clean}} = \prod_{n=1}^{\infty} \left(1 - \frac{\eta_n}{2}\right).$$

Assuming no single term is fatal, $P_{\text{clean}} > 0$ if and only if

$$\sum_{n=1}^{\infty} \eta_n < \infty.$$

Fixed readout noise and harmonic readout-noise decay both give zero probability of a wholly correct infinite advice stream.

noise radius η	per-bit error p	$K_{1/2}$	P_{100}	P_{1000}	P_{10000}
0.01	0.005	138.3	0.6058	0.006654	1.70×10^{-22}
0.001	5.00×10^{-4}	1386	0.9512	0.6065	0.00673
1.00×10^{-4}	5.00×10^{-5}	1.39×10^4	0.995	0.9512	0.6065
1.00×10^{-6}	5.00×10^{-7}	1.39×10^6	1	0.9995	0.995

Table 2: Doubling-map digit extraction with independent uniform readout noise. The per-bit error is $p = \eta/2$, so the probability of recovering all first K shifted digits is $(1 - p)^K$.

noise schedule	$\sum \eta_n < \infty$	infinite clean probability
$\eta_n = 10^{-3}$	no	0
$\eta_n = 10^{-2}/n$	no	0
$\eta_n = 10^{-2}/n^{1.2}$	yes	0.9724
$\eta_n = 10^{-2}/n^2$	yes	0.9918
$\eta_n = 10^{-2}2^{-n}$	yes	0.99

Table 3: Infinite digit extraction obeys the same summability pattern as the earlier reliability law. Fixed or harmonic readout noise gives zero probability of an entirely correct infinite advice stream.

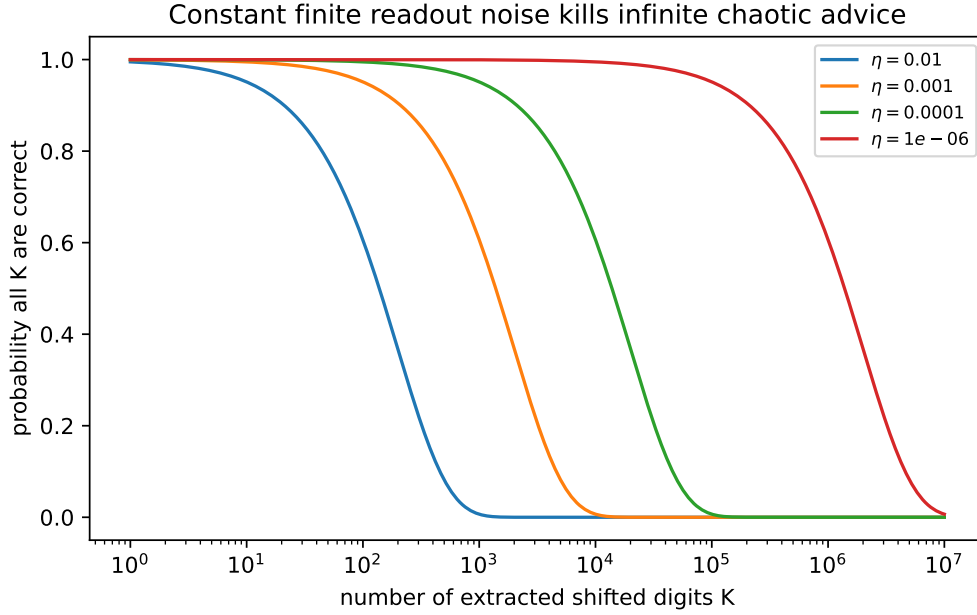


Figure 2: Clean-prefix survival for doubling-map digit extraction with constant readout noise. Even tiny nonzero noise eventually destroys the probability of a completely correct infinite sequence.

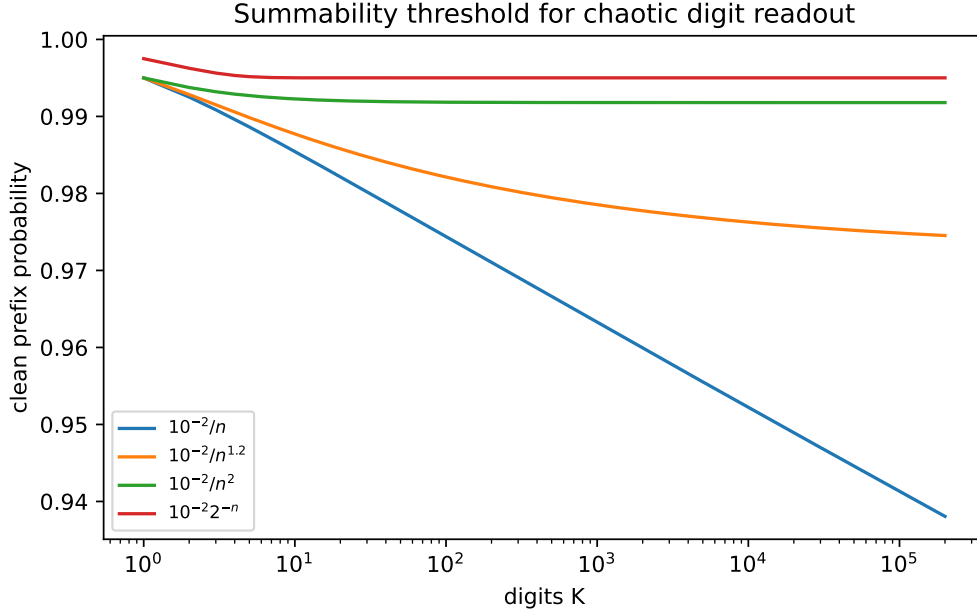


Figure 3: Summability threshold for chaotic digit extraction. Harmonic noise decay continues to erode the clean-prefix probability, while faster-than-harmonic schedules converge to a positive survival probability.

The exceptional point is that chaotic sensitivity and reliability pull in opposite directions. The dynamics amplifies hidden bits, but every bit exposed through a noisy threshold introduces another independent opportunity for irreversible semantic corruption. The same summability law that appeared for Zeno operations reappears for chaotic advice.

15.3 Fractal decision-boundary shadows

Chaotic analog computation can also hide its limit interface in a decision boundary rather than in an initial binary expansion. Let $A \subset [0, 1]^d$ be the set of initial states that lead to answer 1. The exact answer is the membership predicate $x \in A$. With input uncertainty radius η , points within distance η of the boundary ∂A cannot be uniformly certified, because the uncertainty ball contains both answers. Define the fractal gray zone

$$G_\eta(A) = (\partial A)_\eta.$$

For a distribution μ on inputs, $\mu(G_\eta(A))$ is an unavoidable ambiguous mass. If the boundary has Minkowski dimension d_b and finite nonzero Minkowski content, then typically

$$\mu(G_\eta(A)) \asymp \eta^{d-d_b}.$$

A boundary whose dimension approaches the ambient dimension is therefore a nearly maximal limit interface: its gray zone shrinks slowly, and if the boundary has full dimension with positive content, finite precision may leave a persistent ambiguous mass.

For the standard middle-thirds Cantor set $C \subset [0, 1]$, the boundary is C itself and $d_b = \log 2 / \log 3$. The exact η -neighborhood measure is obtained by subtracting the uncovered centers of

complementary gaps:

$$|C_\eta| = 1 - \sum_{k=1}^{\infty} 2^{k-1} (3^{-k} - 2\eta)_+.$$

Consequently

$$|C_\eta| = \Theta(\eta^{1-\log 2/\log 3})$$

up to the usual log-periodic oscillations of a self-similar set.

η	$ C_\eta $	$ C_\eta /\eta^{1-d}$	local slope
0.1	0.8667	2.027	
0.01	0.4363	2.387	0.298
0.001	0.1937	2.479	0.353
1.00×10^{-4}	0.08393	2.513	0.363
1.00×10^{-5}	0.03623	2.538	0.365
1.00×10^{-6}	0.01566	2.565	0.364
1.00×10^{-7}	0.006702	2.569	0.368
1.00×10^{-8}	0.002833	2.54	0.374

Table 4: Gray-zone mass for the standard middle-thirds Cantor boundary, where $d = \log 2/\log 3$. The exact complement-gap formula shows the expected η^{1-d} scaling modulated by log-periodic oscillations.

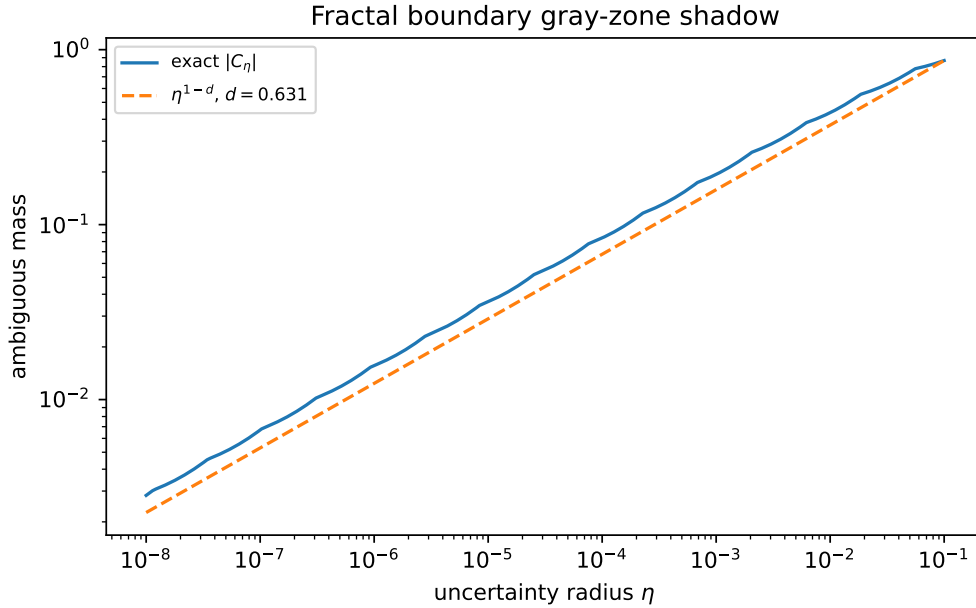


Figure 4: Gray-zone mass around the Cantor decision boundary. The shrinking of ambiguous mass is governed by codimension $1 - d_b$, not by the ambient dimension alone. Fractal boundaries therefore impose a slow precision tax.

This gives a second diagnostic for analog hypercomputation. A proposal may avoid an explicit oracle bitstream but still require exact membership in a fractal basin, exact asymptotic attractor

classification, or exact crossing of a dense boundary. The finite resource is then not the number of simulated steps but the measure and geometry of the gray zone.

15.4 Composite chaotic robust shadow

The chaotic case inherits all previous shadows, but it adds two particularly natural ones: finite integrated expansion and fractal boundary ambiguity. A useful audit equation is

$$N_{\text{chaos}} = \min\{N_{\text{Lip}}, N_{\text{dig}}, N_{\text{frac}}, N_{\text{state}}, N_{\text{energy}}, N_{\text{entropy}}\}.$$

Here N_{Lip} is the advice-packing horizon from finite expansion, N_{dig} is the prefix length surviving noisy digit extraction with a specified confidence, N_{frac} is the scale at which the decision-boundary gray zone is below the accepted residual risk, and N_{state} , N_{energy} , and N_{entropy} are inherited from analog metrology, clock/gradient cost, and finite endpoint capacity.

scenario	N_{Lip}	N_{dig}	N_{frac}	N_{chaos}	active bound
finite expansion	64	2.01×10^7	331	64	Lyapunov
noisy readout	149	2	110	2	digit noise
fractal boundary	183	2.00×10^9	56	56	fractal gray

Table 5: Toy composite chaotic shadows. N_{Lip} is the packing horizon, N_{dig} is the number of shifted digits recoverable with probability at least q , and N_{frac} is the dyadic scale compatible with the target Cantor gray mass.

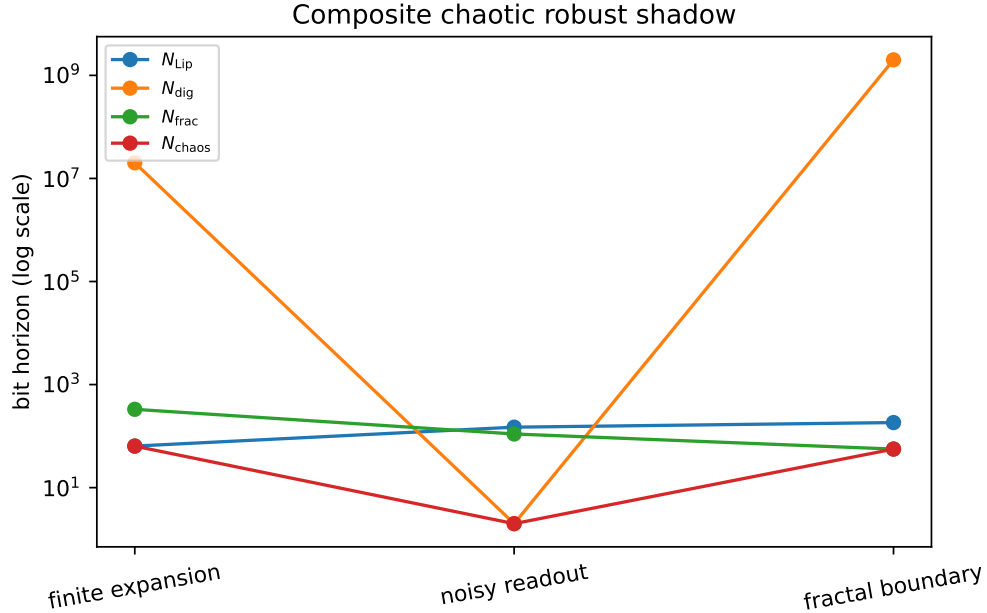


Figure 5: Toy composite chaotic shadows. Different regimes activate different bounds: finite expansion, readout reliability, or fractal boundary ambiguity. This mirrors the earlier robust-shadow pattern for Zeno, spectral, fixed-point, and quantum endpoints.

15.5 New interpretation and applications

The new conceptual claim is the *chaotic limit-interface conservation law*:

$$\begin{aligned} & \text{chaotic hypercomputational power} \\ & \leq \text{ordinary finite-time dynamics} + \text{exact initial condition} \\ & \quad + \text{integrated expansion} + \text{fractal boundary membership} \\ & \quad + \text{summably improving readout} + \text{finite-capacity endpoint decoding.} \end{aligned}$$

Finite positive Lyapunov exponents are not a route around the previous finite-shadow laws; they instantiate a new version of them. The finite resource is now expansion measured in bits. A chaotic system with integrated expansion Λ can reveal on the order of $\Lambda/\ln 2$ aligned bits beyond the measurement floor. It cannot reveal infinitely many aligned bits unless an infinite precision or infinite expansion assumption has been moved into the setup.

The most useful applications appear to be audits rather than impossibility slogans:

- **Analog neural networks and real-weight reservoirs.** Real weights or initial activations should be treated as finite advice channels whose robust semantic capacity is bounded by metrology, expansion, and noise, not by the formal cardinality of the reals.
- **Chaotic cryptography and random-number generation.** Chaotic unpredictability supplies physical entropy, but target-aligned oracle information requires semantic mutual information. The Lyapunov capacity separates entropy amplification from oracle extraction.
- **Hybrid-system verification.** Fractal basin boundaries create gray zones in which finite-precision verification is intrinsically distributional. The gray-zone mass gives an operational residual risk.
- **Theorem-proving and counterexample search.** Chaotic schedulers may be useful as finite portfolio diversifiers, but any claimed super-Turing behavior must identify the exact real parameter, fractal boundary, or infinite expansion that supplies the missing information.
- **Physical simulation of undecidable dynamics.** Moore-type undecidability results motivate exact mathematical questions. The robust experimental question is the finite shadow: what initial precision, observation precision, and expansion are certified?

15.6 Targets for further study

Several formal targets emerge from the chaotic appendage.

1. **Multidimensional Lyapunov packing.** Replace the scalar Lipschitz bound with singular-value growth and Oseledets splittings. A candidate bound is

$$N \lesssim \frac{1}{\ln 2} \sum_i \Lambda_i^+ + d \log_2(D/\eta),$$

where Λ_i^+ are positive finite-time expansion integrals.

2. **Entropy-rate shadows.** Relate robust advice throughput to topological or metric entropy, but with explicit endpoint resolution and semantic alignment terms.
3. **Fractal waterline allocation.** Given a prior over inputs and a measurement budget, allocate resolution to the thickest parts of a fractal boundary and derive the analog of the Bayesian endpoint waterline.

4. **Shadowing-certified analog computation.** Use interval arithmetic and shadowing lemmas to certify finite robust shadows of generalized shifts, recurrent neural networks, and chaotic ODE embeddings.
5. **Noise-summability engineering.** Determine whether physical repeated measurements can make $\sum \eta_n < \infty$ without importing divergent energy, memory, or bandwidth costs.
6. **Fractal-boundary benchmark suite.** Construct benchmark decision problems whose exact answer is known but whose boundary dimension can be tuned, then measure how robust finite-shadow performance scales with $d - d_b$.

The broader pattern is now clearer. Zeno machines, exact-real constants, adiabatic ground states, fixed-point loops, and chaotic amplifiers look different formally. Under finite robust observation, they all reduce to the same diagnostic form: identify the limit interface, compute the active finite shadow, and locate the singular resource that would be needed to push that shadow to infinity.

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16 Version 0.16 appendage: spatial parallelism shadows and aggregation bottlenecks

This appendage tests a route that is orthogonal to the temporal Zeno machine. Instead of compressing an infinite sequence of steps into finite time, a proposed device may try to spread infinitely many tests across space, membranes, agents, cells, processors, optical modes, or chemical compartments, and then aggregate the result.

Membrane-computing literature has explicitly considered accelerated rules and reliable systems in which all possible events happen when enough resources are present [1]. Calude and Paun's bio-inspired accelerated P-system work is a close comparison point because it isolates hypercomputational strength in a massively parallel/accelerated setting [2]. The blindspot pursued here is aggregation: a spatial swarm computes an infinite existential statement only if the absence, presence, and, in auditable variants, location of a witness can be carried to a finite endpoint.

The central finding is that infinite parallelism has its own limit interface. A spatial existential machine must certify silence across all inactive branches, prevent false alarms, carry a global OR through finite fan-in, and optionally return a witness address. Temporal Zeno machines concentrate risk near a horizon; spatial swarms distribute risk over a cut. The same finite-shadow pattern reappears.

16.1 Branch-parallel existential shadows

Consider the bounded existential shadow $F_N(x) = \text{OR}_{\{i < N\}} X_i$, where branch i tests a computable predicate or search region and emits a one-bit alarm X_i . The ideal infinite parallel version is $F_{\infty}(x) = \text{exists } i X_i$.

The physical implementation has at least four resource layers: branch verification reliability; aggregation fan-in and delay; spatial packing, signal propagation, and cooling; and endpoint capacity if a witness address is required.

The key asymmetry is that yes-instances may be witnessed by one successful branch, but no-instances require the silence of all branches. Infinite parallelism therefore turns 'nothing happened' into a global reliability claim.

$$F_N(x) = \bigvee_{i < N} X_i, \quad F_{\infty}(x) = \exists i X_i$$

Theorem 16.1 (Parallel false-positive summability).

On a no-instance, let branch i have false-alarm probability p_i . For independent branch alarms:

$$P_{\text{FP}}^{(N)} = 1 - \prod_{i < N} (1 - p_i)$$

Consequently, if no single branch is fatal:

$$\limsup_{N \rightarrow \infty} P_{\text{FP}}^{(N)} < 1 \iff \sum_{i=0}^{\infty} p_i < \infty$$

Without independence, the sufficient audit condition is the union bound:

$$P_{\text{FP}}^{(N)} \leq \sum_{i < N} p_i, \quad \sum_i p_i \leq \alpha$$

Proof sketch. The product formula is immediate for independent false alarms. The infinite product is positive exactly when the error series converges, assuming no finite prefix is already fatal. The union bound gives the dependence-free sufficient condition.

This theorem is the spatial dual of the earlier temporal reliability law. A Zeno sequence has infinitely many opportunities for a corrupt step. A parallel existential swarm has infinitely many opportunities for a false witness. In both cases, a constant nonzero error floor is fatal.

16.2 The Lambert swarm horizon

Suppose branch false alarms can be suppressed by spending verification cost c_i with $p_i = \exp(-\kappa c_i)$, and the total branch-suppression budget for N active branches is $B = \sum c_i$. Jensen's inequality shows that equal allocation minimizes the sum of false-alarm probabilities.

A no-instance audit with total false-positive budget α therefore requires $B \geq (N/\kappa) \log(N/\alpha)$. Inverting this inequality gives the Lambert swarm horizon. The active reliable branch count grows like $B/\log B$, not like B . The missing factor is the price of certifying silence over a large catalog.

$$\sum_{i < N} e^{-\kappa C_i} \geq N \exp\left(-\frac{\kappa B}{N}\right)$$

$$B \geq \frac{N}{\kappa} \log \frac{N}{\alpha}, \quad N_{\text{rel}}^{\parallel} \leq \frac{\kappa B}{W(\kappa B/\alpha)}$$

total B	N_rel^parallel	B/N	log(N/alpha)	N/B
1000	107	9.346	9.278	0.107
1.00e6	6.38e4	15.67	15.67	0.06382
1.00e9	4.50e7	22.23	22.23	0.04499
1.00e12	3.46e10	28.87	28.87	0.03463

Table 16.1. Parallel false-positive reliability horizon from $N \log(N/\alpha) \leq B$ with $\alpha=10^{-2}$ and $\kappa=1$.

branches N	equal p_i	cost/branch	total cost	total/N
1000	1.01e-5	11.51	1.151e4	11.51
1.00e6	1.01e-8	18.42	1.842e7	18.42
1.00e9	1.01e-11	25.32	2.532e10	25.32
1.00e12	1.01e-14	32.23	3.223e13	32.23

Table 16.2. Equal branch suppression needed for independent no-instance false-positive probability at most 10^{-2} .

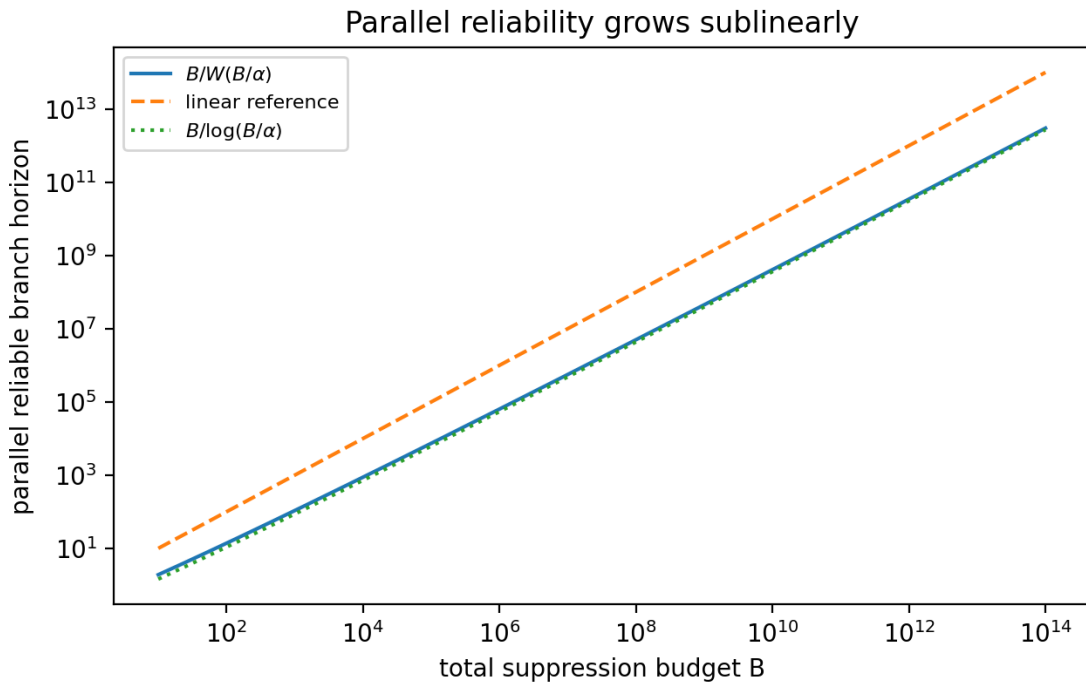


Figure 16.1. Parallel reliability horizon from $N \log(N/\alpha) \leq B$. Even exponentially effective branch verification gives sublinear reliable swarm growth.

$s \text{ in } p_i = 10^{-2}/i^s$	$N=10^2$	$N=10^4$	$N=10^6$	$N=\text{infinity}$
0.8	0.07822	0.2376	0.5268	1
1.0	0.05063	0.09331	0.1341	1
1.2	0.03546	0.04692	0.05146	0.05445
2.0	0.01627	0.01637	0.01637	0.01637

Table 16.3. False-positive avalanche for independent branch alarms. Harmonic or slower decay eventually gives a false yes with probability one.

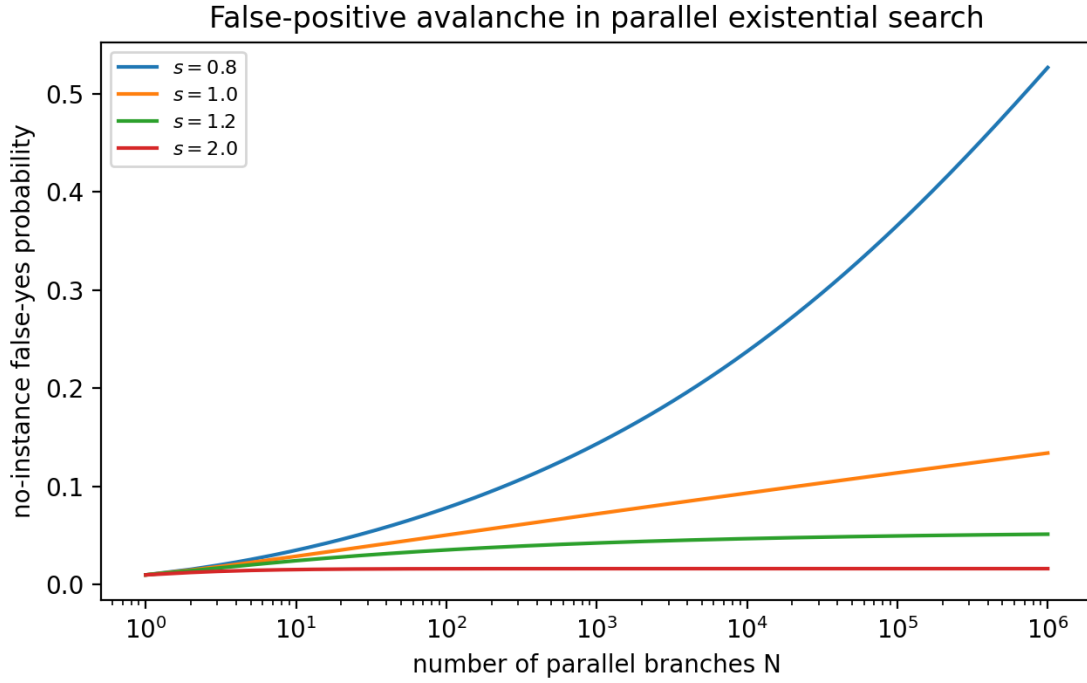


Figure 16.2. The no-instance becomes unreliable unless branch false-positive probabilities are summable.

The exceptional point is that the expensive input for parallel existential search is not the one containing an early witness. It is the no-instance, because no-instance correctness is a claim about the absence of arbitrarily many erroneous local alarms.

16.3 Fan-in latency and aggregation depth

A spatial swarm also needs a global OR. If the endpoint collector is built from gates of fan-in at most f , then a depth- D circuit output can depend on at most f^D input branches. Since OR_N depends on all N inputs, exact aggregation requires $D \geq \lceil \log_f N \rceil$.

With gate delay θ , the aggregation time is at least $\theta \lceil \log_f N \rceil$. Thus finite deadline T and bounded fan-in yield the finite horizon $N_{fan}(T, f, \theta) = f^{\lfloor T/\theta \rfloor}$. To aggregate infinitely many branches in finite time, a design must introduce unbounded fan-in, zero gate delay, nonlocal coupling, or an already-computed global field.

$$D \geq \lceil \log_f N \rceil, \quad T_{agg} \geq \theta \lceil \log_f N \rceil, \quad N_{fan} = f^{\lfloor T/\theta \rfloor}$$

depth D	$f=2$	$f=4$	$f=16$	$f=256$
10	1024	$10^{6.0}$	$10^{12.0}$	$10^{24.1}$
20	$10^{6.0}$	$10^{12.0}$	$10^{24.1}$	$10^{48.2}$
40	$10^{12.0}$	$10^{24.1}$	$10^{48.2}$	$10^{96.3}$
80	$10^{24.1}$	$10^{48.2}$	$10^{96.3}$	$10^{192.7}$

Table 16.4. Maximum number of input branches whose exact OR can reach one collector through fan-in f and depth D : $N \leq f^D$.

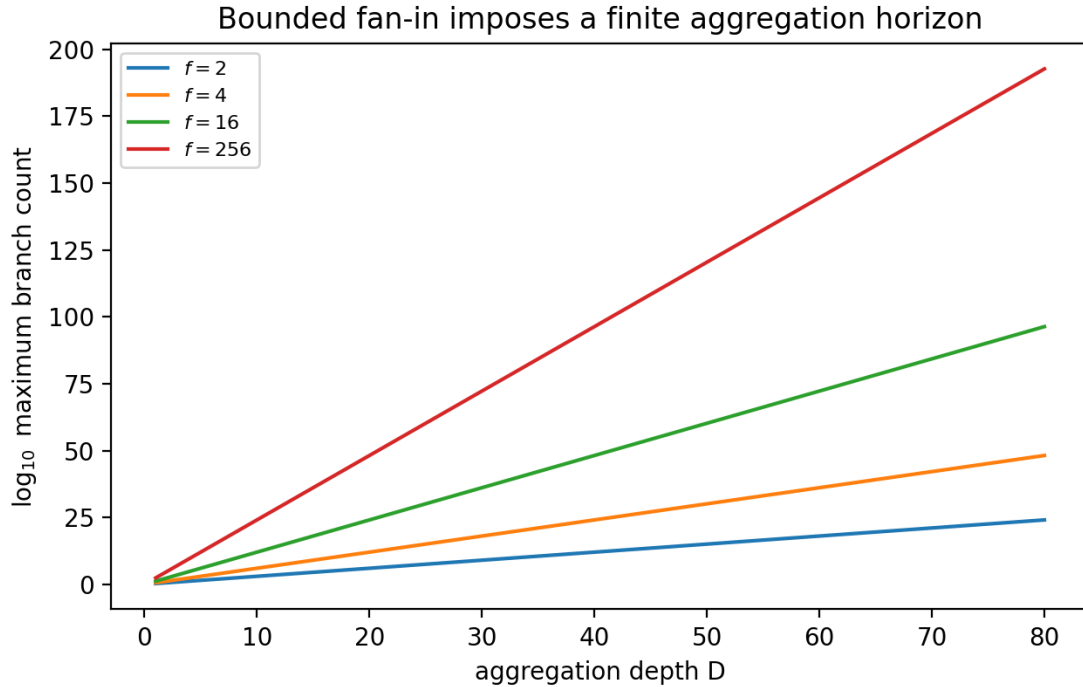


Figure 16.3. Bounded fan-in gives an exponential but finite aggregation horizon.

Candidate law 16.2 (Bounded-depth fan-in energy tax).

Assume an f -input aggregation gate costs at least $Q f^\beta$ with $\beta \geq 1$. A tree aggregating N branches with fan-in f requires at least $(N-1)/(f-1)$ gates. If the allowed depth is at most D , then $f \geq N^{1/D}$, and therefore:

$$E_N(D) \geq Q N^{1 + (\beta - 1)/D}$$

Fixed shallow aggregation is superlinear in N whenever high fan-in is more than linearly costly.

beta	depth D	required f	energy exponent	cost for N=10 ¹²
1	1	1.00e12	1.0000	10 ^{12.00}
1	2	1.00e6	1.0000	10 ^{12.00}
1	4	1000	1.0000	10 ^{12.00}
1	8	31.62	1.0000	10 ^{12.00}
1	16	5.623	1.0000	10 ^{12.00}
1	32	2.371	1.0000	10 ^{12.00}
1.5	1	1.00e12	1.5000	10 ^{18.00}
1.5	2	1.00e6	1.2500	10 ^{15.00}
1.5	4	1000	1.1250	10 ^{13.50}
1.5	8	31.62	1.0625	10 ^{12.75}
1.5	16	5.623	1.0312	10 ^{12.38}
1.5	32	2.371	1.0156	10 ^{12.19}
2	1	1.00e12	2.0000	10 ^{24.00}
2	2	1.00e6	1.5000	10 ^{18.00}
2	4	1000	1.2500	10 ^{15.00}
2	8	31.62	1.1250	10 ^{13.50}
2	16	5.623	1.0625	10 ^{12.75}
2	32	2.371	1.0312	10 ^{12.38}
3	1	1.00e12	3.0000	10 ^{36.00}
3	2	1.00e6	2.0000	10 ^{24.00}
3	4	1000	1.5000	10 ^{18.00}
3	8	31.62	1.2500	10 ^{15.00}
3	16	5.623	1.1250	10 ^{13.50}
3	32	2.371	1.0625	10 ^{12.75}

Table 16.5. Aggregation energy-depth tax under gate cost $Q f^\beta$. Fixed small depth forces high fan-in and superlinear cost when $\beta > 1$.

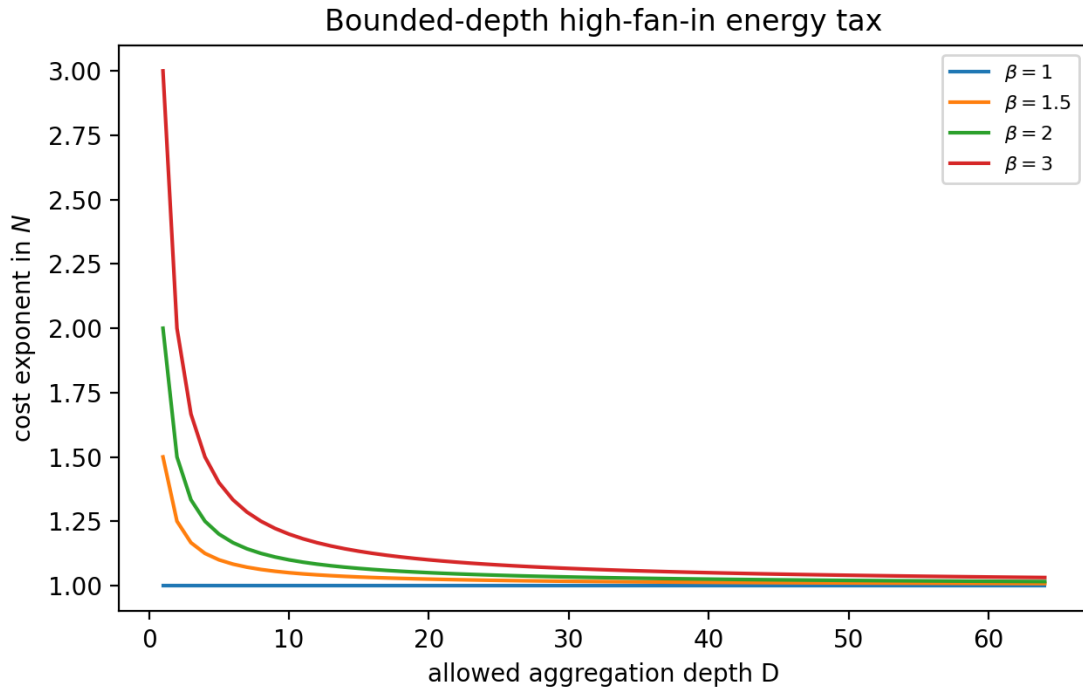


Figure 16.4. Energy exponent in the bounded-depth fan-in tax.

16.4 Spatial packing, cut capacity, and witness return

Finite signal speed and finite feature size give a geometric shadow. In d spatial dimensions, with collector deadline T , signal speed c , and minimum branch separation ℓ , all active branches that can causally affect the collector must lie in a ball of radius cT .

This simple bound is useful because infinite active parallelism in finite time requires zero feature size, infinite dimension, infinite signal speed, infinite spatial volume, or a nonlocal aggregation primitive. VLSI area-time lower-bound traditions make the same kind of warning precise for concrete layouts with finite-width wires and finite bandwidth [3].

If an auditable endpoint must return not merely 'yes' but also a witness address I in $\{0, \dots, N-1\}$ across a cut of capacity Φ bits per unit time during a collection window T_c , Fano's inequality gives a finite witness-address horizon. A yes/no OR endpoint can avoid address capacity only by refusing to return the witness; the cost then moves to a second certificate channel, a persistent branch latch, or an external verification replay.

$$N_{\text{pack}} \leq \phi_d \left(\frac{cT}{\ell} \right)^d$$

$$\log_2 N \leq \frac{\Phi T_c + h_2(\alpha)}{1 - \alpha}$$

A thermal or entropy-export cut gives an analogous finite horizon. If each active branch exports at least h units of entropy or heat and the enclosing surface exports at most $J_{\max}$ S per unit time, then:

$$N_{\text{cool}} \leq \frac{J_{\max} ST}{h}$$

16.5 Composite spatial-parallel shadow equation

The robust branch count of a spatial hypercomputer is not the number of formal branches. It is the smallest certified horizon across independent resource cuts:

$$N_{\parallel} = \min\{N_{\text{pack}}, N_{\text{fan}}, N_{\text{rel}}, N_{\text{cut}}, N_{\text{cool}}, N_{\text{power}}, N_{\text{cert}}\}$$

scenario	N_{pack}	N_{cut}	N_{rel}	N_{fan}	N_{pow}	N_{parallel}	active
lab swarm	1.00e9	5.83e12	6.38e4	1.10e12	1.00e9	6.38e4	reliability
reliability rich, narrow collector	1.00e18	4.96e7	4.09e8	1.05e6	1.00e12	1.05e6	fan-in
high fan-in collector	1.00e8	2.76e25	5.60e5	1.21e24	1.00e12	5.60e5	reliability
packing limited microvolume	1.00e6	2.76e25	3.46e10	2.14e96	1.00e14	1.00e6	packing

Table 16.6. Toy composite parallel shadows. The robust branch count is the smallest certified horizon among packing, cut/address capacity, reliability, fan-in, and power/cooling constraints.

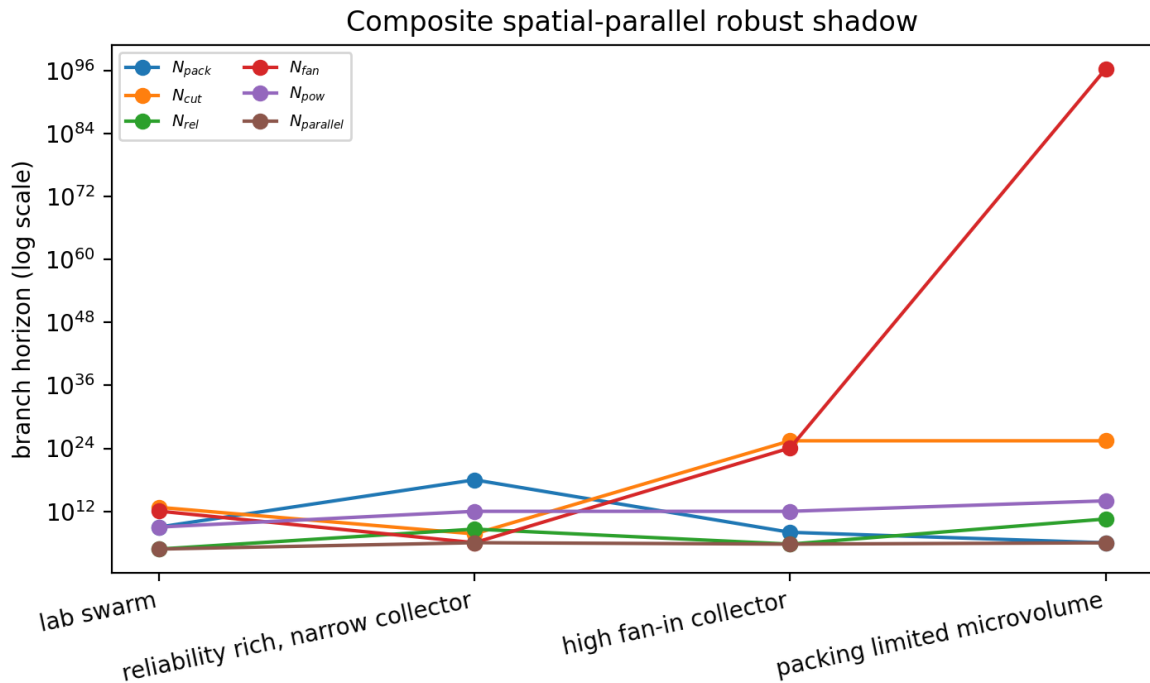


Figure 16.5. The active branch horizon is the minimum over packing, cut/address capacity, reliability, fan-in, and power/cooling constraints.

16.6 Branch-Zeno duality

The appendage suggests a dictionary between temporal and spatial limit interfaces: temporal Zeno tail corresponds to spatial branch catalog; late endpoint readout corresponds to global OR collector; step error summability corresponds to false-alarm summability; latch late-write horizon corresponds to branch alarm/latch horizon; witness-address debt corresponds to cut-capacity address debt; and Zeno de-acceleration after a finite shadow corresponds to ordinary finite parallel search after a finite branch shadow.

The new diagnostic phrase is parallel silence debt: every branch that is allowed to speak on a no-instance contributes to the cost of proving that none did. Infinite parallelism therefore cannot be evaluated only by asking whether all possible witnesses were examined. It must also be evaluated by asking how the global absence of false witnesses is physically certified.

16.7 Applications and targets for further study

Massively parallel theorem search. Finite branch shadows can still be useful: allocate reliable workers to high-prior proof lengths, theorem families, or counterexample classes, and expose the residual blind tail explicitly.

Model checking and counterexample search. The false-positive summability law is directly relevant to bounded but massive portfolios of approximate verifiers: a single spurious counterexample can dominate a global OR.

Membrane and chemical computing. The laws provide finite-resource semantics for systems whose formal description contains large or unbounded numbers of compartments, objects, or reactions.

VLSI and neuromorphic accelerators. Fan-in/depth and cooling shadows give architecture-level audit equations for global alarm buses and event-driven fabrics.

Distributed sensor networks. The no-instance is a silence-certification problem; branch false-positive budgets should be allocated as a summable resource rather than as an independent constant per sensor.

Open targets. Prove VLSI-style area-time-reliability lower bounds for OR aggregation with noisy gates, finite-width wires, and finite heat flux. Develop a spatial Weihrauch or resource-degree account distinguishing temporal limits, parallel existential branching, and finite fan-in aggregation. Solve the nonuniform branch-reliability waterline with simultaneous yes-prior utility and no-instance false-positive constraints. Extend the fan-in energy tax to quantum many-body collectors and analog threshold fields. Build benchmark suites in which formal branch count, reliable branch count, and auditable witness-return count are reported separately.

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